# Practice

MH1200 Linear Algebra I

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## 1. Some Learnings/Things to Remember

Theorem 1

**Theorem** Matrix Products produce Linear Combinations of Columns and Rows

The row vectors of $AB$ are linear combinations of the row vectors of $B$, and the column vectors of $AB$ are linear combinations of the column vectors of $A$.

Note the asymmetry!

For a sketch of the proof of this theorem: Divide the matrix into blocks or simply think about which part of the matrix product we keep fixed as we iterate through rows and columns.

**Theorem** Subspace Dimension is Bounded Above

Theorem 2

Suppose $U, V$ are vector spaces satisfying $U \subseteq V$. Then, $\dim(U) \leq \dim(V)$.

Kind of an obvious one, but it does turn up sometimes. To prove it, consider a basis on $V$ and realize that you can use this basis to build any spanning set of $U$. Thus, any basis on $U$ must be no bigger than the basis on $V$.

Theorem 3

**Theorem** Square Matrix is Invertible iff Column Vectors are Linearly Independent

A square matrix $M$ is invertible if and only if the column vectors of $M$ are linearly independent.

Here's a sketch of the proof: Use the characterization $M\vec{x} = \vec{0}$ only has the trivial solution to conclude that the column vectors of $M$ are linearly independent and vice versa.

Theorem 4

**Theorem** Nonzero Rows of Row Echelon Form form a Basis for the Row Space

The nonzero rows of the row echelon form of matrix $M$ form a basis for $\text{rowspace}(M)$.

Sketch of proof: Show that the elementary row operations preserve the row space. Then Gaussian Elimination immediately gives you the result.

**Theorem** The Linear Independence Equation

Theorem 5

A list of vectors $\vec{v_i}$ is linearly independent iff the only solution to the following equation is the trivial one:

$$\sum_{i=1}^{n} c_i \vec{v_i} = \vec{0}$$

Really basic, but sometimes you forget the basic stuff! Don't forget!

# 2. Other Things to Remember

1. If a problem asks for a careful proof or to prove from the definitions, make sure to actually prove from, e.g., the subspace axioms. Exercise good judgment on how much detail your proof must have.
2. Double check your computations. Your biggest source of errors is some computation mistake in your Gaussian Elimination.

# 3. Problem Set 2

## 3.1. Problem 1

(a)

$$\left(\begin{array}{ccc|c} 7 & 1 & 4 & -3 \\ 3 & 0 & -2 & 5 \\ 0 & -2 & 1 & 7 \end{array}\right)$$

(b)

$$\left(\begin{array}{cccc|c} 7 & 2 & 1 & -3 & 5 \\ 1 & 2 & 4 & 0 & 1 \end{array}\right)$$

(c)

$$\left(\begin{array}{cc|c} 2 & 0 & 0 \\ 3 & -4 & 0 \\ 0 & 1 & 1 \end{array}\right)$$

(d)

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right)$$

## 3.2. Problem 2

(b) and (c) only

## 3.3. Problem 3

$$c{=}2x + y + 3z$$

$$c{=}(x + z) + (x + y + 2z)$$

$$c{=}a+b$$

## 3.4. Problem 4

Augmented matrix:

$$\left(\begin{array}{ccccc|c} 2 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 2 & -5 & 5 \end{array}\right)$$

Parameterization:

$$x_5 = s$$

$$x_4 = \frac{5}{2} + \frac{5}{2}s$$

$$x_3 = t$$

$$x_2 = -\frac{7}{2} - \frac{7}{2}s$$

$$x_1 = \frac{7}{4} + \frac{5}{4}s + \frac{1}{2}t$$

## 3.5. Problem 5

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & 0 & -52 & -104 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

Thus, $x_1 = 3$, $x_2 = 1$, $x_3 = 2$.

## 3.6. Problem 6

$$\left(\begin{array}{cc|c} 2 & -3 & -2 \\ 2 & 1 & 1 \\ 3 & 2 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & -3 & -2 \\ 0 & 4 & 3 \\ 0 & \frac{13}{2} & 4 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & -3 & -2 \\ 0 & 4 & 3 \\ 0 & 13 & 8 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & -3 & -2 \\ 0 & 4 & 3 \\ 0 & 13-13 & 8-3\cdot\frac{13}{4} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & -3 & -2 \\ 0 & 4 & 3 \\ 0 & 0 & -\frac{7}{4} \end{array}\right)$$

This system is inconsistent.

## 3.7. Problem 7

(a)

$$\begin{cases} 2u = 4 \\ 3u - 2v = 3 \\ 6u + 4v + w + z = 0 \end{cases}$$

(b) The above matrix is not in row-echelon form. ~~To convert it to REF, perform~~ $R_1 \leftrightarrow R_3$~~.~~ To convert it to REF, perform:

1. $R_1 \leftarrow \frac{1}{2}R_1$
2. $R_2 \leftarrow R_2 - 3R_1$
3. $R_3 \leftarrow R_3 - 6R_1$
4. $R_3 \leftarrow R_3 + 2R_2$

(c)

$$u = 2$$

$$v = \frac{3}{2}$$

$$w = s$$

$$z = -18 - s$$

## 3.8. Problem 8

(i)

$$\left(\begin{array}{ccc|c} 1 & 0 & -5 & 1 \\ 0 & 10 & 2 & 7 \end{array}\right)$$

(ii)

The matrix above is in REF.

(iii)

$$x_3 = s$$

$$x_2 = \frac{7}{10} - \frac{1}{5}s$$

$$x_1 = 1 + 5s$$

(iv)

The solution set forms a 1-dimensional affine subspace embedded in $\mathbb{R}^3$.

## 3.9. Problem 9

**NAGKAMALI KA SA CALCULATION DITO**

(i)

Substituting in the coordinates gives the following system of linear equations:

$$\begin{cases} a - b + c = -2 \\ c = 1 \end{cases}$$

We can already clearly see that this system has infinitely many solutions, as the corresponding augmented matrix of the system:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Which can be further reduced to RREF as:

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

This matrix is now in RREF, but does not have any rows of zeroes. Thus, the system is consistent. In fact, since $b$ is a free parameter, there are infinitely many such solutions.

For infinitely many of these solutions, $a \neq 0$, so indeed there are infinitely many solutions.

(ii)

We introduce a new equation:

$$a + b + c = 2$$

Which we can encode to the RREF matrix we found in (i) by appending it as a row:

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 2 \end{array}\right)$$

Reducing, we get:

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 0 & 0 & 1 & 1 \\ 0 & 2 & 1 & 5 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Thus, we get:

$$a = -1, b = 2, c = 1$$

As the **unique** solution such that $ax^2 + bx + c$ interpolates through the three points.

Thus, there is only one solution in this case.

(iii)

We now know that $f(x) = -x^2 + 2x + 1$. Also, $f(2) = -4 + 4 + 1 = 1$. Thus, the new point is still on the quadratic. There is still exactly one solution in this case.

Note: In general, $n + 1$ distinct points on an $n$-degree polynomial uniquely determines the polynomial. The proof of this relies on the nonzero determinant of the Vandermonde matrix.

## 3.10. Problem 10

Note to self: be careful of problems like this! It is not asking for the subspace spanned by the three points, but rather the plane that passes through the three points!

In general, a linear equation in $\mathbb{R}^3$ looks like:

$$ax + by + cz = k$$

Substituting in the three points, we get the following augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & k \\ 1 & 0 & 1 & k \\ 0 & 1 & 1 & k \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{k}{2} \\ 0 & 1 & 0 & \frac{k}{2} \\ 0 & 0 & 1 & \frac{k}{2} \end{array}\right)$$

Thus, $a = b = c = \frac{k}{2}$. This gives us the plane:

$$\frac{k}{2}x + \frac{k}{2}y + \frac{k}{2}z = k$$

$$x + y + z = 2$$

## 3.11. Problem 11

Consider:

$$x + y = 4$$

$$x - y = 2$$

$$2x + 2y = 8$$

$$-x - y = -4$$

The augmented matrix is:

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & -1 & 2 \\ 2 & 2 & 8 \\ -1 & -1 & -4 \end{array}\right)$$

Performing Gauss-Jordan Elimination:

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & -1 & 2 \\ 1 & 1 & 4 \\ -1 & -1 & -4 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & -1 & 2 \\ 1 & 1 & 4 \\ 1 & 1 & 4 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & -1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

Notice that there are no free parameters. Thus, the only solution of this system is $y = 1, x = 3$.

The four equations are different, and every entry in the original augmented matrix is non-zero.

This satisfies all four conditions.

## 3.12. Problem 12

$$a + 2b + c = 0$$

$$-a + b + 2c = -3$$

$$a + 4b + 3c = -2$$

$$-a + 3b + 4c = -5$$

## 3.13. Problem 13

Since the solution set is a 1-dimensional Affine Subspace of $\mathbb{R}^3$, we must find two distinct equations that the entries satisfy.

The following equations work:

$$x = y + 4$$

$$z = -2y$$

## 3.14. Problem 14

Substituting four different values for $n$:

$$C(1) = a_1 + a_2 + a_3 + a_4 = 1$$

$$C(2) = 2a_1 + 4a_2 + 8a_3 + 16a_4 = 9$$

$$C(3) = 3a_1 + 9a_2 + 27a_3 + 81a_4 = 36$$

$$C(4) = 4a_1 + 16a_2 + 64a_3 + 256a_4 = 100$$

This corresponds to the following augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 2 & 4 & 8 & 16 & 9 \\ 3 & 9 & 27 & 81 & 36 \\ 4 & 16 & 64 & 256 & 100 \end{array}\right)$$

Reducing this matrix to RREF:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 6 & 24 & 78 & 33 \\ 0 & 12 & 60 & 252 & 96 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 0 & 6 & 36 & 12 \\ 0 & 0 & 24 & 168 & 54 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 0 & 1 & 6 & 2 \\ 0 & 0 & 4 & 28 & 9 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 0 & 1 & 6 & 2 \\ 0 & 0 & 0 & 4 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 0 & 1 & 6 & 2 \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 6 & 14 & 7 \\ 0 & 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1-\frac{1}{4}-\frac{1}{2}-\frac{1}{4} \\ 0 & 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right)$$

Thus, if the sum of the cubes of the first $n$ natural numbers is expressible by a polynomial, that polynomial must be:

$$C(n) = \frac{1}{4}n^2 + \frac{1}{2}n^3 + \frac{1}{4}n^4$$

To prove that this formula indeed works, we proceed by induction.

First, let us show that $C(1)$ gives us the sum of the first cube — namely, 1.

$$C(1) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$$

This proves the base case.

Now, let us show the inductive step. Suppose that $C(n)$ is the sum of the cubes of the first $n$ positive naturals. Then,we want to show that $C(n+1)$ is the sum of the cubes of the first $n+1$ positive naturals.

Since $C(n)$ is the sum of the cubes of the first $n$ positive naturals, we have:

$$C(n) = \frac{1}{4}n^2 + \frac{1}{2}n^3 + \frac{1}{4}n^4 = \sum_{i=1}^{n} i^3$$

Adding $(n+1)^3$ to both sides:

$$\frac{1}{4}n^2 + \frac{1}{2}n^3 + \frac{1}{4}n^4 = \sum_{i=1}^{n} i^3$$

$$(n+1)^3 + \frac{1}{4}n^2 + \frac{1}{2}n^3 + \frac{1}{4}n^4 = (n+1)^3 + \sum_{i=1}^{n} i^3$$

$$(n+1)^3 + \frac{1}{4}n^2 + \frac{1}{2}n^3 + \frac{1}{4}n^4 = \sum_{i=1}^{n+1} i^3$$

$$(n+1)^3 + \frac{1}{4}(n^2 + 2n^3 + n^4) = \sum_{i=1}^{n+1} i^3$$

$$(n+1)^3 + \frac{1}{4}n^2(1 + 2n + n^2) = \sum_{i=1}^{n+1} i^3$$

$$(n+1)^3 + \frac{1}{4}n^2(n+1)^2 = \sum_{i=1}^{n+1} i^3$$

$$(n+1)^2\left(n + 1 + \frac{1}{4}n^2\right) = \sum_{i=1}^{n+1} i^3$$

$$\frac{1}{4}(n+1)^2(n^2 + 4n + 4) = \sum_{i=1}^{n+1} i^3$$

$$\frac{1}{4}(n+1)^2(n+2)^2 = \sum_{i=1}^{n+1} i^3$$

$$\frac{1}{4}(n+1)^2((n+1)+1)^2 = \sum_{i=1}^{n+1} i^3$$

$$\frac{1}{4}(n+1)^2((n+1)^2 + 2(n+1) + 1) = \sum_{i=1}^{n+1} i^3$$

$$\frac{1}{4}((n+1)^4 + 2(n+1)^3 + (n+1)^2) = \sum_{i=1}^{n+1} i^3$$

$$\frac{1}{4}(n+1)^2+\frac{1}{2}(n+1)^3+\frac{1}{4}(n+1)^4=\sum_{i=1}^{n+1} i^3$$

$$C(n+1)=\sum_{i=1}^{n+1} i^3$$

Thus, we have shown that, if $C(n)=\sum_{i=1}^{n} i^3$, then $C(n+1)=\sum_{i=1}^{n+1} i^3$. By the principle of mathematical induction, we have:

$$\sum_{i=1}^{n} i^3 = C(n) = \frac{1}{4}n^2+\frac{1}{2}n^3+\frac{1}{4}n^4$$

## 3.15. Problem 15

**Theorem** Infinitude of Solutions — Theorem 6

A linear system cannot have at least two but finitely many solutions.

To prove this, we will prove some definitions and some intermediate lemmas:

**Definition** Distinct Solutions — Definition 7

$(y_i)_{i=1}^n$ and $(z_i)_{i=1}^n$ are two distinct solutions iff there exists $j$ satisfying the following:
1. $i \leq j \leq n$
2. $y_j \neq z_j$

**Lemma** Solutions Combine in Linear Functions — Lemma 8

Suppose that $(y_i)_{i=1}^n$ and $(z_i)_{i=1}^n$ are two distinct solutions of the following equation:

$$\sum_{i=1}^{n} a_i x_i = b$$

Then, for any $\lambda \in \mathbb{R}$, $(\lambda y_i + (1-\lambda) z_i)_{i=1}^n$ is also a solution of the equation above.

**Proof** Solutions Combine in Linear Functions

Proof 9

We are given that $(y_i)_{i=1}^n$ and $(z_i)_{i=1}^n$ are two distinct solutions of the following equation:

$$\sum_{i=1}^{n} a_i x_i = b$$

Thus:

$$\sum_{i=1}^{n} a_i y_i = b$$

$$\sum_{i=1}^{n} a_i z_i = b$$

Multiplying both sides of the first and second equations by $\lambda$ and $1-\lambda$, respectively:

$$\lambda \sum_{i=1}^{n} a_i y_i = \lambda b$$

$$\sum_{i=1}^{n} a_i (\lambda y_i) = \lambda b$$

$$(1-\lambda) \sum_{i=1}^{n} a_i z_i = (1-\lambda) b$$

$$\sum_{i=1}^{n} a_i ((1-\lambda) z_i) = (1-\lambda) b$$

Thus:

$$\sum_{i=1}^{n} a_i (\lambda y_i) + \sum_{i=1}^{n} a_i ((1-\lambda) z_i) = \lambda b + (1-\lambda) b$$

$$\sum_{i=1}^{n} (a_i (\lambda y_i) + a_i ((1-\lambda) z_i)) = (\lambda + (1-\lambda)) b$$

$$\sum_{i=1}^{n} a_i (\lambda y_i + (1-\lambda) z_i) = b$$

Thus, $(\lambda y_i + (1-\lambda) z_i)_{i=1}^n$ is also a solution of the linear equation, as needed. ■

**Lemma** Combinations of Distinct Solutions are Distinct

Lemma 10

Suppose that $(y_i)_{i=1}^n$ and $(z_i)_{i=1}^n$ are two distinct solutions.

Then, two solutions $(\lambda_1 y_i + (1-\lambda_1) z_i)_{i=1}^n$ and $(\lambda_2 y_i + (1-\lambda_2) z_i)_{i=1}^n$ are distinct iff $\lambda_1 \neq \lambda_2$.

**Proof** Combinations of Distinct Solutions are Distinct

Proof 11

Since $(y_i)_{i=1}^n$ and $(z_i)_{i=1}^n$ are two distinct solutions, there exists $j \in \mathbb{Z}$ satisfying:
1. $1 \leq j \leq n$
2. $y_j \neq z_j$

Now consider the function $f(\lambda) = \lambda y_j + (1-\lambda)z_j = \lambda\big(y_j - z_j\big) + z_j$, which gives the value of the $j$th entry of the solution $(\lambda y_i + (1-\lambda)z_i)_{i=1}^n$ as a function of $\lambda$.

Since $y_j \neq z_j$, $y_j - z_j \neq 0$, and thus $f(\lambda)$ is a non-constant linear function of $\lambda$. Thus, $f(\lambda)$ is a bijective function from $\mathbb{R}$ to itself.

Therefore, if $\lambda_1 \neq \lambda_2$, then $f(\lambda_1) \neq f(\lambda_2)$, so the solutions $(\lambda_1 y_i + (1-\lambda_1)z_i)_{i=1}^n$ and $(\lambda_2 y_i + (1-\lambda_2)z_i)_{i=1}^n$ are distinct.

Furthermore, if $\lambda_1 = \lambda_2$, the two solutions $(\lambda_1 y_i + (1-\lambda_1)z_i)_{i=1}^n$ and $(\lambda_2 y_i + (1-\lambda_2)z_i)_{i=1}^n$ are not distinct.

Thus, the two solutions are distinct iff $\lambda_1 \neq \lambda_2$, as needed. ■

We are now ready to prove the main theorem:

**Proof** Infinitude of Solutions

Proof 12

Suppose not. Then, the linear system has at least two, but finitely many solutions.

Let two of these solutions be $(y_i)_{i=1}^n$ and $(z_i)_{i=1}^n$. Both solutions satisfy all linear equations in the system. Since these solutions are distinct, by Lemma 8, $(\lambda y_i + (1-\lambda)z_i)_{i=1}^n$ is also a solution of the linear system for any $\lambda \in \mathbb{R}$.

Next, consider the set:

$$S = \{\lambda y_i + (1-\lambda)z_i \in \mathbb{R} \mid \lambda \in \mathbb{R}\}$$

By Lemma 8, all elements of the set are solutions of the system. By Lemma 10, there exists a bijection $f$ from $S$ to $\mathbb{R}$. Thus, $|S| = |\mathbb{R}|$, so $S$ is an infinite set, contradiction.

Therefore, there cannot be at least two but finitely many solutions of a linear system, as needed. ■

## 3.16. Problem 16

**Theorem** $(M_{n\times m}(\mathbb{R}), \sim)$ is a Setoid

Theorem 13

The set of $n \times m$ real matrices equipped with the row equivalence relation $\sim$ forms a setoid. (That is, $\sim$ is an equivalence relation over the set $M_{n\times m}(\mathbb{R})$)

Proof $(M_{n\times m}(\mathbb{R}), \sim)$ is a Setoid

Proof 14

For $(M_{n\times m}(\mathbb{R}), \sim)$, we have to verify that the axioms defining a setoid hold.

Suppose $X, Y, Z \in M_{n\times m}(\mathbb{R})$ are three particular but arbitrarily chosen matrices of the same shape.

Then, for any matrix $X$, we note that, clearly, $X \sim X$ since all matrices are row-equivalent to themselves.

Now, suppose that $X \sim Y$. Then, there is a sequence of row operations $R_1, R_2, R_3, ..., R_k$ that takes $X$ to $Y$:

$$X \overset{R_1}{\to} X_1 \overset{R_2}{\to} X_2 \overset{R_3}{\to} X_3 \overset{R_4}{\to} \cdots \overset{R_k}{\to} Y$$

Since row operations are invertible by other row operations, each row operation $R_i$ has a corresponding inverse $R_i^{-1}$ that is also a row operation (this was proven in lectures). Thus, there exists a sequence of row operations that takes $Y$ to $X$:

$$Y \overset{R_k^{-1}}{\to} \cdots \overset{R_4^{-1}}{\to} X_3 \overset{R_3^{-1}}{\to} X_2 \overset{R_2^{-1}}{\to} X_1 \overset{R_1^{-1}}{\to} X$$

Thus, $Y \sim X$. Therefore, if $X \sim Y$, then $Y \sim X$, implying that $\sim$ is a symmetric relation.

Finally, suppose $X \sim Y$ and $Y \sim Z$. Then, there exist row operations $R_1, R_2, R_3, ..., R_k$ and $S_1, S_2, S_3, ..., S_j$ such that:

$$X \overset{R_1}{\to} X_1 \overset{R_2}{\to} X_2 \overset{R_3}{\to} \cdots \overset{R_k}{\to} Y$$

And:

$$Y \overset{S_1}{\to} Y_1 \overset{S_2}{\to} Y_2 \overset{S_3}{\to} \cdots \overset{S_j}{\to} Z$$

Notice that we can concatenate both sequences together as follows:

$$X \overset{R_1}{\to} X_1 \overset{R_2}{\to} X_2 \overset{R_3}{\to} \cdots \overset{R_k}{\to} Y \overset{S_1}{\to} Y_1 \overset{S_2}{\to} Y_2 \overset{S_3}{\to} \cdots \overset{S_j}{\to} Z$$

Since $R_i$ and $S_i$ are both elementary row operations for all $i$, there exists a sequence of elementary row operations that take $X$ to $Z$. Therefore, $X \sim Y \wedge Y \sim Z \to X \sim Z$, implying that $\sim$ is a transitive relation.

Since $\sim$ is a reflexive, symmetric, and transitive relation over $M_{n\times m}(\mathbb{R})$, $\sim$ is an equivalence relation over $M_{n\times m}(\mathbb{R})$. Thus, $(M_{n\times m}(\mathbb{R}), \sim)$ is a setoid, as needed. ■

# 4. Problem Set 3

## 4.1. Problem 1

The corresponding augmented matrix is:

$$\left(\begin{array}{cccc|c} 0 & 10 & -4 & 1 & 1 \\ 1 & 4 & -1 & 1 & 2 \\ 3 & 2 & 1 & 2 & 5 \\ -2 & -8 & 2 & -2 & -4 \\ 1 & -6 & 3 & 0 & 1 \end{array}\right)$$

Reducing:

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 1 \\ 1 & 4 & -1 & 1 & 2 \\ 3 & 2 & 1 & 2 & 5 \\ -2 & -8 & 2 & -2 & -4 \\ 0 & 10 & -4 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 1 \\ 0 & 10 & -4 & 1 & 1 \\ 0 & 20 & -8 & 2 & 2 \\ 0 & -20 & 8 & -2 & -2 \\ 0 & 10 & -4 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 1 \\ 0 & 10 & -4 & 1 & 1 \\ 0 & 10 & -4 & 1 & 1 \\ 0 & 10 & -4 & 1 & 1 \\ 0 & 10 & -4 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 1 \\ 0 & 10 & -4 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 1 \\ 0 & 1 & -\frac{2}{5} & \frac{1}{10} & \frac{1}{10} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & \frac{3}{5} & \frac{3}{5} & \frac{8}{5} \\ 0 & 1 & -\frac{2}{5} & \frac{1}{10} & \frac{1}{10} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

## 4.2. Problem 2

The corresponding augmented matrix is:

$$\left(\begin{array}{ccc|c} 5 & -2 & 6 & 0 \\ -2 & 1 & 3 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & -\frac{4}{5} & \frac{12}{5} & 0 \\ -2 & 1 & 3 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & -\frac{4}{5} & \frac{12}{5} & 0 \\ 0 & \frac{1}{5} & \frac{27}{5} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 5 & -2 & 6 & 0 \\ 0 & 1 & 27 & 5 \end{array}\right)$$

Thus, a parameterization of this system is:

$$x_3 = t$$

$$x_2 = 5 - 27t$$

$$x_1 = 2 - 12t$$

## 4.3. Problem 3

$$\left(\begin{array}{cccc|c} 0 & 1 & -2 & 1 & 1 \\ 2 & -1 & 0 & -1 & 0 \\ 4 & 1 & -6 & 1 & 3 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 2 & -1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 1 & 1 \\ 4 & 1 & -6 & 1 & 3 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 2 & -1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 1 & 1 \\ 0 & 3 & -6 & 3 & 3 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 2 & -1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 2 & 0 & -2 & 0 & 1 \\ 0 & 1 & -2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & \frac{1}{2} \\ 0 & 1 & -2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

## 4.4. Problem 4

This system corresponds to the following augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 1 & a & 1 & 0 \\ a & 1 & 1 & 0 \end{array}\right)$$

When $a = 1$, this matrix reduces to:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

And thus there are infinitely many solutions in this case.

When $a \neq 1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 0 & a-1 & 1-a & 0 \\ 0 & 1-a & 1-a^2 & 0 \end{array}\right)$$

Since $a - 1 \neq 0$:

$$\left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & a+1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & a & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & a+2 & 0 \end{array}\right)$$

When $a = -2$, $a + 2 = 0$ and thus there are infinitely many solutions. Otherwise, back-substitution guarantees a unique solution.

In summary:

When $a = 1$ or $a = -2$, there are infinitely many solutions. Otherwise, there is exactly one solution.

## 4.5. Problem 5

This system corresponds to the following augmented matrix:

$$\left(\begin{array}{ccc|c} a & 9 & 2 & 2 \\ 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ a & 9 & 2 & 2 \\ 0 & a & 2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & 9-a^2 & 2-4a & 2-4a \\ 0 & a & 2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 9-a^2 & 2-4a & 2-4a \end{array}\right)$$

When $a = 0$, the matrix becomes:

$$\left(\begin{array}{ccc|c} 1 & 0 & 4 & 4 \\ 0 & 0 & 2 & 0 \\ 0 & 9 & 2 & 2 \end{array}\right)$$

This matrix has exactly one solution because it can be reduced to:

$$\left(\begin{array}{ccc|c} 1 & 0 & 4 & 4 \\ 0 & 9 & 2 & 2 \\ 0 & 0 & 2 & 0 \end{array}\right)$$

Which has no free parameters.

When $a \neq 0$, the matrix is:

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 9-a^2 & 2-4a & 2-4a \end{array}\right)$$

This can be further reduced with $R_3 \leftarrow R_3 - \dfrac{9-a^2}{a} R_2$

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 0 & 2-4a-\frac{18-2a^2}{a} & 2-4a \end{array}\right)$$

Since $a \neq 0$, we can multiply the bottom row by $a$:

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 0 & 2a-4a^2-(18-2a^2) & 2a-4a^2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 0 & -2a^2+2a-18 & 2a-4a^2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & a & 4 & 4 \\ 0 & a & 2 & 0 \\ 0 & 0 & a^2-a+9 & 2a^2-a \end{array}\right)$$

$a^2 - a + 9 = \left(a^2 - a + \dfrac{1}{4}\right) + \dfrac{35}{4} = \left(a - \dfrac{1}{2}\right)^2 + \dfrac{35}{4} \geq \dfrac{35}{4} > 0$, so the bottom row is never a zero row.

Thus, this system always has exactly one solution.

## 4.6. Problem 6

The augmented matrix corresponding to this system is:

$$\left(\begin{array}{ccc|c} 1 & 3 & 2a^4-32 & 37+2a \\ 0 & 4 & 0 & 28 \\ 2 & 6 & 5a^4-80 & 76+5a \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 3 & 2a^4-32 & 37+2a \\ 0 & 4 & 0 & 28 \\ 0 & 0 & a^4-16 & 76+5a \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 3 & 2a^4-32 & 37+2a \\ 0 & 1 & 0 & 7 \\ 0 & 0 & a^4-16 & 76+5a \end{array}\right)$$

Notice that when $a = \pm 2$, $76 + 5a \neq 0$ but $a^4 - 16 = 0$. Thus, the system has no solutions if and only if $|a| = 2$ and has exactly one solution otherwise (as back-substitution guarantees a unique solution).

## 4.7. Problem 7

The augmented matrix corresponding to this system is:

$$\left(\begin{array}{cc|c} a & b & f \\ c & d & g \end{array}\right)$$

If $a = 0$ and $c = 0$, then the system either has infinitely many solutions or is inconsistent.

If $a = 0$ and $c \neq 0$, then the system has a unique solution exactly when $b \neq 0$.

If $c = 0$ and $a \neq 0$, then the system has a unique solution exactly when $d \neq 0$.

If $a \neq 0$ and $c \neq 0$, then the matrix reduces to:

$$\left(\begin{array}{cc|c} a & b & f \\ c - a \cdot \frac{c}{a} & d - b \cdot \frac{c}{a} & g - f \cdot \frac{c}{a} \end{array}\right)$$

$$\left(\begin{array}{cc|c} a & b & f \\ 0 & d - b \cdot \frac{c}{a} & g - f \cdot \frac{c}{a} \end{array}\right)$$

Notice that, when $d - b \cdot \dfrac{c}{a} = 0$, then either the system has a free parameter or is inconsistent. Thus, the solution is unique iff $d - b \cdot \dfrac{c}{a} \neq 0$. Since, in this case, $a \neq 0$, the solution is unique iff $ad - bc \neq 0$.

In fact, if you check all cases, the solution is unique iff $ad - bc \neq 0$. The astute reader may observe that this expression is exactly the determinant of the $2 \times 2$ matrix.

## 4.8. Problem 8

The system corresponds to the following augmented matrix:

$$\left(\begin{array}{ccccc|c} 3 & 6 & 9 & 12 & (a^5+5) & a^4-1 \\ 1 & a+3 & a+4 & a+6 & 1 & 0 \\ 0 & 0 & 0 & 0 & a^2-1 & a+1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & a+3 & a+4 & a+6 & 1 & 0 \\ 3 & 6 & 9 & 12 & (a^5+5) & a^4-1 \\ 0 & 0 & 0 & 0 & a^2-1 & a+1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & a+3 & a+4 & a+6 & 1 & 0 \\ 0 & 6-3a-9 & 9-3a-12 & 12-3a-18 & (a^5+5)-3 & a^4-1 \\ 0 & 0 & 0 & 0 & a^2-1 & a+1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & a+3 & a+4 & a+6 & 1 & 0 \\ 0 & -3a-3 & -3a-3 & -3a-6 & a^5+2 & a^4-1 \\ 0 & 0 & 0 & 0 & a^2-1 & a+1 \end{array}\right)$$

We note that at least one of the middle three entries of the second row is nonzero because $-3a-3$ and $-3a-6$ cannot both be simultaneously be zero. Therefore, this augmented matrix is always in row-echelon form for any choice $a \in \mathbb{R}$. Furthermore, the last row is composed of all zeroes (except possibly the number to the right of the bar) exactly when $a = \pm 1$.

When $a = 1$, then the last row takes on the form $0, 0, 0, 0, 0 \mid 2$, which is clearly indicative of an inconsistent system.

When $a = -1$, all entries of the bottom row — including those to the right of the bar — become zero. Therefore, the system is consistent in this case.

Furthermore, when $a = -1$, then the system has three free parameters. Otherwise, if it is consistent, then it has two free parameters.

## 4.9. Problem 9

(i)

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + x_2 + x_3 = 1$$

$$x_1 + x_2 + x_3 = 2$$

$$x_1 + x_2 + x_3 = 3$$

(ii)

$$x_1 = 1$$

$$x_2 = 2$$

$$x_3 = 3$$

$$x_1 + x_2 + x_3 = 6$$

(iii)

$$x_1 + x_2 + x_3 = 6$$

$$x_2 + x_3 = 5$$

$$x_1 + 2x_2 + 2x_3 = 11$$

$$x_1 + 3x_2 + 3x_3 = 16$$

## 4.10. Problem 10

We can substitute in the given function values to arrive at a certain system of linear equations. The system corresponds to the following augmented matrix:

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 & 1 & 3 \\ 81 & 27 & 9 & 3 & 1 & 31 \\ 256 & 64 & 16 & 4 & 1 & 133 \end{array}\right)$$

Applying a modified version of Gauss-Jordan Elimination:

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 16 & 8 & 4 & 2 & 0 & 2 \\ 81 & 27 & 9 & 3 & 0 & 30 \\ 256 & 64 & 16 & 4 & 0 & 132 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 8 & 4 & 2 & 1 & 0 & 1 \\ 27 & 9 & 3 & 1 & 0 & 10 \\ 64 & 16 & 4 & 1 & 0 & 33 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 7 & 3 & 1 & 0 & 0 & 1 \\ 26 & 8 & 2 & 0 & 0 & 10 \\ 63 & 15 & 3 & 0 & 0 & 33 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 7 & 3 & 1 & 0 & 0 & 1 \\ 12 & 2 & 0 & 0 & 0 & 8 \\ 42 & 6 & 0 & 0 & 0 & 30 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 7 & 3 & 1 & 0 & 0 & 1 \\ 12 & 2 & 0 & 0 & 0 & 8 \\ 6 & 0 & 0 & 0 & 0 & 6 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 7 & 3 & 1 & 0 & 0 & 1 \\ 6 & 1 & 0 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 7 & 3 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

Thus, $f(x) = x^4 - 2x^3 + x + 1$

## 4.11. Problem 11

The following system is linear-in-form with the substitution $X = x^2$, $Y = y^2$ and $Z = z^2$.

The augmented matrix corresponding to this system is:

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 2 & 2 & -5 & 3 \\ 2 & 5 & 1 & 9 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 0 & 4 & -9 & -9 \\ 0 & 7 & -3 & -3 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 0 & 28 & -63 & -63 \\ 0 & 28 & -12 & -12 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 0 & 28 & -63 & -63 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 6 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Thus, $x = \pm 2$, $y = 0$, and $z = \pm 1$

## 4.12. Problem 12

There are either no solutions or infinitely many solutions.

If there are two equations that contradict one another, then there are no solutions as this system would result in a row with zeros (except for the number on the right of the bar, which will be nonzero).

Furthermore, if there are any solutions, since the augmented matrix is wider than it is tall (because it has more variables than equations), then there will be at least one column with a free parameter. Thus, the solution set will have infinitely many members in this case.

It is not possible for there to be one unique solution. If there were one unique solution, then there should be no free parameters. That is, every column (corresponding to a variable) should have a leading entry. This implies that there are at least as many equations as there are variables, which is not the case in this problem.

## 4.13. Problem 13

(i)

The augmented matrix corresponding to this system is:

$$\left(\begin{array}{ccccc|c} 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array}\right)$$

(ii)

Applying Gaussian Elimination to reduce this matrix into row-echelon form:

$$\left(\begin{array}{ccccc|c} 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array}\right)$$

(iii) Applying further elementary row operations to reduce this matrix to reduced row-echelon form:

$$\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array}\right)$$

(iv)

$$x_5 = s$$

$$x_4 = r$$

$$x_3 = 1 - s$$

$$x_2 = 1 - r - s$$

$$x_1 = -1 + s$$

(v) Swapping two columns in an augmented matrix is equivalent to swapping two coordinates of each solution in the solution set.

(vi) It is equivalent to dividing the entry corresponding to the column we multiplied of every possible solution by $c$.

(vii) This is equivalent to introducing a new free parameter to the system. Geometrically, one can think of this operation as extruding the set of solutions along a new dimension.

## 4.14. Problem 14

(i) $(1,1,1),(2,2,2),(3,3,3),(1,1,2),(2,2,3)$ (ii) For $(3,3,3)$ to be in the set, the following system has to have at least one solution:

$$\left(\begin{array}{cc|c} a & 1 & 3 \\ 1 & 1 & 3 \\ 1 & b & 3 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ a & 1 & 3 \\ 1 & b & 3 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1-a & 3-3a \\ 0 & b-1 & 0 \end{array}\right)$$

Notice that, if $b \neq 1$, then $b-1 \neq 0$ and thus $s = 0$. Therefore, $0 = (1-a)s = 3-3a$ and thus $a = 1$.

Further, if $a \neq 1$, then $1-a \neq 0$ and $3-3a \neq 0$. Therefore, $(1-a)s = 3-3a \rightarrow s = 1$. Thus, $b-1 = (b-1)s = 0$, implying $b = 1$.

We have thus shown that:

$$a \neq 1 \rightarrow b = 1$$

$$b \neq 1 \rightarrow a = 1$$

When $a = b = 1$, the system becomes:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

Which is clearly consistent.

When $a = 1 \vee b = 1$, the system is consistent.

Otherwise, it is inconsistent.

Thus, $(3,3,3) \in P_{a,b}$ exactly when $a = 1 \vee b = 1$.

The set of solutions forms a 2d affine subspace of $\mathbb{R}^3$ when $a \neq 1$ or $b \neq 1$ and forms a 1d affine subspace of $\mathbb{R}^3$ otherwise.

## 4.15. Problem 15

We can set-up the following system of four linear equations in four variables:

$$\begin{cases} 4T_1 = 10 + 20 + T_2 + T_4 \\ 4T_2 = 20 + 40 + T_1 + T_3 \\ 4T_3 = 40 + 30 + T_2 + T_4 \\ 4T_4 = 30 + 10 + T_1 + T_3 \end{cases}$$

Which is equivalent to:

$$\begin{cases} 4T_1 - T_2 - T_4 = 30 \\ 4T_2 - T_1 - T_3 = 60 \\ 4T_3 - T_2 - T_4 = 70 \\ 4T_4 - T_1 - T_3 = 40 \end{cases}$$

This corresponds to the following augmented matrix:

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 4 & -1 & 70 \\ -1 & 0 & -1 & 4 & 40 \end{array}\right)$$

Reducing this matrix into REF:

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ -4 & 16 & -4 & 0 & 240 \\ 0 & -1 & 4 & -1 & 70 \\ -4 & 0 & -4 & 16 & 160 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & -1 & 4 & -1 & 70 \\ 0 & -1 & -4 & 15 & 190 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & -15 & 60 & -15 & 1050 \\ 0 & -15 & -60 & 225 & 2850 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & 0 & 56 & -16 & 1320 \\ 0 & 0 & -64 & 224 & 3120 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & 0 & 7 & -2 & 165 \\ 0 & 0 & -4 & 14 & 195 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & 0 & 28 & -8 & 660 \\ 0 & 0 & -28 & 98 & 1365 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & 0 & 28 & -8 & 660 \\ 0 & 0 & 0 & 90 & 2025 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & 0 & 28 & -8 & 660 \\ 0 & 0 & 0 & 2 & 45 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & 0 & 28 & 0 & 840 \\ 0 & 0 & 0 & 2 & 45 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & 2 & 45 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 30 & -8 & -2 & 540 \\ 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & 2 & 45 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 30 & 0 & 0 & 825 \\ 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & 2 & 45 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & -1 & 0 & -1 & 30 \\ 0 & 1 & 0 & 0 & \frac{55}{2} \\ 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & 1 & \frac{45}{2} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 4 & 0 & 0 & 0 & 80 \\ 0 & 1 & 0 & 0 & \frac{55}{2} \\ 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & 1 & \frac{45}{2} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 20 \\ 0 & 1 & 0 & 0 & \frac{55}{2} \\ 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & 1 & \frac{45}{2} \end{array}\right)$$

Thus, $T_1 = 20, T_2 = \dfrac{55}{2}, T_3 = 30, T_4 = \dfrac{45}{2}$

# 5. Problem Set 4

## 5.1. Problem 1

We note that:

$$A = \begin{pmatrix} -1 & -4 & -7 & -10 \\ 1 & -2 & -5 & -8 \\ 3 & 0 & -3 & -6 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$D = \begin{pmatrix} -1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

$$E = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}$$

$$F = \begin{pmatrix} 5 & -1 \\ 9 & 1 \\ 2 & 0 \end{pmatrix}$$

$$G = \begin{pmatrix} 1 \\ -1 \\ 3 \\ 2 \end{pmatrix}$$

(a) $\begin{pmatrix} -6 & 6 & -6 \\ -6 & 6 & -6 \\ -6 & 6 & -6 \end{pmatrix}$

(b) $\begin{pmatrix} -4 & 2 & 5 & 8 \\ 1 & -5 & -5 & -8 \\ -1 & 2 & 2 & 8 \\ 1 & -2 & -5 & -11 \end{pmatrix}$

(c) Not defined

(d) $\begin{pmatrix} 1 & 3 & 6 \\ 0 & 4 & 10 \\ 0 & 0 & 9 \end{pmatrix}$

(e) $\begin{pmatrix} -3 & 3 & -4 \\ 3 & -3 & 4 \\ -3 & 3 & -4 \\ 3 & -3 & 4 \end{pmatrix}$

(f) $\begin{pmatrix} 3 & -6 & -15 & -24 \\ 8 & -4 & -16 & -28 \\ 9 & 0 & -9 & -18 \end{pmatrix}$

(g) Not defined

(h) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

(i) $\begin{pmatrix} -38 \\ -28 \\ -18 \end{pmatrix}$

(j) Not defined

(k) $\begin{pmatrix} 16 & 0 \\ 22 & 2 \\ 6 & 0 \end{pmatrix}$

(l) $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

(m) $\begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & -2 & 0 \end{pmatrix}$

(n) Undefined

(o) $\begin{pmatrix} 1 & -1 & 3 & 2 \\ -1 & 1 & -3 & -2 \\ 3 & -3 & 9 & 6 \\ 2 & -2 & 6 & 4 \end{pmatrix}$

(p) (15) (careful, make sure to explicitly write the brackets of a $1 \times 1$ matrix)

## 5.2. Problem 2

1.i. $(AB)_{3,4}$

1.ii. $(AB)_{4,1}$

1.iii. $(BA)_{3,2}$

1.iv. $(BA)_{2,5}$

2.a. $\sum_{i=1}^{n} a_{3i} b_{i2}$

2.b. $\sum_{i=1}^{m} b_{4i} a_{i1}$

## 5.3. Problem 3

**Reminder to Self:** When showing that two matrices are equal, make sure to **state that both matrices are of the same size and show that every entry is equal.**

We will only show $(A + B)C = AC + BC$

First, for all operations to be defined, $A$ and $B$ must be of the same shape (say, $n \times m$), and $C$ has to be of the shape $m \times k$.

Then, the resulting matrix on both the left and right sides of the equation to prove will be $n \times k$.

To show that the left- and right-hand sides of the equation are equal, we will show that each entry of the two matrices are equal.

$$\begin{aligned}((A+B)C)_{i,j} &= \sum_{i=1}^{m} (A+B)_{im} C_{mj} \\ &= \sum_{i=1}^{m} (A_{im} + B_{im}) C_{mj} \\ &= \sum_{i=1}^{m} \left(A_{im} C_{mj} + B_{im} C_{mj}\right) \\ &= \sum_{i=1}^{m} \left(A_{im} C_{mj}\right) + \sum_{i=1}^{m} \left(B_{im} C_{mj}\right) \\ &= (AC)_{ij} + (BC)_{ij} \\ &= (AC + BC)_{ij}\end{aligned}$$

Thus, $(A+B)C = (AC+BC)$, as needed. ■

## 5.4. Problem 4

$$\begin{aligned}\sum_{m=1}^{5} \sum_{n=1}^{3} f(m,n) &= \sum_{m=1}^{5} (f(m,1) + f(m,2) + f(m,3)) \\ &= (f(1,1) + f(1,2) + f(1,3)) \\ &+ (f(2,1) + f(2,2) + f(2,3)) \\ &+ (f(3,1) + f(3,2) + f(3,3)) \\ &+ (f(4,1) + f(4,2) + f(4,3)) \\ &+ (f(5,1) + f(5,2) + f(5,3)) \\ &= (f(1,1) + f(2,1) + f(3,1) + f(4,1) + f(5,1)) \\ &+ (f(1,2) + f(2,2) + f(3,2) + f(4,2) + f(5,2)) \\ &+ (f(1,3) + f(2,3) + f(3,3) + f(4,3) + f(5,3)) \\ &= \sum_{m=1}^{5} f(m,1) + \sum_{m=1}^{5} f(m,2) + \sum_{m=1}^{5} f(m,3) \\ &= \sum_{n=1}^{3} \sum_{m=1}^{5} f(m,n)\end{aligned}$$

## 5.5. Problem 5

### 5.5.1. 1(a)

$$\begin{aligned}\operatorname{tr}(A+B) &= \sum_{i=1}^{n} (A+B)_{ii} \\ &= \sum_{i=1}^{n} (A_{ii} + B_{ii}) \\ &= \sum_{i=1}^{n} A_{ii} + \sum_{i=1}^{n} B_{ii}\end{aligned}$$

$$=\operatorname{tr}(A)+\operatorname{tr}(B)$$

#### 5.5.2. 1(b)

$$\begin{aligned}\operatorname{tr}(rA)&=\sum_{i=1}^{n}(rA)_{ii}\\&=\sum_{i=1}^{n}r(A_{ii})\\&=r\sum_{i=1}^{n}A_{ii}\\&=r\operatorname{tr}(A)\end{aligned}$$

#### 5.5.3. 2

$$\begin{aligned}\operatorname{tr}(CD)&=\sum_{i=1}^{p}(CD)_{ii}\\&=\sum_{i=1}^{p}\sum_{j=1}^{q}C_{ij}D_{ji}\\&=\sum_{j=1}^{q}\sum_{i=1}^{p}D_{ji}C_{ij}\\&=\sum_{j=1}^{q}(DC)_{jj}\\&=\operatorname{tr}(DC)\end{aligned}$$

#### 5.5.4. 3

Suppose there exist square matrices $A$ and $B$ such that:

$$AB-BA=I$$

Note that $\operatorname{tr}(I)>0$.

Furthermore:

$$\begin{aligned}\operatorname{tr}(AB-BA)&=\operatorname{tr}(I)\\\operatorname{tr}(AB)-\operatorname{tr}(BA)&=\operatorname{tr}(I)\\\operatorname{tr}(AB)-\operatorname{tr}(AB)&=\operatorname{tr}(I)\\0&=\operatorname{tr}(I)\end{aligned}$$

This contradicts $\operatorname{tr}(I)>0$. Thus, there exist no square matrices $A$ and $B$ such that $AB-BA=I$.

## 5.6. Problem 6

Again, to prove this matrix equation, we simply prove that every entry matches.

$$\begin{aligned}\left(AI_n\right)_{ij}&=\sum_{k=1}^{n}A_{ik}\left(I_n\right)_{kj}\\&=A_{ij}\left(I_n\right)_{jj}\\&=A_{ij}\cdot 1\end{aligned}$$

$$
\begin{aligned}
&= A_{ij} \\
(I_m A)_{ij} &= \sum_{k=1}^{m} (I_m)_{ik} A_{kj} \\
&= (I_m)_{ii} A_{ij} \\
&= 1 \cdot A_{ij} \\
&= A_{ij}
\end{aligned}
$$

Thus,

$$I_m A = A I_n = A$$

## 5.7. Problem 7

Simplifying the left- and right-hand sides:

$$
\begin{aligned}
a\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + b\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} &= c\begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix} + d\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\
\begin{pmatrix} a & -a+2b \\ -a+b & a \end{pmatrix} &= \begin{pmatrix} 3c+d & c+d \\ -c & 3c+d \end{pmatrix}
\end{aligned}
$$

Thus, we have the following system of linear equations:

$$a = 3c + d$$

$$-a + 2b = c + d$$

$$-a + b = -c$$

$$a = 3c + d$$

Which corresponds to the following augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 1 & -2 & 1 & 1 & 0 \\ 1 & -1 & -1 & 0 & 0 \\ 1 & 0 & -3 & -1 & 0 \end{array}\right)$$

Reducing into RREF:

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & -2 & 4 & 2 & 0 \\ 1 & -1 & -1 & 0 & 0 \\ 1 & 0 & -3 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & -2 & 4 & 2 & 0 \\ 0 & -1 & 4 & 1 & 0 \\ 1 & 0 & -3 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & -2 & 4 & 2 & 0 \\ 0 & -1 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & 1 & -2 & -1 & 0 \\ 0 & -1 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & 1 & -2 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & 1 & -2 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Thus, the set of all possible solutions of the equation is:

$$d = r$$

$$c = 0$$

$$a = b = r$$

For any $r \in \mathbb{R}$.

## 5.8. Problem 8

### 5.8.1. (a)

Since $A = -A^T$, $A + A^T = 0$.

Further, note that $\operatorname{tr}(A) = \operatorname{tr}(A^T)$ and $\operatorname{tr}(0) = 0$

Thus:

$$\begin{aligned} 0 &= \operatorname{tr}(A + A^T) \\ &= \operatorname{tr}(A) + \operatorname{tr}(A^T) \\ &= \operatorname{tr}(A) + \operatorname{tr}(A) \\ &= 2\operatorname{tr}(A) \end{aligned}$$

Therefore, $2\operatorname{tr}(A) = 0$.

Thus, $\operatorname{tr}(A) = 0$, as needed. ■

### 5.8.2. (b)

$X = B^T$ works.

$$\operatorname{tr}(B^T A) = \sum_{i=1}^{n} (B^T A)_{ii}$$

$$
\begin{aligned}
&= \sum_{i=1}^{n} \left(B^T A\right)_{ii} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} \left(B^T\right)_{ij} A_{ji} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} B_{ji} A_{ji} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ji} B_{ji} \\
&= \sum_{j=1}^{n} \sum_{i=1}^{n} A_{ji} B_{ji} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} B_{ij}
\end{aligned}
$$

As needed. ■

## 5.9. Problem 9

### 5.9.1. (a)

$$
\begin{aligned}
\begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}^2 &= \begin{pmatrix} 1 \cdot 1 + -1 \cdot 2 & 2 \cdot 1 + -1 \cdot 2 \\ 1 \cdot -1 + -1 \cdot -1 & 2 \cdot -1 + -1 \cdot -1 \end{pmatrix} \\
&= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}
\end{aligned}
$$

Let $A = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}$. We note that $A^2 = -I$.

### 5.9.2. (b)

Let us organize the matrix into blocks as follows:

$$
B = \begin{pmatrix} A & I \\ I & A \end{pmatrix}
$$

Then:

$$
\begin{aligned}
\begin{pmatrix} A & I \\ I & A \end{pmatrix}^2 &= \begin{pmatrix} A^2 + I^2 & 2A \\ 2A & A^2 + I^2 \end{pmatrix} \\
&= \begin{pmatrix} -I + I & 2A \\ 2A & -I + I \end{pmatrix} \\
&= \begin{pmatrix} 0 & 2A \\ 2A & 0 \end{pmatrix}
\end{aligned}
$$

Thus:

$$
B^2 = \begin{pmatrix} 0 & 0 & 2 & 4 \\ 0 & 0 & -2 & -2 \\ 2 & 4 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{pmatrix}
$$

## 5.10. Problem 10

### 5.10.1. (a)

$$\begin{aligned}\operatorname{tr}(AX^T) &= \sum_{i=1}^{m} (AX^T)_{ii} \\ &= \sum_{i=1}^{m} (AX^T)_{ii} \\ &= \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}(X^T)_{ji} \\ &= \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}X_{ij}\end{aligned}$$

### 5.10.2. (b)

Suppose FTSOC $X \neq 0$, then there exists $i, j \in \mathbb{Z}$ satisfying:
1. $1 \leq i \leq m$
2. $1 \leq j \leq n$
3. $X_{ij} \neq 0$

Next, consider the matrix $A$ defined as:

$$A_{pq} = \begin{cases} 1 \text{ if } (p,q) = (i,j) \\ 0 \text{ otherwise} \end{cases}$$

Notice that:

$$\operatorname{tr}(AX^T) = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}X_{ij} = A_{ij}X_{ij} > 0$$

Contradiction. Therefore, $X = 0$, as needed. ■

## 5.11. Problem 11

Let us first show the forward direction: If a matrix $B$ with the property $AB = I$ exists, then $a_{ii} \neq 0$ for all $i$.

This is equivalent to showing that, if $a_{ii} = 0$ for some $i$, then no matrix $B$ satisfies $AB = I$.

Suppose not. Then, $a_{ii} = 0$ for some $i$ **and** there exists a matrix $B$ such that $AB = I$. Then, the $i$th row of $A$ is composed of all zeroes. Consider any entry in the $i$th row of $AB$:

$$\begin{aligned}(AB)_{ij} &= \sum_{k=1}^{n} A_{ik}B_{kj} \\ &= \sum_{k=1}^{n} A_{ik}B_{kj} \\ &= 0\end{aligned}$$

Since $AB = I$, we have deduced that the $i$th row of $I$ is composed of all zeroes, contradiction. Thus, if $a_{ii} = 0$ for some $i$, then no matrix $B$ satisfies $AB = I$, as needed.

Let us now show the backward direction: If $a_{ii} \neq 0$ for all $i$, then there is a matrix $B$ such that $AB = I$.

The proof is by direct construction. Let $B$ be the matrix such that:

$$b_{ij} = \begin{cases} \frac{1}{a_{ii}} \text{ if } i = j \\ 0 \text{ otherwise} \end{cases}$$

Then, we have:

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

Notice that when $i \neq j$, $A_{ik}$ and $B_{kj}$ cannot both be nonzero simultaneously for any choice of $k$. Thus, $\sum_{k=1}^{n} A_{ik} B_{kj} = 0$ in these cases.

On the other hand, when $i = j$, $A_{ik}$ and $B_{kj}$ are both nonzero exactly when $k = i = j$. Thus, $\sum_{k=1}^{n} A_{ik} B_{kj} = A_{ii} B_{ii} = \frac{A_{ii}}{A_{ii}} = 1$ in these cases.

In summary, we have found:

$$(AB)_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ otherwise} \end{cases}$$

Thus, $AB = I$. So indeed, if $a_{ii} \neq 0$ for all $i$, there exists a matrix $B$ such that $AB = I$, as needed. ■

## 5.12. Problem 12

### 5.12.1. (i)

For $AB$ to be defined, $A \in M_{n \times m}(\mathbb{R})$ and $B \in M_{m \times n}(\mathbb{R})$ must both hold for some $n, m \in \mathbb{Z}^+$.

Notice that $B^T A^T$ is also defined since $B^T \in M_{n \times m}(\mathbb{R})$ and $A^T \in M_{m \times n}(\mathbb{R})$.

Now, to prove that $(AB)^T = B^T A^T$, we will show that every entry of the left matrix is equal to every entry of the right matrix.

$$\begin{aligned} \left((AB)^T\right)_{ij} &= (AB)_{ji} \\ &= \sum_{k=1}^{m} A_{jk} B_{ki} \\ &= \sum_{k=1}^{m} \left(A^T\right)_{kj} \left(B^T\right)_{ik} \\ &= \sum_{k=1}^{m} \left(B^T\right)_{ik} \left(A^T\right)_{kj} \\ &= \left(B^T A^T\right)_{ij} \end{aligned}$$

Thus,

$$(AB)^T = B^T A^T$$

As needed. ■

### 5.12.2. (ii)

First, we will show that if $A$ were invertible, then $A^T$ is also invertible.

Since $A$ is invertible, there exists a matrix $B$ such that $AB = I = BA$.

Taking the transpose on all sides of the equation:

$$B^T A^T = (AB)^T = I^T = I = I^T = (BA)^T = A^T B^T$$

Thus:

$$B^T A^T = A^T B^T = I$$

Implying that $B^T$ is the inverse of $A^T$ and thus that $A^T$ is invertible.

Now, we will show that if $A^T$ were invertible, then $A$ is also invertible.

By our work already done, if $A^T$ were invertible, then $\left(A^T\right)^T$ is also invertible. However, $\left(A^T\right)^T = A$ for any $A$. Thus, if $A^T$ were invertible, then so is $A$. Since we have shown both sides of the identity, we have proven that $A$ is invertible exactly when $A^T$ is invertible, as needed. ■

## 5.13. Problem 13

We will consider $(AB)_{ij}$ for $j > i$.

Thus, notice that for any $k$, if $k \geq j$, then $k > i$ by transitivity. Thus, when $k \geq j$, $A_{ik} = 0$ because $k \geq j \to k > i$.

Further, when $k < j$, $B_{kj} = 0$.

Note that:

$$\begin{aligned}
(AB)_{ij} &= \sum_{k=1}^{n} A_{ik} B_{kj} \\
&= \sum_{k=1}^{n} A_{ik} B_{kj} \\
&= \sum_{k=1,k<j}^{n} A_{ik} B_{kj} + \sum_{k=1,k\geq j}^{n} A_{ik} B_{kj} \\
&= \sum_{k=1,k<j}^{n} A_{ik} \cdot 0 + \sum_{k=1,k\geq j}^{n} 0 \cdot B_{kj} \\
&= 0
\end{aligned}$$

## 5.14. Problem 14

Suppose $A$ were an involutory matrix. Thus, $A^2 = I$.

Notice that

$$\left(\frac{A+I}{2}\right)^2 = \frac{1}{4}(A^2 + AI + IA + I^2) = \frac{1}{4}(2I + 2A) = \frac{A+I}{2}$$

Thus, $\dfrac{A+I}{2}$ is idempotent.

Similarly,

$$\left(\frac{I-A}{2}\right)^2 = \frac{1}{4}(A^2 - AI - IA + I^2) = \frac{1}{4}(2I - 2A) = \frac{I-A}{2}$$

Thus, $\dfrac{I-A}{2}$ is idempotent.

Finally, $\frac{A+I}{2} - \frac{I-A}{2} = A$, so indeed, $A$ can be expressed as the difference of two idempotents, as needed. ■

## 5.15. Problem 15

Note to self: "show the steps in your working" = just show the intermediate steps. It does not necessarily mean show every computation — just show the intermediate matrices.

In the interest of time, I will only solve (iii). We show a generalization of this property for arbitrary positive integer exponents $n$.

Define a permutation matrix as a matrix where each row is composed of all 0s except for exactly one entry, which is 1. We note that $P^3 = I_3$. An easy way to see this is to consider the isomorphism between the group consisting of permutation matrices of order $n$ under the operation of matrix multiplication and the symmetric group over $n$ elements, $S_n$. A more elementary argument is presented here:

Since $P^3 = I_3$, I claim that the two matrices commute with one another.

Notice that,

$$\begin{aligned} I_3 P &= P^3 P \\ &= P^4 \\ &= P P^3 \\ &= P I_3 \end{aligned}$$

Since the matrices commute with each other, they satisfy the binomial theorem.

Thus, we have:

$$\begin{aligned} \operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^n\right) &= \operatorname{tr}\left((aI_3 + bP)^n\right) \\ &= \operatorname{tr}\left((aP^3 + bP)^n\right) \\ &= \operatorname{tr}\left(\sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i} P^{3i} P^{n-i}\right) \\ &= \operatorname{tr}\left(\sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i} P^{n+2i}\right) \end{aligned}$$

Note that $\operatorname{tr}\left(P^{0 \bmod 3}\right) = 3$, and $\operatorname{tr}\left(P^{1 \bmod 3}\right) = \operatorname{tr}\left(P^{2 \bmod 3}\right) = 0$. Therefore,

$$\begin{aligned} \operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^n\right) &= \operatorname{tr}\left(\sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i} P^{n+2i}\right) \\ &= \operatorname{tr}\left(\sum_{i=0, i \equiv n \bmod 3}^{n} \binom{n}{i} a^i b^{n-i} P^{n+2i}\right) \\ &= 3 \sum_{j=0, j \equiv n \bmod 3}^{n} \binom{n}{j} a^j b^{n-j} \end{aligned}$$

To simplify this sum, we will use generating functions and complex numbers. The appearance of the $n \bmod 3$ restriction motivates the introduction of complex numbers to handle the cycling behavior of modular arithmetic.

Let $P(x) = \sum_{j=0}^{n} \binom{n}{j} a^j b^{n-j} x^j = (ax+b)^n$ be the ordinary power series generating function describing our sequence. Further, let $\zeta_1, \zeta_2, \zeta_3$ be the three roots of $x^3 - 1 = 0$ (i.e., the third roots of unity).

Notice that $\zeta_1^n + \zeta_2^n + \zeta_3^n = \begin{cases} 0 \text{ if } n \not\equiv 0 \bmod 3 \\ 3 \text{ if } n \equiv 0 \bmod 3 \end{cases}$. Therefore, we have:

$$\begin{aligned}
\operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^n\right) &= 3 \sum_{j=0, j \equiv n \bmod 3}^{n} \binom{n}{j} a^j b^{n-j} \\
&= \zeta_1^{2n} P(\zeta_1) + \zeta_2^{2n} P(\zeta_2) + \zeta_3^{2n} P(\zeta_3) \\
&= (a\zeta_1 + b)^n + \zeta_2^{2n} (a\zeta_2 + b)^n + \zeta_3^{2n} (a\zeta_3 + b)^n \\
&= (a+b)^n + \left(\frac{-1+i\sqrt{3}}{2}\right)^{2n} \left(a\frac{-1+i\sqrt{3}}{2} + b\right)^n + \\
&\quad \left(\frac{-1-i\sqrt{3}}{2}\right)^{2n} \left(a\frac{-1-i\sqrt{3}}{2} + b\right)^n \\
&= (a+b)^n + \left(a + \left(\frac{-1-i\sqrt{3}}{2}\right)b\right)^n + \left(a + \left(\frac{-1+i\sqrt{3}}{2}\right)b\right)^n
\end{aligned}$$

Thus, we have:

$$\operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^n\right) = (a+b)^n + \left(a + \left(\frac{-1-i\sqrt{3}}{2}\right)b\right)^n + \left(a + \left(\frac{-1+i\sqrt{3}}{2}\right)b\right)^n$$

Where $i = \sqrt{-1}$.

Note to self: It's kinda crazy that complex numbers show up, but apparently the answer is related to a deeper fact about the trace of a matrix — that the trace of a matrix $M$ is the sum of its eigenvalues.

Eigenvalues, in general, need not be real numbers. This is why we get complex numbers as the result.

## 5.16. Problem 16

### 5.16.1. (a)

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

As you can see, both matrices on the left and right-hand sides are strictly upper triangular.

### 5.16.2. (b)

Define a $k$-strictly upper triangular matrix to be a matrix such that $a_{ij} = 0$ whenever $i + k > j$. Thus, a 1-strictly upper triangular matrix is equivalent to a normal upper-triangular matrix.

Further, notice that an $n \times n$ matrix that is $n$-strictly upper triangular satisfies:

$$a_{ij} = 0 \text{ whenever } i + n > j$$

However, we also have $i + n > n \geq j$, so $i + n \geq j$ always holds. Therefore, if an $n \times n$ matrix is $n$-strictly upper triangular, it must be the zero matrix.

Before we prove the whole statement, I will first prove the following intermediate lemma:

**Lemma** Product of Strictly Upper Triangular Matrices — Lemma 15

Suppose that $A$ and $B$ are square upper triangular matrices of the same size. Further suppose that $A$ is 1-strictly upper triangular and that $B$ is $k$-strictly upper triangular. Then, $AB$ is $(k+1)$-strictly upper triangular.

**Proof** Product of Strictly Upper Triangular Matrices — Proof 16

Since $A$ is 1-strictly upper triangular and $B$ is $k$-strictly upper triangular, the following hold:

$$i + 1 > j \to A_{ij} = 0$$

$$i + k > j \to B_{ij} = 0$$

We want to show that $AB$ is $(k+1)$-strictly upper triangular. That is, whenever $i + k + 1 > j$, $(AB)_{ij} = 0$.

Suppose that $i$ and $j$ are particular but arbitrarily chosen integers such that $1 \leq i, j \leq n$ and $i + k + 1 > j$.

Then:

$$(AB)_{ij} = \sum_{l=1}^{n} A_{il} B_{lj}$$

Whenever $i + 1 > l$, $A_{il} = 0$. Further, whenever $l + k > j$, $B_{lj} = 0$.

I claim that if $i + k + 1 > j$, then either $i + 1 > l$ or $l + k > j$.

Suppose not. Then, $i + k + 1 > j$ and $i + 1 \leq l$ and $l + k \leq j$.

Thus, $i + k + 1 \leq l + k$. Also, $l + k \leq j$. By transitivity, $l + k + 1 \leq j$, contradiction.

Thus, if $i + k + 1 > j$, then either $i + 1 > l$ or $l + k > j$.

Applying modus ponens, if $i + k + 1 > j$, then either $A_{il} = 0$ or $B_{lj} = 0$. By the zero-product property, if $i + k + 1 > j$, then $A_{il} B_{lj} = 0$.

Therefore, if $i + k + 1 > j$, $(AB)_{ij} = 0$, proving that $AB$ is $(k+1)$-strictly upper triangular, as needed. ■

Now, we will prove the main theorem. I claim that setting $N = n$ works. To show this, consider the following predicate:

$$P(k) : A^k \text{ is } k\text{-strictly upper triangular}$$

First, let us consider the base case $P(1)$. Clearly, $A$ is 1-strictly upper triangular because $A$ is strictly upper triangular. This proves the base case.

Now, let us consider the inductive step.

Suppose $A^k$ is $k$-strictly upper triangular for $k < n$. We want to show that $A^{k+1}$ is $(k+1)$-strictly upper triangular.

We note that $A^{k+1} = AA^k$, and that $A$ is 1-strictly upper triangular (given) and $A^k$ is $k$-strictly upper triangular (inductive hypothesis). By Lemma 15, $A^{k+1}$ must be $(k+1)$-strictly upper triangular as it is the product of a 1-strictly upper triangular and a $k$-strictly upper triangular matrix.

This proves that, for $1 \leq k \leq n$, $A^k$ is $k$-strictly upper triangular.

Thus, $A^n$ is $n$-strictly upper triangular. Since $A^n \in M_{n\times n}(\mathbb{R})$ and is $n$-strictly upper triangular, $A^n = 0$.

Therefore, there exists $N$ (particularly $N = n$) such that $A^N = 0$, as needed. ■

## 5.17. Problem 17

First, we will show:

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^k \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} F_k \\ F_{k+1} \end{pmatrix}$$

The base case is trivially true:

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$$

Suppose:

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^k \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} F_k \\ F_{k+1} \end{pmatrix}$$

Then:

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^{k+1} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^k \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} F_k \\ F_{k+1} \end{pmatrix} = \begin{pmatrix} F_{k+1} \\ F_k + F_{k+1} \end{pmatrix} = \begin{pmatrix} F_{k+1} \\ F_{k+2} \end{pmatrix}$$

As needed. ■

Next, we will compute the inverse of $P$:

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ \phi_+ & \phi_- & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & \phi_- - \phi_+ & -\phi_+ & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -\sqrt{5} & -\phi_+ & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{\phi_+}{\sqrt{5}} & -\frac{\sqrt{5}}{5} \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 - \frac{\phi_+}{\sqrt{5}} & \frac{\sqrt{5}}{5} \\ 0 & 1 & \frac{\phi_+}{\sqrt{5}} & -\frac{\sqrt{5}}{5} \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 0 & \frac{5-\sqrt{5}}{10} & \frac{\sqrt{5}}{5} \\ 0 & 1 & \frac{5+\sqrt{5}}{10} & -\frac{\sqrt{5}}{5} \end{array}\right)$$

Thus,

$$P^{-1} = \begin{pmatrix} \frac{5-\sqrt{5}}{10} & \frac{\sqrt{5}}{5} \\ \frac{5+\sqrt{5}}{10} & -\frac{\sqrt{5}}{5} \end{pmatrix}$$

And finally:

$$\begin{aligned}
P\begin{pmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{pmatrix}P^{-1} &= \begin{pmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{pmatrix}\begin{pmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{pmatrix}\begin{pmatrix} \frac{5-\sqrt{5}}{10} & \frac{\sqrt{5}}{5} \\ \frac{5+\sqrt{5}}{10} & -\frac{\sqrt{5}}{5} \end{pmatrix} \\
&= \begin{pmatrix} \phi_+ & \phi_- \\ \phi_+^2 & \phi_-^2 \end{pmatrix}\begin{pmatrix} \frac{5-\sqrt{5}}{10} & \frac{\sqrt{5}}{5} \\ \frac{5+\sqrt{5}}{10} & -\frac{\sqrt{5}}{5} \end{pmatrix} \\
&= \begin{pmatrix} \phi_+ & \phi_- \\ \phi_+ + 1 & \phi_- + 1 \end{pmatrix}\begin{pmatrix} \frac{5-\sqrt{5}}{10} & \frac{\sqrt{5}}{5} \\ \frac{5+\sqrt{5}}{10} & -\frac{\sqrt{5}}{5} \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} \phi_+ & \phi_- \\ \phi_+ + 1 & \phi_- + 1 \end{pmatrix}\begin{pmatrix} \frac{5-\sqrt{5}}{2} & \sqrt{5} \\ \frac{5+\sqrt{5}}{2} & -\sqrt{5} \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} \phi_+ & \phi_- \\ \phi_+ + 1 & \phi_- + 1 \end{pmatrix}\begin{pmatrix} 2+\phi_- & \phi_+ - \phi_- \\ 2+\phi_+ & \phi_- - \phi_+ \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} (2+\phi_-)\phi_+ + (2+\phi_+)\phi_- & (\phi_+ - \phi_-)\phi_+ + (\phi_- - \phi_+)\phi_- \\ (2+\phi_-)(\phi_+ + 1) + (2+\phi_+)(\phi_- + 1) & (\phi_+ - \phi_-)(\phi_+ + 1) + (\phi_- - \phi_+)(\phi_- + 1) \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} (2\phi_+ - 1) + (2\phi_- - 1) & (\phi_+ - \phi_-)^2 \\ (2\phi_+ + \phi_-) + (2\phi_- + \phi_+) + 2 & (\phi_+ - \phi_-)^2 \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} 2(\phi_+ + \phi_-) - 2 & (\phi_+^2 - 2\phi_+\phi_- + \phi_-^2) \\ 3(\phi_+ + \phi_-) + 2 & (\phi_+^2 - 2\phi_+\phi_- + \phi_-^2) \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} 2 - 2 & (\phi_+^2 + \phi_-^2 + 2) \\ 3 + 2 & (\phi_+^2 + \phi_-^2 + 2) \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} 0 & (\phi_+ + \phi_- + 4) \\ 5 & (\phi_+ + \phi_- + 4) \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} 0 & (\phi_+ + \phi_- + 4) \\ 5 & (\phi_+ + \phi_- + 4) \end{pmatrix} \\
&= \frac{1}{5}\begin{pmatrix} 0 & 5 \\ 5 & 5 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}
\end{aligned}$$

Finally, we have:

$$\begin{aligned}
\begin{pmatrix} F_k \\ F_{k+1} \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^k \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
&= \left( P \begin{pmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{pmatrix} P^{-1} \right)^k \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
&= P \begin{pmatrix} \phi_+^k & 0 \\ 0 & \phi_-^k \end{pmatrix} P^{-1} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
&= \frac{1}{5} \begin{pmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{pmatrix} \begin{pmatrix} \phi_+^k & 0 \\ 0 & \phi_-^k \end{pmatrix} \begin{pmatrix} 2+\phi_- & \phi_+ - \phi_- \\ 2+\phi_+ & \phi_- - \phi_+ \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
&= \frac{1}{5} \begin{pmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{pmatrix} \begin{pmatrix} \phi_+^k & 0 \\ 0 & \phi_-^k \end{pmatrix} \begin{pmatrix} \phi_+ - \phi_- \\ \phi_- - \phi_+ \end{pmatrix} \\
&= \frac{1}{5} \begin{pmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{pmatrix} \begin{pmatrix} \phi_+^{k+1} - \phi_- \phi_+^k \\ \phi_-^{k+1} - \phi_+ \phi_-^k \end{pmatrix}
\end{aligned}$$

Thus,

$$F_k = \frac{1}{5}(\phi_+^{k+1} - \phi_- \phi_+^k + \phi_-^{k+1} - \phi_+ \phi_-^k)$$

$$F_k = \frac{1}{5}(\phi_+^k(\phi_+ - \phi_-) + \phi_-^k(\phi_- - \phi_+))$$

$$F_k = \frac{1}{5}(\phi_+^k(\phi_+ - \phi_-) - \phi_-^k(\phi_+ - \phi_-))$$

$$F_k = \frac{1}{5}(\phi_+^k - \phi_-^k)(\phi_+ - \phi_-)$$

$$F_k = \frac{\sqrt{5}}{5}(\phi_+^k - \phi_-^k)$$

$$F_k = \frac{1}{\sqrt{5}}(\phi_+^k - \phi_-^k)$$

Finally, we notice that:

$$\begin{aligned}
\lim_{n \to +\infty} \left( F_n - \frac{1}{\sqrt{5}} \phi_+^n \right) &= \lim_{n \to +\infty} \left( \frac{1}{\sqrt{5}}(\phi_+^n - \phi_-^n) - \frac{1}{\sqrt{5}} \phi_+^n \right) \\
&= \lim_{n \to +\infty} \left( -\frac{1}{\sqrt{5}} \phi_-^n \right) \\
&= 0
\end{aligned}$$

As needed. ■

# 6. Problem Set 5

## 6.1. Problem 1

TODO

## 6.2. Problem 2

TODO

### 6.3. Problem 3
TODO

### 6.4. Problem 4
TODO

### 6.5. Problem 5
TODO

### 6.6. Problem 6
TODO

### 6.7. Problem 7
TODO

### 6.8. Problem 8
TODO

### 6.9. Problem 9
TODO

### 6.10. Problem 10
TODO

### 6.11. Problem 11
TODO

### 6.12. Problem 12
TODO

### 6.13. Problem 13
TODO

### 6.14. Problem 14
In general, the $n \times n$ matrix can be reduced to the identity as follows:

1. $R_1 \to R_1 + R_2$
2. $R_2 \to R_2 + R_3$
3. $R_3 \to R_3 + R_4$
4. $\vdots$
5. $R_{n-1} \to R_{n-1} + R_n$

Applying these operations to the identity matrix, we obtain the following general form for $A^{-1}$:

$$A^{-1} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 1 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

## 6.15. Problem 15

Suppose $A$ were singular. Then, there must exist a nonzero vector $\vec{x}$ such that $A\vec{x} = \vec{0}$.

Then, clearly $(BA)\vec{x} = B(A\vec{x}) = B\vec{0} = \vec{0}$, implying that $BA$ is singular.

To show that $AB$ is singular, we do some casework on $B$.

Suppose $B$ were singular. Then, there exists a nonzero vector $\vec{y}$ such that $B\vec{y} = \vec{0}$. Clearly, $(AB)\vec{y} = A(B\vec{y}) = A\vec{0} = \vec{0}$. Thus, $AB$ is singular in this case.

Now suppose $B$ were invertible. Then, let $\vec{y} = B^{-1}\vec{x}$. Since $\vec{x}$ is a nonzero vector, and $B^{-1}$ is invertible, $\vec{y} = B^{-1}\vec{x} \neq \vec{0}$.

Now, notice that:

$$(AB)\vec{y} = A(B\vec{y}) = A(BB^{-1}\vec{x}) = A(BB^{-1})\vec{x} = A\big(I_q\big)\vec{x} = A\vec{x} = \vec{0}$$

Thus, $AB$ is also singular in this case.

Since we have exhausted all cases, if $B$ is singular, then so are $AB$ and $BA$, as needed. ■

## 6.16. Problem 16

### 6.16.1. (a)

Let $A = \big(A_{ij}\big)_{p\times q}$ and $B = \big(B_{ij}\big)_{q\times r}$.

First, notice that $R(AB)$ and $R(A)B$ are both $p \times r$ matrices.

Thus, it remains to be shown that each entry of both matrix expressions match.

#### 6.16.1.1. Case 1: R is a Row Swap

Suppose we swap rows $p$ and $q$

Notice that $R(A)_{ij} = \begin{cases} A_{qj} \text{ if } i = p \\ A_{pj} \text{ if } i = q \\ A_{ij} \text{ otherwise} \end{cases}$

By definition,

$$(R(A)B)_{ij} = \sum_{k=1}^{q} R(A)_{ik}B_{kj}$$

Thus:

$$(R(A)B)_{ij} = \begin{cases} \sum_{k=1}^{q} A_{qk}B_{kj} \text{ if } i = p \\ \sum_{k=1}^{q} A_{pk}B_{kj} \text{ if } i = q \\ \sum_{k=1}^{q} A_{ik}B_{kj} \text{ otherwise} \end{cases}$$

By the definition of matrix multiplication, this is:

$$(R(A)B)_{ij} = \begin{cases} (AB)_{qj} \text{ if } i = p \\ (AB)_{pj} \text{ if } i = q \\ (AB)_{ij} \text{ otherwise} \end{cases}$$

Thus, by the definition of $R$, we have:

$$(R(A)B)_{ij} = R(AB)_{ij}$$

Thus, we have $R(A)B = R(AB)$ in this case.

### 6.16.2. Case 2: Row Addition

Suppose $R$ corresponds to the row operation of $R_p \to R_p + cR_q$ for $p \neq q$.

Notice that $R(A)_{ij} = \begin{cases} A_{ij} + cA_{qj} \text{ if } i = p \\ A_{ij} \text{ otherwise} \end{cases}$.

By definition,

$$(R(A)B)_{ij} = \sum_{k=1}^{q} R(A)_{ik} B_{kj}$$

Thus:

$$\begin{aligned}(R(A)B)_{ij} &= \begin{cases} \sum_{k=1}^{q} (A_{ik} + cA_{qk}) B_{kj} \text{ if } i = p \\ \sum_{k=1}^{q} A_{ik} B_{kj} \text{ otherwise} \end{cases} \\ &= \begin{cases} \sum_{k=1}^{q} (A_{ik} B_{kj}) + c \sum_{k=1}^{q} (A_{qk} B_{kj}) \text{ if } i = p \\ \sum_{k=1}^{q} A_{ik} B_{kj} \text{ otherwise} \end{cases} \\ &= \begin{cases} (AB)_{ij} + c(AB)_{qj} \text{ if } i = p \\ (AB)_{ij} \text{ otherwise} \end{cases}\end{aligned}$$

By the definition of $R$, the above is just $R(AB)_{ij}$. Thus:

$$\begin{aligned}(R(A)B)_{ij} &= \begin{cases} (AB)_{ij} + c(AB)_{qj} \text{ if } i = p \\ (AB)_{ij} \text{ otherwise} \end{cases} \\ &= R(AB)_{ij}\end{aligned}$$

Thus, $R(A)B = R(AB)$ too in this case.

### 6.16.3. Case 3: Row Scaling

Suppose $R$ corresponds to the row operation $R_p \leftarrow cR_p$ with $c \neq 0$.

Thus,

$$R(A)_{ij} = \begin{cases} cA_{ij} \text{ if } i = p \\ A_{ij} \text{ otherwise} \end{cases}$$

By definition,

$$(R(A)B)_{ij} = \sum_{k=1}^{q} R(A)_{ik} B_{kj}$$

Thus,

$$\begin{aligned}(R(A)B)_{ij} &= \begin{cases} \sum_{k=1}^{q} cA_{ik} B_{kj} \text{ if } i = p \\ \sum_{k=1}^{q} A_{ik} B_{kj} \text{ otherwise} \end{cases} \\ &= \begin{cases} c \sum_{k=1}^{q} A_{ik} B_{kj} \text{ if } i = p \\ \sum_{k=1}^{q} A_{ik} B_{kj} \text{ otherwise} \end{cases}\end{aligned}$$

$$=\begin{cases} c(AB)_{ij} \text{ if } i = p \\ (AB)_{ij} \text{ otherwise} \end{cases}$$

By the definition of $R$, we have:

$$(R(A)B)_{ij}=\begin{cases} c(AB)_{ij} \text{ if } i = p \\ (AB)_{ij} \text{ otherwise} \end{cases}$$

$$=R(AB)_{ij}$$

Thus, we also have $R(A)B = R(AB)$ in this case.

Since all cases have been exhausted, $R(A)B = R(AB)$ for any elementary row operation, as needed. ■

#### 6.16.4. (b)

This deduction is quite trivial.

We notice that:

$$R(A) = R\big(I_p A\big) = R\big(I_p\big)A = EA$$

As needed. ■

## 6.17. Problem 17

No, it is not possible.

Since every column has the numbers $0, 1, 2, -3$ in some order, the sum of each row of $A^T$ is 0.

Thus,

$$A^T \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Always holds, and thus $A^T$ is singular by the characterizations of invertibility. Thus, $A$ is also singular, as needed. ■

## 6.18. Problem 18

Notice that:

$$ME = \left(E^T M^T\right)^T$$

Now, $E^T$ is an elementary matrix corresponding to the row operation $R_j \to R_j + cR_i$.

Thus, $E^T M^T$ is equivalent to performing the row operation $R_j \to R_j + cR_i$ on $M^T$.

This corresponds to the following column operation on $M$, the transpose of $M^T$:

$$C_j \to C_j + cC_i$$

## 6.19. Problem 19

#### 6.19.1. (a)

We will show that both $A$ and $B$ are invertible. Suppose not. Then either $A$ or $B$ is singular.

First, suppose that $B$ were singular. Then, by the characterizations of invertibility, there exists a nonzero vector $\vec{x}$ such that $B\vec{x} = \vec{0}$. We also note that:

$$\vec{0} = A\vec{0} = A(B\vec{x}) = (AB)\vec{x}$$

Thus, there exists a nonzero vector $\vec{x}$ such that $(AB)\vec{x} = \vec{0}$. By the characterizations of invertibility, this implies that $(AB)\vec{x} = \vec{0}$ has a nontrivial solution and thus that $AB$ is singular.

Next, suppose that $B$ were invertible but $A$ were singular.

Since $A$ is singular, there exists a nonzero vector $\vec{x}$ such that $A\vec{x} = \vec{0}$.

I claim that there also exists a vector $\vec{v}$ such that $(AB)\vec{v} = \vec{0}$ in this case.

Let $\vec{v} = B^{-1}\vec{x}$.

Notice that:

$$(AB)\vec{v} = (AB)B^{-1}\vec{x} = A(BB^{-1})\vec{x} = AI\vec{x} = A\vec{x} = \vec{0}$$

This implies, by the characterizations of invertibility, that $AB$ is singular in this case too.

Since we have exhausted all cases consistent with the hypotheses, $AB$ is singular. But $AB = I_n$. Therefore, $I_n$ is also singular, contradiction.

Thus, $A$ and $B$ are invertible, as needed. ■

#### 6.19.2. (b)

Since $B$ is invertible, there must be a matrix $C$ such that $BC = CB = I_n$.

Multiplying both sides of $AB = I_n$ by $C$:

$$\begin{aligned} AB &= I_n \\ ABC &= I_nC \\ A(BC) &= C \\ AI_n &= C \\ A &= C \end{aligned}$$

Thus, $A = C$ and $I_n = BC = BA$, as needed. ■

## 6.20. Problem 20

#### 6.20.1. (a)

$$\begin{pmatrix} 2 & -3 \\ 3 & -5 \end{pmatrix}$$

#### 6.20.2. (b)

Reducing into RREF.

$$\begin{pmatrix} 2 & -3 \\ 3 & -5 \end{pmatrix}$$

$$\downarrow \quad R_2 \leftarrow R_2 - \frac{3}{2}R_1$$

$$\begin{pmatrix} 2 & -3 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

$$\downarrow \qquad R_1 \leftarrow R_1 - 6R_2$$

$$\begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

$$\downarrow \qquad R_1 \leftarrow \frac{1}{2}R_1$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

$$\downarrow \qquad R_2 \leftarrow -2R_2$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus, $C = \begin{pmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{pmatrix} \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$

**6.20.3. (c)**

$$R_2 \leftarrow -\frac{1}{2}R_2$$

$$R_1 \leftarrow 2R_1$$

$$R_1 \leftarrow R_1 + 6R_2$$

$$R_2 \leftarrow R_2 + \frac{3}{2}R_1$$

# 7. Problem Set 6

## 7.1. Problem 1

TODO

## 7.2. Problem 2

TODO

## 7.3. Problem 3

TODO

## 7.4. Problem 4

TODO

## 7.5. Problem 5

TODO

## 7.6. Problem 6

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## 7.7. Problem 7
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## 7.8. Problem 8
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## 7.9. Problem 9
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## 7.10. Problem 10
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## 7.11. Problem 11
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## 7.12. Problem 12
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## 7.13. Problem 13
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## 7.14. Problem 14
TODO

## 7.15. Problem 15
TODO

## 7.16. Problem 16
TODO

## 7.17. Problem 17
TODO

## 7.18. Problem 18
The determinant of a $3 \times 3$ matrix is given by:

$$\begin{aligned}\det\left(\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}\right) =& a\det\left(\begin{pmatrix} e & f \\ h & i \end{pmatrix}\right) - b\det\left(\begin{pmatrix} d & f \\ g & i \end{pmatrix}\right) + c\det\left(\begin{pmatrix} d & e \\ g & h \end{pmatrix}\right) \\ =& a(ei - hf) - b(di - gf) + c(dh - ge)\end{aligned}$$

We note that $a, b, c, d, e, f, g, h, i \in \{0, 1\}$.

Further, $ei, hf, di, gf, dh, ge \in \{0, 1\}$.

Also, $ei - hf, di - gf, dh - ge \in \{-1, 0, 1\}$.

Finally, $a(ei - hf), -b(di - gf), c(dh - ge) \in \{-1, 0, 1\}$.

Since

$$-1 \leq a(ei - hf), -b(di - gf), c(dh - ge) \leq 1$$

We have:

$$-3 \leq a(ei - hf) - b(di - gf) + c(dh - ge) \leq 3$$

Now, suppose, WLOG, that

$$a(ei - hf) - b(di - gf) + c(dh - ge) = 3$$

The argument for $a(ei - hf) - b(di - gf) + c(dh - ge) = -3$ is similar.

I claim that $a(ei - hf) - b(di - gf) + c(dh - ge) < 3$. Suppose not, then

$$a(ei - hf) = -b(di - gf) = c(dh - ge) = 1$$

This implies that $a = c = 1$ and $b = -1$. Further, $ei - hf = dh - ge = 1$ and $di - gf = -1$. this further implies that $ei = 1$, $hf = 0$, $dh = 1$, $ge = 0$, $di = 0$, and $gf = 1$. Thus, $e = i = d = h = g = f = 1$.

However, this implies that $0 = hf = 1 \cdot 1 = 1$, contradiction. Therefore, 3 is not an attainable value.

Furthermore, notice that the assignment $a = c = e = i = d = h = 1$ and $b = f = g = 0$ gives the following value of the determinant:

$$a(ei - hf) - b(di - gf) + c(dh - ge) = 1(1 \cdot 1 - 1 \cdot 0) - 0 \cdot (1 \cdot 1 - 0 \cdot 0) + 1 \cdot (1 \cdot 1 - 0 \cdot 1) = 2$$

Therefore, $\det(A) \leq 2$. By a similar argument, $\det(A) \geq -2$.

Therefore, the largest and smallest values of the determinant are 2 and $-2$, respectively.

## 7.19. Problem 19

$$\begin{aligned}
\det(A) &= \det\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{pmatrix}\right) \\
&= \det\left(\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b^2-a^2 & c^2-a^2 & d^2-a^2 \\ 0 & b^3-a^3 & c^3-a^3 & d^3-a^3 \end{pmatrix}\right) \\
&= (-1)^{1+1} \cdot 1 \cdot \det\left(\begin{pmatrix} b-a & c-a & d-a \\ b^2-a^2 & c^2-a^2 & d^2-a^2 \\ b^3-a^3 & c^3-a^3 & d^3-a^3 \end{pmatrix}\right) \\
&= \det\left(\begin{pmatrix} b-a & c-a & d-a \\ b^2-a^2 & c^2-a^2 & d^2-a^2 \\ b^3-a^3 & c^3-a^3 & d^3-a^3 \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a) \det\left(\begin{pmatrix} 1 & 1 & 1 \\ b+a & c+a & d+a \\ b^2+ab+a^2 & c^2+ac+a^2 & d^2+ad+a^2 \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a) \\
&\cdot \det\left(\begin{pmatrix} 1 & 1 & 1 \\ b+a & c+a & d+a \\ b^2+ab+a^2-a(b+a) & c^2+ac+a^2-a(c+a) & d^2+ad+a^2-a(d+a) \end{pmatrix}\right)
\end{aligned}$$

$$
\begin{aligned}
&= (b-a)(c-a)(d-a) \cdot \det\left(\begin{pmatrix} 1 & 1 & 1 \\ b+a & c+a & d+a \\ b^2 & c^2 & d^2 \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a) \cdot \det\left(\begin{pmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2 & c^2 & d^2 \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a) \cdot \det\left(\begin{pmatrix} 1 & 1 & 1 \\ 0 & c-b & d-b \\ 0 & c^2-b^2 & d^2-b^2 \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a) \cdot (-1)^{1+1} \cdot 1 \det\left(\begin{pmatrix} c-b & d-b \\ c^2-b^2 & d^2-b^2 \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a) \det\left(\begin{pmatrix} c-b & d-b \\ c^2-b^2 & d^2-b^2 \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a)(c-b)(d-b) \det\left(\begin{pmatrix} 1 & 1 \\ c+b & d+b \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a)(c-b)(d-b) \det\left(\begin{pmatrix} 1 & 1 \\ c & d \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a)(c-b)(d-b) \det\left(\begin{pmatrix} 1 & 1 \\ 0 & d-c \end{pmatrix}\right) \\
&= (b-a)(c-a)(d-a)(c-b)(d-b)(-1)^{1+1} \cdot 1 \cdot \det((d-c)) \\
&= (b-a)(c-a)(d-a)(c-b)(d-b) \det((d-c)) \\
&= (b-a)(c-a)(d-a)(c-b)(d-b)(d-c)
\end{aligned}
$$

Thus, $A$ is invertible when $\det(A) \neq 0$, that is, when all $a, b, c, d$ are pairwise distinct.

## 7.20. Problem 20

Let

$$
A = \begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & a_4 & \cdots & a_n \\ a_1^2 & a_2^2 & a_3^2 & a_4^2 & \cdots & a_n^2 \\ a_1^3 & a_2^3 & a_3^3 & a_4^3 & \cdots & a_n^3 \\ a_1^4 & a_2^4 & a_3^4 & a_4^4 & \cdots & a_n^4 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & a_3^{n-1} & a_4^{n-1} & \cdots & a_n^{n-1} \end{pmatrix}
$$

We want to show that $\det(A) = \prod_{i=1}^{n-1} \prod_{j=i+1}^{n} (a_j - a_i)$.

First, we will show the base case. When $n = 1$, the matrix $A$ becomes $A = (1)$. So, $\det(A) = 1$ in this case.

Furthermore, $\prod_{i=1}^{1-1} \prod_{j=i+1}^{1} (a_j - a_i) = 1$, since the product contains no product terms.

$$\det(A)=\det\left(\begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & a_4 & \cdots & a_n \\ a_1^2 & a_2^2 & a_3^2 & a_4^2 & \cdots & a_n^2 \\ a_1^3 & a_2^3 & a_3^3 & a_4^3 & \cdots & a_n^3 \\ a_1^4 & a_2^4 & a_3^4 & a_4^4 & \cdots & a_n^4 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & a_3^{n-1} & a_4^{n-1} & \cdots & a_n^{n-1} \end{pmatrix}\right)$$

$$=\det\left(\begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 0 & a_2-a_1 & a_3-a_1 & a_4-a_1 & \cdots & a_n-a_1 \\ 0 & a_2^2-a_1^2 & a_3^2-a_1^2 & a_4^2-a_1^2 & \cdots & a_n^2-a_1^2 \\ 0 & a_2^3-a_1^3 & a_3^3-a_1^3 & a_4^3-a_1^3 & \cdots & a_n^3-a_1^3 \\ 0 & a_2^4-a_1^4 & a_3^4-a_1^4 & a_4^4-a_1^4 & \cdots & a_n^4-a_1^4 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a_2^{n-1}-a_1^{n-1} & a_3^{n-1}-a_1^{n-1} & a_4^{n-1}-a_1^{n-1} & \cdots & a_n^{n-1}-a_1^{n-1} \end{pmatrix}\right)$$

$$=(-1)^{1+1}\cdot 1\cdot\det\left(\begin{pmatrix} a_2-a_1 & a_3-a_1 & a_4-a_1 & \cdots & a_n-a_1 \\ a_2^2-a_1^2 & a_3^2-a_1^2 & a_4^2-a_1^2 & \cdots & a_n^2-a_1^2 \\ a_2^3-a_1^3 & a_3^3-a_1^3 & a_4^3-a_1^3 & \cdots & a_n^3-a_1^3 \\ a_2^4-a_1^4 & a_3^4-a_1^4 & a_4^4-a_1^4 & \cdots & a_n^4-a_1^4 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_2^{n-1}-a_1^{n-1} & a_3^{n-1}-a_1^{n-1} & a_4^{n-1}-a_1^{n-1} & \cdots & a_n^{n-1}-a_1^{n-1} \end{pmatrix}\right)$$

To simplify this, we note the following algebraic identity:

$$\begin{aligned} b^n-a^n-a\big(b^{n-1}-a^{n-1}\big) &= (b-a)\left(\sum_{i=0}^{n-1}a^ib^{n-1-i}-a\sum_{i=0}^{n-2}a^ib^{n-2-i}\right) \\ &= (b-a)\left(\sum_{i=0}^{n-1}a^ib^{n-1-i}-\sum_{j=1}^{n-1}a^jb^{n-2-j+1}\right) \\ &= (b-a)\left(\sum_{i=0}^{n-1}a^ib^{n-1-i}-\sum_{j=1}^{n-1}a^jb^{n-1-j}\right) \\ &= (b-a)\big(b^{n-1}\big) \end{aligned}$$

Thus,

$$\det(A)=(-1)^{1+1}\cdot 1\cdot\det\left(\begin{pmatrix} a_2-a_1 & a_3-a_1 & a_4-a_1 & \cdots & a_n-a_1 \\ a_2^2-a_1^2 & a_3^2-a_1^2 & a_4^2-a_1^2 & \cdots & a_n^2-a_1^2 \\ a_2^3-a_1^3 & a_3^3-a_1^3 & a_4^3-a_1^3 & \cdots & a_n^3-a_1^3 \\ a_2^4-a_1^4 & a_3^4-a_1^4 & a_4^4-a_1^4 & \cdots & a_n^4-a_1^4 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_2^{n-1}-a_1^{n-1} & a_3^{n-1}-a_1^{n-1} & a_4^{n-1}-a_1^{n-1} & \cdots & a_n^{n-1}-a_1^{n-1} \end{pmatrix}\right)$$

$$=\det\left(\begin{pmatrix} a_2-a_1 & a_3-a_1 & \cdots & a_n-a_1 \\ a_2^2-a_1^2 & a_3^2-a_1^2 & \cdots & a_n^2-a_1^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_2^{n-1}-a_1^{n-1} & a_3^{n-1}-a_1^{n-1} & \cdots & a_n^{n-1}-a_1^{n-1} \end{pmatrix}\right)$$

$$
\begin{aligned}
=&\det\left(\begin{pmatrix}
a_2-a_1 & a_3-a_1 & \cdots & a_n-a_1\\
a_2^2-a_1^2 & a_3^2-a_1^2 & \cdots & a_n^2-a_1^2\\
\vdots & \vdots & \ddots & \vdots\\
a_2^{n-1}-a_1^{n-1} & a_3^{n-1}-a_1^{n-1} & \cdots & a_n^{n-1}-a_1^{n-1}\\
-a_1(a_2^{n-2}-a_1^{n-2}) & -a_1(a_3^{n-2}-a_1^{n-2}) & & -a_1(a_n^{n-2}-a_1^{n-2})
\end{pmatrix}\right)\\
=&\det\left(\begin{pmatrix}
a_2-a_1 & a_3-a_1 & \cdots & a_n-a_1\\
a_2^2-a_1^2 & a_3^2-a_1^2 & \cdots & a_n^2-a_1^2\\
\vdots & \vdots & \ddots & \vdots\\
(a_2-a_1)a_2^{n-2} & (a_3-a_1)a_3^{n-2} & \cdots & (a_n-a_1)a_n^{n-2}
\end{pmatrix}\right)\\
=&\det\left(\begin{pmatrix}
a_2-a_1 & a_3-a_1 & \cdots & a_n-a_1\\
(a_2-a_1)a_2 & (a_3-a_1)a_3 & \cdots & (a_n-a_1)a_n\\
\vdots & \vdots & \ddots & \vdots\\
(a_2-a_1)a_2^{n-2} & (a_3-a_1)a_3^{n-2} & \cdots & (a_n-a_1)a_n^{n-2}
\end{pmatrix}\right)\\
=&\left(\prod_{j=2}^{n}(a_j-a_1)\right)\cdot\det\left(\begin{pmatrix}
1 & 1 & \cdots & 1\\
a_2 & a_3 & \cdots & a_n\\
\vdots & \vdots & \ddots & \vdots\\
a_2^{n-2} & a_3^{n-2} & \cdots & a_n^{n-2}
\end{pmatrix}\right)\\
=&\left(\prod_{j=2}^{n}(a_j-a_1)\right)\cdot\prod_{i=2}^{n-1}\prod_{j=i+1}^{n}(a_j-a_i)\\
=&\left(\prod_{i=1}^{1}\prod_{j=i+1}^{n}(a_j-a_i)\right)\cdot\prod_{i=2}^{n-1}\prod_{j=i+1}^{n}(a_j-a_i)\\
=&\prod_{i=1}^{n-1}\prod_{j=i+1}^{n}(a_j-a_i)
\end{aligned}
$$

Thus, by the principle of mathematical induction,

$$\det(A)=\prod_{i=1}^{n-1}\prod_{j=i+1}^{n}(a_j-a_i)$$

As needed. ■

## 7.21. Problem 21

### 7.21.1. (i)

Cofactor expansion on the leftmost column gives:

$$
\begin{aligned}
\det(A)=&a\det\left(\begin{pmatrix}d & g & h\\ 0 & i & j\\ 0 & k & l\end{pmatrix}\right)-c\det\left(\begin{pmatrix}b & e & f\\ 0 & i & j\\ 0 & k & l\end{pmatrix}\right)\\
=&a\left(d\det\left(\begin{pmatrix}i & j\\ k & l\end{pmatrix}\right)\right)-c\left(b\det\left(\begin{pmatrix}i & j\\ k & l\end{pmatrix}\right)\right)\\
=&ad\det\left(\begin{pmatrix}i & j\\ k & l\end{pmatrix}\right)-cb\det\left(\begin{pmatrix}i & j\\ k & l\end{pmatrix}\right)\\
=&(ad-cb)\det\left(\begin{pmatrix}i & j\\ k & l\end{pmatrix}\right)
\end{aligned}
$$

$$=\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)\det\left(\begin{pmatrix} i & j \\ k & l \end{pmatrix}\right)$$

Thus,

$$\det(A)=\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)\det\left(\begin{pmatrix} i & j \\ k & l \end{pmatrix}\right)$$

As needed. ■

**7.21.2. (ii)**

The proof is by induction on $l$.

First, let us consider the base case $l = 1$.

Then, the matrix is of the following form:

$$B=\begin{pmatrix} x & Y \\ 0_{k\times 1} & Z \end{pmatrix}$$

Cofactor expansion on the leftmost column gives:

$$\det(B)=x\det(Z)$$

Since the determinant of the $1 \times 1$ matrix $X = (x)$ is $\det(X) = x$, we have:

$$\det(B)=\det(X)\det(Z)$$

This proves the formula in the base case.

Now, we will show the inductive step. Suppose that $\det(B) = \det(X)\det(Z)$ holds for block-upper triangular matrices $B'$ whose upper-left block is of size $l \times l$.

We want to show that it holds for block-upper triangular matrices $B$ whose upper-left block is of size $(l + 1) \times (l + 1)$.

Let us consider a cofactor expansion on the first column of $B$. Since $l + 1 > 0$, the first column should consist of $l + 1$ potentially nonzero entries on top and all zero entries on the bottom.

Thus,

$$\begin{aligned}
\det(B)&=\sum_{k=1}^{l+1}(-1)^k B_{k1}\det(\text{minormatrix}(B,k,1))\\
&=\sum_{k=1}^{l+1}(-1)^k B_{k1}\det(\text{minormatrix}(X,k,1))\det(Z)\\
&=\left(\sum_{k=1}^{l+1}(-1)^k B_{k1}\det(\text{minormatrix}(X,k,1))\right)\det(Z)\\
&=\det(X)\det(Z)
\end{aligned}$$

Thus, we have shown that $\det(B) = \det(X)\det(Z)$. This completes the inductive step.

Therefore, for any $l, k \in \mathbb{Z}^+$, we have:

$$\det(B)=\det(X)\det(Z)$$

As needed. ■

## 7.22. Problem 22

### 7.22.1. (a)

Since $A$ is invertible, we have:

$$\begin{aligned}\det(I_n + AB) &= \det(A(A^{-1} + B)) \\ &= \det((A^{-1} + B)A) \\ &= \det(A^{-1}A + BA) \\ &= \det(I_n + BA)\end{aligned}$$

Thus:

$$\det(I_n + AB) = \det(I_n + BA)$$

### 7.22.2. (b)

We note that:

$$\begin{aligned}(I_n + X)(I_n - X + X^2 - X^3 + \cdots + (-1)^n X^n) &= I_n - X + X^2 - X^3 + \cdots + (-1)^n X^n \\ &\quad + X - X^2 + X^3 - \cdots + (-1)^n X^{n+1} \\ &= I_n + (-1)^n X^{n+1} \\ &= I_n + (-1)^n X X^n \\ &= I_n + (-1)^n X 0_{n\times n} \\ &= I_n\end{aligned}$$

Thus, $I_n + X$ is invertible, with $(I_n + X)^{-1} = I_n - X + X^2 - X^3 + \cdots + (-1)^n X^n$.

Therefore, by the first big determinant theorem,

$$\det(I_n + X) \neq 0$$

As needed. ■

## 7.23. Problem 23

Let $\pi : \{1, 2, 3, ..., n\} \to \{1, 2, 3, ..., n\}$ be a permutation of $\{1, 2, 3, ..., n\}$, such that $i$ is mapped to $\pi(i)$.

Let the identity permutation $e$ be the unique permutation satisfying $e(i) = i$.

The sign of a permutation $\pi$ is equal to $(-1)^{\text{swaps}}$, where swaps is the number of swaps required to obtain $\pi$ from the identity permutation $e$.

Let $\pi_m$ be the unique permutation on $\{1, 2, 3, ..., n\}$ such that:

$$\pi_m(i) = \begin{cases} m \text{ if } i = 1 \\ i - 1 \text{ if } i \leq m \\ i \text{ otherwise} \end{cases}$$

I claim that $\text{sign}(\pi_m) = (-1)^{m-1}$.

The proof is by construction.

The table of values of $\pi_m$ is:

| $i$ | 1 | 2 | $\cdots$ | $m$ | $m+1$ | $m+2$ | $\cdots$ | $n$ |
|---|---|---|---|---|---|---|---|---|
| $\pi_m(i)$ | $m$ | 1 | ... | $m-1$ | $m+1$ | $m+2$ | $\cdots$ | $n$ |

Meanwhile, the identity is:

| $i$ | 1 | 2 | $\cdots$ | $m$ | $m+1$ | $m+2$ | $\cdots$ | $n$ |
|---|---|---|---|---|---|---|---|---|
| $e(i)$ | 1 | 2 | ... | $m$ | $m+1$ | $m+2$ | $\cdots$ | $n$ |

We can obtain $\pi_m$ from the identity $e$ through the following sequence of swaps:

1. $i = m-1$, $i = m$
2. $i = m-2$, $i = m-1$
3. $\vdots$
4. $i = 1$, $i = 2$

In total, there are $m-1$ of these swaps. Thus, $\text{sign}(\pi_m) = (-1)^{m-1}$, as needed.

Notice that, once we fix $m$ so that $\pi(i) = m$, the rest of the permutation $\pi$ is isomorphic (i.e., could be relabeled) to some permutation over $\{1, 2, 3, ..., n-1\}$.

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## 7.24. Problem 24

### 7.24.1. (b)

Yes, $J_{2N}$ is invertible. Its inverse is:

$$J_{2N}^{-1} = \begin{pmatrix} 0_{N\times N} & -I_N \\ I_N & 0_{N\times N} \end{pmatrix}$$

### 7.24.2. (a)

Since $J_{2N}$ is invertible, $\det(J_{2N}) \neq 0$.

Furthermore, we have:

$$\begin{aligned} \det(S^T J_{2N} S) &= \det(J_{2N}) \\ \det(S^T)\det(J_{2N})\det(S) &= \det(J_{2N}) \\ \det(S^T)\det(S) &= 1 \\ \det(S)^2 &= 1 \end{aligned}$$

Thus, $\det(S) = \pm 1$.

### 7.24.3. (c)

Since the determinant of any symplectic matrix is nonzero, all symplectic matrices are invertible.

### 7.24.4. (e)

Multiplying both sides of the equation by $J_{2N}^{-1}$ to the left:

$$\begin{aligned} S^T J_{2N} S &= J_{2N} \\ J_{2N}^{-1} S^T J_{2N} S &= J_{2N}^{-1} J_{2N} \\ -J_{2N} S^T J_{2N} S &= I_{2N} \end{aligned}$$

$$-\begin{pmatrix}0_{N\times N} & I_N\\ -I_N & 0_{N\times N}\end{pmatrix}\begin{pmatrix}A^T & C^T\\ B^T & D^T\end{pmatrix}\begin{pmatrix}0_{N\times N} & I_N\\ -I_N & 0_{N\times N}\end{pmatrix}\begin{pmatrix}A & B\\ C & D\end{pmatrix}=I_{2N}$$

$$-\begin{pmatrix}0_{N\times N} & I_N\\ -I_N & 0_{N\times N}\end{pmatrix}\begin{pmatrix}-C^T & A^T\\ -D^T & B^T\end{pmatrix}\begin{pmatrix}A & B\\ C & D\end{pmatrix}=I_{2N}$$

$$-\begin{pmatrix}-D^T & B^T\\ C^T & -A^T\end{pmatrix}\begin{pmatrix}A & B\\ C & D\end{pmatrix}=I_{2N}$$

$$\begin{pmatrix}D^T & -B^T\\ -C^T & A^T\end{pmatrix}\begin{pmatrix}A & B\\ C & D\end{pmatrix}=I_{2N}$$

$$\begin{pmatrix}D & -C\\ -B & A\end{pmatrix}^T\begin{pmatrix}A & B\\ C & D\end{pmatrix}=I_{2N}$$

Thus, $S^{-1}=\begin{pmatrix}D^T & -B^T\\ -C^T & A^T\end{pmatrix}=\begin{pmatrix}D & -C\\ -B & A\end{pmatrix}^T$

**7.24.5. (d)**

**Note to self:** READ the problem statement! You didn't see the $2\times 2$ matrix condition and thus thought this problem was hard.

Suppose $S=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$. We want to show that $\det(S)=1$.

We know that:

$$S^TJ_{2N}S=J_2N$$

Thus:

$$\begin{pmatrix}a & c\\ b & d\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$$

$$\begin{pmatrix}a & c\\ b & d\end{pmatrix}\begin{pmatrix}c & d\\ -a & -b\end{pmatrix}=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$$

$$\begin{pmatrix}0 & ad-bc\\ bc-ad & 0\end{pmatrix}=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$$

Thus, $1=ad-bc=\det(S)$, as needed.

Now suppose $\det(S)=1$. We want to show that $S$ is symplectic.

Since $\det(S)=1$, $ad-bc=1$.

Furthermore, by our work already done:

$$S^TJ_{2N}S=\begin{pmatrix}a & c\\ b & d\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}=\begin{pmatrix}0 & ad-bc\\ bc-ad & 0\end{pmatrix}=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}=J_{2N}$$

Thus, $S$ is symplectic, as needed.

Therefore, a $2\times 2$ matrix $S$ is symplectic if and only if $\det(S)=1$. ■

# 8. Problem Set 7

## 8.1. Problem 1

$$\begin{aligned}
\begin{pmatrix} 2 & 4 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{pmatrix}^{-1} &= \frac{1}{\begin{vmatrix} 2 & 4 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{vmatrix}} \begin{pmatrix} \begin{vmatrix} 1 & 4 \\ 0 & 3 \end{vmatrix} & -\begin{vmatrix} 0 & 4 \\ 0 & 3 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} \\ -\begin{vmatrix} 4 & 1 \\ 0 & 3 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix} & -\begin{vmatrix} 2 & 4 \\ 0 & 0 \end{vmatrix} \\ \begin{vmatrix} 4 & 1 \\ 1 & 4 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 0 & 4 \end{vmatrix} & \begin{vmatrix} 2 & 4 \\ 0 & 1 \end{vmatrix} \end{pmatrix}^T \\
&= \frac{1}{6} \begin{pmatrix} 3 & -0 & 0 \\ -12 & 6 & -0 \\ 15 & -8 & 2 \end{pmatrix}^T \\
&= \frac{1}{6} \begin{pmatrix} 3 & 0 & 0 \\ -12 & 6 & 0 \\ 15 & -8 & 2 \end{pmatrix}^T \\
&= \frac{1}{6} \begin{pmatrix} 3 & -12 & 15 \\ 0 & 6 & -8 \\ 0 & 0 & 2 \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{2} & -2 & \frac{5}{2} \\ 0 & 1 & -\frac{4}{3} \\ 0 & 0 & \frac{1}{3} \end{pmatrix}
\end{aligned}$$

To save on time, I just compute the cofactors directly for the second and third matrices:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 0 \\ 3 & 0 & 3 \end{pmatrix}^{-1} = \frac{1}{\begin{vmatrix} 1 & 2 & 3 \\ 2 & -1 & 0 \\ 3 & 0 & 3 \end{vmatrix}} \begin{pmatrix} +(-3) & -(6) & +(3) \\ -(6) & +(-6) & -(-6) \\ +(3) & -(-6) & +(-5) \end{pmatrix}^T = -\frac{1}{6} \begin{pmatrix} -3 & -6 & 3 \\ -6 & -6 & 6 \\ 3 & 6 & -5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 5 \\ 2 & 1 & 4 \\ -1 & 6 & 2 \end{pmatrix}^{-1} = \frac{1}{\begin{vmatrix} 1 & 3 & 5 \\ 2 & 1 & 4 \\ -1 & 6 & 2 \end{vmatrix}} \begin{pmatrix} -22 & -8 & 13 \\ 24 & 7 & -9 \\ 7 & 6 & -5 \end{pmatrix}^T = \frac{1}{19} \begin{pmatrix} -22 & 24 & 7 \\ -8 & 7 & 6 \\ 13 & -9 & -5 \end{pmatrix}$$

## 8.2. Problem 2

We know that:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

In particular, this means that:

$$\left(A^{-1}\right)_{ij} = \frac{1}{\det(A)} \cdot (-1)^{i+j} \cdot \det\left(M_{ji}\right)$$

Where $M_{ij}$ is the matrix obtained by removing the $i$th row and $j$th column of $A$.

Let us first show the backward direction. Suppose that $\det(A) = \pm 1$. Then, $\frac{1}{\det(A)} \in \mathbb{Z}$, $(-1)^{i+j} \in \mathbb{Z}$, and $\det\left(M_{ij}\right) \in \mathbb{Z}$ (the last one can be proven either by inducting on matrices with integer entries or using the permutation definition of the determinant).

Since $\mathbb{Z}$ is closed under multiplication, $\frac{1}{\det(A)} \cdot (-1)^{i+j} \cdot \det\big(M_{ji}\big) \in \mathbb{Z}$, thus $\big(A^{-1}\big)_{ij} \in \mathbb{Z}$, as needed.

Next, let us show the forward direction. Suppose that all entries of $A^{-1}$ are integers. Then, $\det(A^{-1}) \in \mathbb{Z}$. We also know that all entries of $A$ are integers. Therefore, $\det(A) \in \mathbb{Z}$.

However, we know that $\det(A^{-1})\det(A) = \det(A^{-1}A) = \det(I) = 1$ and that $\det(A^{-1}), \det(A) \in \mathbb{Z}$. Therefore, $\det(A) \mid 1$, which implies that $\det(A) = \pm 1$, as needed. ■

## 8.3. Problem 3

### 8.3.1. (i)

This corresponds to the following matrix equation:

$$\begin{pmatrix} 1 & 1 & -1 \\ 3 & -1 & 1 \\ 3 & 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}$$

We note that $\begin{vmatrix} 1 & 1 & -1 \\ 3 & -1 & 1 \\ 3 & 2 & 4 \end{vmatrix} = (-4-2) - (12-3) - (6+3) = -6-9-9 = -24$

Thus:

$$x = \frac{\begin{vmatrix} 2 & 1 & -1 \\ 5 & -1 & 1 \\ 0 & 2 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & -1 \\ 3 & -1 & 1 \\ 3 & 2 & 4 \end{vmatrix}} = \frac{2 \cdot (-4-2) - 5 \cdot (4+2)}{-24} = \frac{7}{4}$$

$$y = \frac{\begin{vmatrix} 1 & 2 & -1 \\ 3 & 5 & 1 \\ 3 & 0 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & -1 \\ 3 & -1 & 1 \\ 3 & 2 & 4 \end{vmatrix}} = \frac{-2 \cdot (12-3) + 5 \cdot (4+3)}{-24} = -\frac{17}{24}$$

$$z = \frac{\begin{vmatrix} 1 & 1 & 2 \\ 3 & -1 & 5 \\ 3 & 2 & 0 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & -1 \\ 3 & -1 & 1 \\ 3 & 2 & 4 \end{vmatrix}} = \frac{2 \cdot (6+3) - 5 \cdot (2-3)}{-24} = -\frac{23}{24}$$

### 8.3.2. (ii)

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$4 \times 4$ determinants T_T

## 8.4. Problem 4

### 8.4.1. (i)

$$\det(A) = 8(64) + a(a^2) = 512 + a^3$$

Thus, $A$ is invertible iff $a \neq -8$.

**8.4.2. (ii)**

$$x_1 = \frac{\begin{vmatrix} b & a & 0 \\ c & 8 & a \\ d & 0 & 8 \end{vmatrix}}{\begin{vmatrix} 8 & a & 0 \\ 0 & 8 & a \\ a & 0 & 8 \end{vmatrix}} = \frac{64b - 8ac + a^2 d}{a^3 + 512}$$

$$x_2 = \frac{\begin{vmatrix} 8 & b & 0 \\ 0 & c & a \\ a & d & 8 \end{vmatrix}}{\begin{vmatrix} 8 & a & 0 \\ 0 & 8 & a \\ a & 0 & 8 \end{vmatrix}} = \frac{a^2 b + 64c - 8ad}{a^3 + 512}$$

$$x_3 = \frac{\begin{vmatrix} 8 & a & b \\ 0 & 8 & c \\ a & 0 & d \end{vmatrix}}{\begin{vmatrix} 8 & a & 0 \\ 0 & 8 & a \\ a & 0 & 8 \end{vmatrix}} = \frac{-8ab + a^2 c + 64d}{a^3 + 512}$$

## 8.5. Problem 5

**8.5.1. (i)**

$$\det(A) = a^2(a^2 - b^2) - b^2(a^2 - b^2) = (a^2 - b^2)^2$$

$A$ is invertible exactly when the determinant is not zero. This happens exactly when $a \neq \pm b$.

**8.5.2. (ii)**

No.

**8.5.3. (iii)**

When $A$ is invertible (i.e., when $a \neq \pm b$) , we can solve for $X$ by multiplying along the left and right by the inverse of $A$:

$$A^{-1}(AXA)A^{-1} = A^{-1}IA^{-1}$$

$$(A^{-1}A)X(AA^{-1}) = (A^{-1})^2$$

$$X = (A^{-1})^2$$

Otherwise, when $A$ is not invertible (i.e., when $a = \pm b$), there is no $X$ that satisfies the equation, because:

$$\det(AXA) = \det(A)\det(X)\det(A) = 0 \cdot \det(X) \cdot 0 = 0 \neq 1 = \det(I)$$

So

$$\det(AXA) \neq \det(I)$$

Therefore,

$$AXA \neq I$$

## 8.6. Problem 6

### 8.6.1. (a)

The determinant of $A$ is:

$$\det(A) = 2$$

Thus, $A$ is always invertible.

### 8.6.2. (b)

$$\operatorname{adj}(A) = \begin{pmatrix} +2 & -0 & +(-2a) \\ -(1+a) & +1 & -(a-a(1+a)) \\ +0 & -0 & +2 \end{pmatrix}^T = \begin{pmatrix} 2 & 0 & -2a \\ -1-a & 1 & a^2 \\ 0 & 0 & 2 \end{pmatrix}^T = \begin{pmatrix} 2 & -1-a & 0 \\ 0 & 1 & 0 \\ -2a & a^2 & 2 \end{pmatrix}$$

### 8.6.3. (c)

We have:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) = \frac{1}{2} \operatorname{adj}(A)$$

Thus:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{2} \operatorname{adj}(A) \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & -1-a & 0 \\ 0 & 1 & 0 \\ -2a & a^2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} a+3 \\ -1 \\ -a^2-2a+2 \end{pmatrix}$$

Finally, we have:

$$x = \frac{1}{2}a + \frac{3}{2}$$

$$y = -\frac{1}{2}$$

$$z = -\frac{1}{2}a^2 - a + 1$$

## 8.7. Problem 7

Exploiting the linearity of the determinant:

$$\begin{aligned} \det(B) &= \begin{vmatrix} a_{11} & \frac{1}{2}a_{12} & \frac{1}{3}a_{13} \\ 2a_{21} & a_{22} & \frac{2}{3}a_{23} \\ 3a_{31} & \frac{3}{2}a_{32} & a_{33} \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} a_{11} & a_{12} & \frac{1}{3}a_{13} \\ 2a_{21} & 2a_{22} & \frac{2}{3}a_{23} \\ 3a_{31} & 3a_{32} & a_{33} \end{vmatrix} \\ &= \frac{1}{2} \cdot \frac{1}{3} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 2a_{21} & 2a_{22} & 2a_{23} \\ 3a_{31} & 3a_{32} & 3a_{33} \end{vmatrix} \\ &= \frac{1}{2} \cdot \frac{1}{3} \cdot 2 \cdot 3 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \end{aligned}$$

$$\begin{aligned}&=\frac{1}{2}\cdot\frac{1}{3}\cdot 2\cdot 3\cdot\det(A)\\&=\det(A)\end{aligned}$$

Thus, $\det(A)=\det(B)$.

## 8.8. Problem 8

Since $B$ is an $m\times n$ matrix, $B^T$ is an $n\times m$ matrix. Thus, $B^TB$ is a $n\times n$ matrix.

We want to show that this matrix is symmetric. That is, for any $i,j\in\mathbb{Z}\cap[1,n]$, $(B^TB)_{ij}=(B^TB)_{ji}$.

We achieve this by directly computing the entry using the definition of matrix multiplication:

$$\begin{aligned}(B^TB)_{ij}&=\sum_{k=1}^{m}(B^T)_{ik}(B)_{kj}\\&=\sum_{k=1}^{m}(B)_{ki}(B^T)_{jk}\\&=\sum_{k=1}^{m}(B^T)_{jk}(B)_{ki}\\&=(B^TB)_{ji}\end{aligned}$$

Thus, the matrix is symmetric, as needed. ■

Comment: There is an easier way to show this. To show that a matrix is symmetric, we can show that a matrix is equal to its transpose.

This leads us to the following one-line proof:

$$(B^TB)^T=(B^T)(B^T)^T=(B^T)B=B^TB$$

### 8.8.1. (ii)

No, it is not. First, note that $A$ is a square matrix since it is symmetric.

Consider the matrix $A=\begin{pmatrix}1&0\\0&-1\end{pmatrix}$, which has a negative determinant ($\det(A)=-1$).

However, note that for any $B$, $\det(B^TB)=\det(B^T)\det(B)=\det(B)^2\geq 0$. Thus, we cannot express every symmetric matrix $A$ as a product of the form $B^TB$.

## 8.9. Problem 9

### 8.9.1. (a)

This is quite trivial. Divide the matrix $A$ into blocks, where each block is a single column:

$$A=(\boldsymbol{C_1}\ \boldsymbol{C_2}\ \boldsymbol{C_3}\ \cdots\ \boldsymbol{C_n})$$

Now, we have:

$$AX=(\boldsymbol{C_1}\ \boldsymbol{C_2}\ \boldsymbol{C_3}\ \cdots\ \boldsymbol{C_n})\begin{pmatrix}x_1\\x_2\\\vdots\\x_n\end{pmatrix}=\sum_{i=1}^{n}x_i\boldsymbol{C_i}$$

As needed. ■

### 8.9.2. (b)

We can express $\boldsymbol{A}_j$ using blocks:

$$\boldsymbol{A}_j = \begin{pmatrix} \boldsymbol{C}_1 & \boldsymbol{C}_2 & \cdots & \boldsymbol{C}_{j-1} & \sum_{i=1}^{n} x_i \boldsymbol{C}_i & \boldsymbol{C}_{j+1} & \cdots & \boldsymbol{C}_n \end{pmatrix}$$

By the linearity of the determinant, we have:

$$\det(\boldsymbol{A}_j) = \sum_{i=1}^{n} x_i \begin{vmatrix} \boldsymbol{C}_1 & \boldsymbol{C}_2 & \cdots & \boldsymbol{C}_{j-1} & \boldsymbol{C}_i & \boldsymbol{C}_{j+1} & \cdots & \boldsymbol{C}_n \end{vmatrix}$$

When $i \neq j$, there is a repeated row so the inner determinant is zero. Otherwise, when $i = j$, the matrix $\begin{pmatrix} \boldsymbol{C}_1 & \boldsymbol{C}_2 & \cdots & \boldsymbol{C}_{j-1} & \boldsymbol{C}_i & \boldsymbol{C}_{j+1} & \cdots & \boldsymbol{C}_n \end{pmatrix}$ becomes $\boldsymbol{A}$. Therefore, we get:

$$\det(\boldsymbol{A}_j) = x_j \begin{vmatrix} \boldsymbol{C}_1 & \boldsymbol{C}_2 & \cdots & \boldsymbol{C}_{j-1} & \boldsymbol{C}_j & \boldsymbol{C}_{j+1} & \cdots & \boldsymbol{C}_n \end{vmatrix} = x_j \det(\boldsymbol{A})$$

### 8.9.3. (c)

Using basic algebra,

$$x_j = \frac{\det(\boldsymbol{A}_j)}{\det(\boldsymbol{A})}$$

## 8.10. Problem 10

### 8.10.1. (a)

Since determinants respect multiplication:

$$\det(\boldsymbol{A}^m) = \det(\boldsymbol{0}_{n \times n})$$

$$\det(\boldsymbol{A})^m = 0$$

This immediately implies $\det(\boldsymbol{A}) = 0$.

### 8.10.2. (b)

Recall that:

$$B \operatorname{adj}(B) = \det(B) I$$

Thus:

$$\det(B \operatorname{adj}(B)) = \det(\det(B) I)$$

Since $\det(B)I$ is a diagonal matrix, its determinant is the product of the diagonal entries. Since its diagonal consists of $n$ repetitions of $\det(B)$, $\det(\det(B)I) = \det(B)^m$.

Thus:

$$\det(B \operatorname{adj}(B)) = \det(B)^m$$

$$\det(B) \det(\operatorname{adj}(B)) = \det(B)^m$$

There are two cases to consider.

The first case is when

$$\det(B) \neq 0$$

. In this case:

$$\det(B)\det(\text{adj}(B)) = \det(B)^m \to \det(\text{adj}(B)) = \det(B)^{m-1}$$

Thus, $\det(\text{adj}(B)) = \det(B)^{m-1}$.

In the second case, $\det(B) = 0$.

I claim that $\text{adj}(B)$ is singular. Suppose not, then there exists a matrix $\text{adj}(B)^{-1}$ such that $\text{adj}(B)\text{adj}(B)^{-1} = \text{adj}(B)^{-1}\ \text{adj}(B) = I$.

Multiplying both sides of $B\ \text{adj}(B) = \det(B)I$ by $\text{adj}(B)^{-1}$:

$$B\ \text{adj}(B) = \det(B)I$$

$$B\ \text{adj}(B)\text{adj}(B)^{-1} = \det(B)I\text{adj}(B)^{-1}$$

$$B = \det(B)\text{adj}(B)^{-1}$$

$$B = 0 \cdot \text{adj}(B)^{-1}$$

$$B = \mathbf{0}$$

Thus:

$$\text{adj}(B) = \mathbf{0}$$

But this contradicts the fact that $\text{adj}(B)$ is invertible. Thus, $\text{adj}(B)$ is singular.

This implies that $\det(\text{adj}(B)) = 0$, as needed.

In both cases, the formula $\det(\text{adj}(B)) = \det(B)^{m-1}$ holds. ■

## 8.11. Problem 11

### 8.11.1. (a)

We prove this by checking that every entry is the same. The following calculation verifies this identity:

$$\begin{aligned}
\left((AB)^T\right)_{ij} &= (AB)_{ji} \\
&= \sum_{k=1}^{n} A_{jk}B_{ki} \\
&= \sum_{k=1}^{n} \left(A^T\right)_{kj}\left(B^T\right)_{ik} \\
&= \sum_{k=1}^{n} \left(B^T\right)_{ik}\left(A^T\right)_{kj} \\
&= \left(B^T A^T\right)_{ij}
\end{aligned}$$

This proves the identity.

### 8.11.2. (b)

Since $A$ is symmetric,

$$A^T = A$$

Thus:

$$A^T A^{-1} = AA^{-1} = I$$

Taking the transpose of both sides:

$$(A^T A^{-1})^T = I^T$$

$$(A^{-1})^T A = I = A^{-1} A$$

Thus,

$$\left((A^{-1})^T - A^{-1}\right) A = \mathbf{0}$$

Multiplying both sides by $A^{-1}$:

$$\left((A^{-1})^T - A^{-1}\right) A = \mathbf{0}$$

$$\left((A^{-1})^T - A^{-1}\right) A A^{-1} = \mathbf{0} A^{-1}$$

$$(A^{-1})^T - A^{-1} = \mathbf{0}$$

$$(A^{-1})^T = A^{-1}$$

Thus, $A^{-1}$ is transpose.

### 8.11.3. (c)

Yes. The $(i, j)$-minor of $A$ is just the transpose of the $(j, i)$-minor of $A^T$. But since $A^T = A$, the $(i, j)$-minor of $A$ is the transpose of the $(j, i)$-minor of $A$. Finally, since $\det(B) = \det(B^T)$ for any square matrix $B$, we conclude that the $(i, j)$-entry of $\text{adj}(A)$ (which is $(-1)^{i+j} \det(\text{minor}(A, (i, j)))$) is equal to its $(j, i)$-entry (which is $(-1)^{j+i} \det(\text{minor}(A, (j, i)))$). Therefore, $\text{adj}(A)$ is also symmetric.

Note: In the actual exam, try explicitly computing the $i, j$ entry.

## 8.12. Problem 12

### 8.12.1. (a)

Since $AA^T = I_n$, we have:

$$\det(AA^T) = \det(I_n)$$

$$\det(AA^T) = 1$$

$$\det(A) \det(A^T) = 1$$

$$\det(A)^2 = 1$$

$$(\det(A) - 1)(\det(A) + 1) = 0$$

Thus, $\det(A) = 1$ or $\det(A) = -1$.

### 8.12.2. (b)

Suppose $A$ and $B$ were both orthogonal. Notice that:

$$(AB)(AB)^T = (AB)(B^T A^T) = A(BB^T)A^T = AI_n A^T = AA^T = I_n$$

$$(AB)^T(AB) = (B^T A^T)(AB) = B^T(A^T A)B = B^T I_n B = B^T B = I_n$$

**8.12.3. (c)**
Yes, the inverse of orthogonal matrix $A$ is $A^T$.

**8.12.4. (d)**

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

or even more simply:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

**8.12.5. (e)**
Since $A(t)$ is always orthogonal, we have:

$$A^T(t)A(t) = I_n$$

Thus, by the definition of matrix multiplication:

$$\begin{aligned} \left(A^T(t)A(t)\right)_{ij} &= (I_n)_{ij} \\ \sum_{k=1}^{n} \left(A^T(t)\right)_{ik} (A(t))_{kj} &= (I_n)_{ij} \\ \sum_{k=1}^{n} a_{ki}(t)a_{kj}(t) &= (I_n)_{ij} \end{aligned}$$

Differentiating both sides with respect to time:

$$\begin{aligned} \frac{d}{dt}\sum_{k=1}^{n} a_{ki}(t)a_{kj}(t) &= \frac{d}{dt}(I_n)_{ij} \\ \sum_{k=1}^{n} \left(a_{ki}'(t)a_{kj}(t) + a_{ki}(t)a_{kj}'(t)\right) &= 0 \\ \sum_{k=1}^{n} \left(a_{ki}'(0)a_{kj}(0) + a_{ki}(0)a_{kj}'(0)\right) &= 0 \\ a_{ji}'(0)a_{jj}(0) + a_{ii}(0)a_{ij}'(0) &= 0 \\ a_{ji}'(0) \cdot 1 + 1 \cdot a_{ij}'(0) &= 0 \\ a_{ji}'(0) + a_{ij}'(0) &= 0 \\ a_{ji}'(0) &= -a_{ij}'(0) \end{aligned}$$

Thus, $D$ is indeed anti-symmetric, as needed. ■

## 8.13. Problem 13

It helps to organize the matrices into blocks:

$$X_k = \begin{pmatrix} (A^{-1}B)_{k+1,1} & \cdots & (A^{-1}B)_{k+1,n-k} \\ \vdots & \ddots & \vdots \\ (A^{-1}B)_{n,1} & \cdots & (A^{-1}B)_{n,n-k} \end{pmatrix}$$

$$Y_k = \begin{pmatrix} \boldsymbol{A}_{:,1} & \cdots & \boldsymbol{A}_{:,k} & \boldsymbol{B}_{:,1} & \cdots & \boldsymbol{B}_{:,n-k} \end{pmatrix}$$

Where : denotes picking all entries (eg., $A_{:,1}$ denotes picking all elements along column 1).

Let us left-multiply $Y_k$ by $A^{-1}$:

$$A^{-1}Y_k = \begin{pmatrix} (I_n)_{[1,n],[1,k]} & X_{[1,n],[1,n-k]} \end{pmatrix}$$

We can further decompose the left and right blocks into smaller blocks:

$$A^{-1}Y_k = \begin{pmatrix} I_k & X_{[1,k],[1,n-k]} \\ \mathbf{0}_{(n-k)\times k} & X_{[k+1,n],[1,n-k]} \end{pmatrix}$$

But notice that $X_{[k+1,n],[1,n-k]}$ is just the result of removing the first $k$ rows and the last $k$ columns of $X$. Thus, by definition, $X_k = X_{[k+1,n],[1,n-k]}$. We therefore have:

$$A^{-1}Y_k = \begin{pmatrix} I_k & X_{[1,k],[1,n-k]} \\ \mathbf{0}_{(n-k)\times k} & X_k \end{pmatrix}$$

Finally, we recognize that the matrix to the right is block upper triangular. Therefore, the determinant of the matrix is simply the product of the determinants of the diagonal blocks. We therefore have:

$$\begin{aligned} \det(Y_k) &= \frac{1}{1} \cdot \det(Y_k) \\ &= \frac{1}{\det(A)} \cdot \det(Y_k) \\ &= \det(A^-1) \det(Y_k) \\ &= \det(A^-1 Y_k) \\ &= \begin{vmatrix} I_k & X_{[1,k],[1,n-k]} \\ \mathbf{0}_{(n-k)\times k} & X_k \end{vmatrix} \\ &= \det(I_k) \cdot \det(X_k) \\ &= 1 \cdot \det(X_k) \\ &= \det(X_k) \end{aligned}$$

This proves the result. ■

# 9. Problem Set 8

## 9.1. Problem 1

1. Subspace: Nonempty, closed under vector addition, closed under scalar multiplication.
2. Not a subspace: Add $(1, 1, 1)$ to itself, the result is not in the set.
3. Not a subspace: Same reason as 2.
4. Not a subspace: $(0, 0, 1)$ in set but scalar multiple $(0, 0, \pi)$ is not.

5. Not a subspace: $(0,0,1)$ in set but scalar multiple $(0,0,-1)$ is not.
6. Subspace: Span of $(0,0,1)$.
7. Not a subspace: Zero vector is not in set.
8. Not a subspace: $(3,2,1)$ in set but not scalar multiple $(-3,-2,-1)$.
9. Subspace: Homogenous System.
10. Subspace: Span of $(1,0,0),(0,1,1)$.
11. Not a subspace: $(2,2,4)$ in set but not scalar multiple $(4,4,8)$.
12. Not a subspace: $(1,1,1)$ in set but not scalar multiple $(-1,-1,-1)$.
13. Subspace: $f(x) = x^3$ is bijective over $\mathbb{R}$, so this is just $\mathbb{R}^3$.
14. Subspace: Homogenous System.
15. Not a subspace: Inhomogenous in variables $x-1, y, z$.
16. Subspace: Homogenous in $x, y-1, z$.
17. Subspace: Homogenous System.

## 9.2. Problem 2

Not closed under vector addition: $(0,1,1),(0,1,-1)$ in the set, but the sum $(0,2,0)$ is not.

Closed under scalar multiplication:

$$x^4 + y^4 - z^4 = 0 \to (kx)^4 + (ky)^4 - (kz)^4 = k^4(x^4 + y^4 - z^4) = 0$$

So $\forall k \in \mathbb{R}, (x,y,z) \in S \to (kx, ky, kz) \in S$

Nonempty: $(0,0,0) \in S$

## 9.3. Problem 3

$S$ is just equal to:

$$S = \{(u+v, 2u-v, u+2v) \in \mathbb{R}^3; u, v \in \mathbb{R}\}$$

It is closed under vector addition:

$$(u_1+v_1, 2u_1-v_1, u_1+2v_1) + (u_2+v_2, 2u_2-v_2, u_2+2v_2)$$

$$= ((u_1+u_2)+(v_1+v_2), 2(u_1+u_2)-(v_1+v_2), (u_1+u_2)+2(v_1+v_2))$$

Since $u_1+u_2, v_1+v_2 \in \mathbb{R}$, by definition, the sum is in the set $S$.

$S$ is nonempty because $(0,0,0) \in S$ (set $u = v = 0$ to get the zero vector).

## 9.4. Problem 4

For any element of the set $S$, $a = -b$, so $a(1,1,0) + b(0,1,1) = (a, a-a, -a) = a(1,0,-1)$.

Thus:

$$S = \{a(1,0,-1); a \in \mathbb{R}\}$$

Consider two arbitrary elements of $S$, $a_1(1,0,-1)$ and $a_2(1,0,-1)$.

By definition, since $a_1 + a_2 \in \mathbb{R}$, $(a_1+a_2)(1,0,-1) \in S$.

However, $(a_1+a_2)(1,0,-1) = a_1(1,0,-1) + a_2(1,0,-1)$, so the set $S$ is closed under vector addition.

Further, for any $k \in \mathbb{R}$, $ka_1 \in \mathbb{R}$. Thus, $(ka_1)(1,0,-1) \in S$. This shows that $S$ is closed under scalar multiplication.

Finally, it is nonempty, since $(0,0,0) = 0(1,0,-1) \in S$.

Thus, $S$ is a subspace.

## 9.5. Problem 5

### 9.5.1. 1

$U$ and $V$ contain the origin.

### 9.5.2. 2

It the list obtained by concatenating $L_1$ and $L_2$.

### 9.5.3. 3

#### 9.5.3.1. $U \cap V$

The list of equations with this set of solutions is:

$$2x - y + 3z = 0$$

$$3x + 2y - z = 0$$

This corresponds to the following augmented matrix:

$$\left(\begin{array}{ccc|c} 2 & -1 & 3 & 0 \\ 3 & 2 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 6 & -3 & 9 & 0 \\ 6 & 4 & -2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 6 & -3 & 9 & 0 \\ 0 & 7 & -11 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 42 & -21 & 63 & 0 \\ 0 & 21 & -33 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 42 & 0 & 30 & 0 \\ 0 & 21 & -33 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 7 & 0 & 5 & 0 \\ 0 & 7 & -11 & 0 \end{array}\right)$$

A parameterization for the set of solutions is thus:

$$(-5r, 11r, 7r) = r(-5, 11, 7)$$

#### 9.5.3.2. $V \cap W$

The list of equations whose set of solutions is $V \cap W$ is:

$$3x + 2y - z = 0$$

$$x - 3y - 2z = 1$$

This corresponds to the following augmented matrix:

$$\left(\begin{array}{ccc|c} 3 & 2 & -1 & 0 \\ 1 & -3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -3 & -2 & 1 \\ 3 & 2 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} -3 & 9 & 6 & -3 \\ 3 & 2 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} -3 & 9 & 6 & -3 \\ 0 & 11 & 5 & -3 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} -33 & 99 & 66 & -33 \\ 0 & 99 & 45 & -27 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} -33 & 0 & 21 & -6 \\ 0 & 99 & 45 & -27 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 11 & 0 & -7 & 2 \\ 0 & 11 & 5 & -3 \end{array}\right)$$

A parameterization for the set of solutions is thus:

$$\left(\frac{2}{11} + 7r, -\frac{3}{11} - 5r, 11r\right) = \left(\frac{2}{11}, -\frac{3}{11}, 0\right) + r(7, -5, 11)$$

#### 9.5.4. 4

$U \cap V$ is a subspace because it can be expressed as the span of $(-5, 11, 7)$.

$V \cap W$, on the other hand, is not a subspace because $(0, 0, 0)$ is not in the set (you can easily check this).

## 9.6. Problem 6

#### 9.6.1. (a)

$a + d$

#### 9.6.2. (b)

Subspace: Homogenous System.

#### 9.6.3. (c)

Not a subspace: $(1, 1, 1, 1)$ is in the set but not its scalar multiple $(-1, -1, -1, -1)$.

#### 9.6.4. (d)

We have:

$$\operatorname{tr}\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}^2\right) = (a^2 + bc) + (d^2 + bc) = 0$$

So, $(1, 1, -1, 1), (-1, 1, -1, 1) \in S_3$. However, notice that the sum $(0, 2, -2, 0) \notin S_3$.

#### 9.6.5. (e)

Notice that

$$\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}^T\right) = \det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}^T\right) = \det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)^2$$

This is always $\geq 0$, so $S_4 = \mathbb{R}^4$, which is clearly a subspace.

## 9.7. Problem 7

### 9.7.1. (i)

Not a subspace. Inhomogenous.

### 9.7.2. (ii)

Subspace. Determinant reduces to homogenous system.

### 9.7.3. (iii)

Subspace. The determinant is $c - b$. This reduces to a homogenous system.

### 9.7.4. (iv)

Subspace. Homogenous system.

### 9.7.5. (v)

Subspace. Determinant is always zero, so $S_5 = \mathbb{R}^3$.

### 9.7.6. (vi)

Subspace. Determinant is zero because three of the vectors are linearly dependent. Thus, the rank is at most 3. Therefore, the RREF contains at least one row of zeros, so the RREF is not the identity matrix. Therefore, the matrix is noninvertible, which implies zero determinant.

Therefore, the condition is always true and $S_6 = \mathbb{R}^3$.

## 9.8. Problem 8

### 9.8.1. (a)

Not a subspace. $(1, 1, 0), (0, 0, 1) \in S$, but their sum $(1, 1, 1)$ is not.

### 9.8.2. (b)

$\mathbb{Z}^2$

$(1, 1)$ is in the set but scalar multiple $(\pi, \pi)$ is not.

## 9.9. Problem 9

Applying cofactor expansion on the last row, the determinant condition becomes:

$$\det(M_{\vec{v}}) = \sum_{i=1}^{n} \det(\text{minor}(M_{\vec{v}}, (i, n)))(\vec{v})_i = 0$$

This is clearly a homogenous system in terms of the entries of $\vec{v}$. Thus, we conclude that the set $S$ is a subspace of $\mathbb{R}^n$. ■

NOTE: In the actual exam, this is **not enough**. If it asks for a careful proof, we must check the axioms.

## 9.10. Problem 10

### 9.10.1. (i)

Suppose $(x_1, z_1), (x_2, z_2) \in W$ are two particular but arbitrarily-chosen elements of $W$.

By definition, there must exist $y_1, y_2 \in \mathbb{R}$ such that $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$.

Since $V$ is a subspace, $((x_1 + x_2), (y_1 + y_2), (z_1 + z_2)) = (x_1, y_1, z_1) + (x_2, y_2, z_2) \in V$. Thus, by definition, $(x_1, z_1) + (x_2, z_2) = (x_1 + x_2, z_1 + z_2) \in W$.

Thus, $W$ is closed under vector addition.

Further, since $(x_1, y_1, z_1) \in V$, $(kx_1, ky_1, kz_1) = k(x_1, y_1, z_1) \in V$. Therefore, by definition, $k(x_1, z_1) = (kx_1, kz_1) \in W$.

Thus, $W$ is closed under scalar multiplication.

Finally, $W$ is nonempty since $V$ is nonempty. Formally, suppose $(x_1, y_1, z_1) \in V$ is some element. Then, $(x_1, z_1) \in W$.

#### 9.10.2. (ii)

$$W = \{(x, 0); x \in \mathbb{R}\}$$

## 9.11. Problem 11

#### 9.11.1. 1

It is definitely closed. Suppose $\vec{v_1}, \vec{v_2} \in V$. Then, there exist $\vec{u_1}, \vec{u_2} \in \mathbb{R}^n$ such that:

$$\vec{v_1} = A\vec{u_1}$$

$$\vec{v_2} = A\vec{u_2}$$

Since $\vec{u_1}, \vec{u_2} \in \mathbb{R}^n$, $\vec{u_1} + \vec{u_2} \in \mathbb{R}^n$.

Thus, $A(\vec{u_1} + \vec{u_2}) \in V$ by definition. However, $A(\vec{u_1} + \vec{u_2}) = A\vec{u_1} + A\vec{u_2} = \vec{v_1} + \vec{v_2}$. Thus, $\vec{v_1} + \vec{v_2} = A(\vec{u_1} + \vec{u_2}) \in V$. This proves that $V$ is closed under vector addition.

Furthermore, let $k \in \mathbb{R}$ and $\vec{v} \in V$ be particular but arbitrarily chosen elements of their respective sets.

By definition, there exists a $\vec{u} \in \mathbb{R}^n$ such that $\vec{v} = A\vec{u}$.

Since $\vec{u} \in \mathbb{R}$, $k\vec{u} \in \mathbb{R}$. Thus, by definition, $k\vec{v} = k(A\vec{u} = A(k\vec{u}) \in V$. Thus, $V$ is closed under scalar multiplication.

Finally, $V$ is clearly nonempty as $A\vec{0} \in V$.

#### 9.11.2. 2

(i) $(x_1 + 2x_2 + 3x_3, x_2 + x_3)$

(ii) $(x_1, 2x_1 + x_2, 3x_1 + x_2)$

## 9.12. Problem 12

Suppose not. Then, there exist $\vec{v_1}, \vec{v_2} \in \mathbb{R}^n$ such that $\vec{v_1} \in V_1 \wedge \vec{v_1} \notin V_2$ and $\vec{v_2} \notin V_1 \wedge \vec{v_2} \in V_2$.

Since $V_1 \cup V_2$ is a subspace and $\vec{v_1}, \vec{v_2} \in V_1 \cup V_2$, $\vec{v_1} + \vec{v_2} \in V_1 \cup V_2$.

Let us consider both cases.

Suppose $\vec{v_1} + \vec{v_2} \in V_1$. Since $\vec{v_1} \in V_1$, $\vec{v_2} = (\vec{v_1} + \vec{v_2}) - \vec{v_1} \in V_1$, contradiction.

Now suppose $\vec{v_1} + \vec{v_2} \in V_2$. Since $\vec{v_2} \in V_2$, $\vec{v_1} = (\vec{v_1} + \vec{v_2}) - \vec{v_2} \in V_2$, contradiction.

In both cases, we have a contradiction. Therefore, we conclude that $V_1 \subset V_2 \vee V_2 \subset V_1$.

## 9.13. Problem 13

### 9.13.1. 1

Suppose $\vec{u_1}, \vec{u_2} \in V_1 + V_2$ are particular but arbitrarily chosen elements of the set. By definition, there exist $\vec{w_1}, \vec{w_2} \in V_1$ and $\vec{v_1}, \vec{v_2} \in V_2$ such that:

$$\vec{u_1} = \vec{w_1} + \vec{v_1}$$

$$\vec{u_2} = \vec{w_2} + \vec{v_2}$$

Let us first verify that $V_1 + V_2$ is closed under vector addition. That is, that $\vec{u_1} + \vec{u_2} \in V_1 + V_2$

Since $V_1$ and $V_2$ are subspaces, $\vec{w_1} + \vec{w_2} \in V_1$ and $\vec{v_1} + \vec{v_2} \in V_2$.

Thus, by definition,

$$(\vec{w_1} + \vec{w_2}) + (\vec{v_1} + \vec{v_2}) \in V_1 + V_2$$

Further,

$$\vec{u_1} + \vec{u_2} = \vec{w_1} + \vec{v_1} + \vec{w_2} + \vec{v_2} = (\vec{w_1} + \vec{w_2}) + (\vec{v_1} + \vec{v_2})$$

Thus,

$$\vec{u_1} + \vec{u_2} \in V_1 + V_2$$

This proves that $V_1 + V_2$ is closed under vector addition.

Now, we want to show that $V_1 + V_2$ is closed under scalar multiplication. That is, that $k\vec{u_1} \in V_1 + V_2$

Since $\vec{w_1} \in V_1$ and $\vec{v_1} \in V_2$ and both $V_1, V_2$ are subspaces, for any $k \in \mathbb{R}$, $k\vec{w_1} \in V_1$ and $k\vec{v_1} \in V_2$. Thus, by definition,

$$(k\vec{w_1}) + (k\vec{v_1}) \in V_1 + V_2$$

Further,

$$k\vec{u_1} = k(\vec{w_1} + k\vec{v_1}) = (k\vec{w_1}) + (k\vec{v_1})$$

Thus,

$$k\vec{u_1} \in V_1 + V_2$$

As needed. This shows that $V_1 + V_2$ is closed under scalar multiplication.

Finally, we must show that $V_1 + V_2$ is nonempty.

Since $V_1, V_2$ are both subspaces, $\vec{0} \in V_1 \wedge \vec{0} \in V_2$.

Therefore, by definition, $\vec{0} = \vec{0} + \vec{0} \in V_1 + V_2$.

Thus, $V_1 + V_2$ is nonempty. Since $V_1 + V_2$ satisfies all subspace axioms, $V_1 + V_2$ is a subspace, as needed. ■

### 9.13.2. 2

Let $\vec{u}$ be a particular but arbitrarily-chosen element of $V_1 + V_2$. By definition, there exist $\vec{v_1} \in V_1$ and $\vec{v_2} \in V_2$ such that $\vec{u} = \vec{v_1} + \vec{v_2}$.

Since $V_1 \subset W$ and $V_2 \subset W$ and $\vec{v_1} \in V_1$ and $\vec{v_2} \in V_2$, we conclude that both $\vec{v_1} \in W$ and $\vec{v_2} \in W$ hold.

Finally, since $W$ is a subspace, $\vec{v_1} + \vec{v_2} \in W$. But $\vec{u} = \vec{v_1} + \vec{v_2}$. Therefore, $\vec{u} \in W$.

Thus, $V_1 + V_2 \subset W$.

## 9.14. Problem 14

### 9.14.1. (a)

First, let us show that the set $S$ is closed under vector addition. Suppose that $\vec{s}, \vec{t} \in S$ are two particular but arbitrarily-chosen elements of $S$. We want to show that $\vec{s} + \vec{t} \in S$

By definition, we have the following equalities:

$$\sum_{i=1}^{n} s_i v_i = 0$$

$$\sum_{i=1}^{n} t_i v_i = 0$$

Notice that:

$$\sum_{i=1}^{n} (s_i + t_i) v_i = \sum_{i=1}^{n} (s_i v_i + t_i v_i) = \sum_{i=1}^{n} s_i v_i + \sum_{i=1}^{n} t_i v_i = 0 + 0 = 0$$

However, $\left(\vec{s} + \vec{t}\right) \cdot \vec{v} = \sum_{i=1}^{n} (s_i + t_i) v_i$ by definition, so $\left(\vec{s} + \vec{t}\right) \cdot \vec{v} = 0$.

By definition, this implies that $\vec{s} + \vec{t} \in S$, as needed.

Next, let us show that the set $S$ is closed under scalar multiplication. Suppose that $\vec{s} \in S$ is a particular but arbitrarily-chosen element of $S$. For any arbitrary constant $k \in \mathbb{R}$, we want to show that $k\vec{s} \in S$.

Since $\vec{s} \in S$, by definition, we have:

$$\sum_{i=1}^{n} s_i v_i = 0$$

Now, notice that:

$$(k\vec{s}) \cdot \vec{v} = \sum_{i=1}^{n} (k s_i) v_i = \sum_{i=1}^{n} k(s_i v_i) = k \sum_{i=1}^{n} (s_i v_i) = k \cdot 0 = 0$$

Thus, by definition, $k\vec{s} \in S$.

Finally, $S$ is nonempty since $\vec{v} \cdot \vec{0} = \sum_{i=1}^{n} v_i \cdot 0 = \sum_{i=1}^{n} 0 = 0$, so $\vec{0} \in S$.

Since $S$ satisfies all three subspace axioms, $S$ is a subspace.

### 9.14.2. (b)

I claim that setting $c = \dfrac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}}$ and $\vec{z} = \vec{w} - \dfrac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}} \vec{v}$ works.

First, clearly $c \in \mathbb{R}$ and $\vec{w} = \left(\dfrac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}}\right) \vec{v} + \left(\vec{w} - \dfrac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}} \vec{v}\right) = c\vec{v} + \vec{z}$.

We thus only need to show that $\vec{z} \in S$, that is, that $\vec{v} \cdot \vec{z} = 0$.

We notice that:

$$\begin{aligned}
\vec{v}\cdot\vec{z} &= \sum_{i=1}^{n} v_i z_i \\
&= \sum_{i=1}^{n} v_i \left( w_i - \frac{\vec{v}\cdot\vec{w}}{\vec{v}\cdot\vec{v}} v_i \right) \\
&= \sum_{i=1}^{n} v_i w_i - \sum_{i=1}^{n} \frac{\vec{v}\cdot\vec{w}}{\vec{v}\cdot\vec{v}} v_i^2 \\
&= \sum_{i=1}^{n} v_i w_i - \frac{\vec{v}\cdot\vec{w}}{\vec{v}\cdot\vec{v}} \sum_{i=1}^{n} v_i^2 \\
&= \sum_{i=1}^{n} v_i w_i - \frac{\sum_{i=1}^{n} v_i w_i}{\sum_{i=1}^{n} v_i^2} \sum_{i=1}^{n} v_i^2 \\
&= \sum_{i=1}^{n} v_i w_i - \left( \sum_{i=1}^{n} v_i w_i \right) \\
&= 0
\end{aligned}$$

Thus, $\vec{v}\cdot\vec{z} = 0$. By definition, we conclude $\vec{z} \in S$, so indeed this construction is valid. We thus prove the original statement. ■

# 10. Problem Set 9

## 10.1. Problem 1

### 10.1.1. (1)

(Note: I accidentally swapped the order of w, x, y, z. oops)

We can do this by directly verifying the subspace axioms.

First, suppose that $(x_1, y_1, z_1, w_1), (x_2, y_2, z_2, w_2) \in V$ are particular but arbitrarily-chosen (not necessarily distinct) elements of $V$.

By definition,

$$w_1 + x_1 = y_1 + z_1 = -x_1 + 2y_1$$

$$w_2 + x_2 = y_2 + z_2 = -x_2 + 2y_2$$

Adding these equations together, we get:

$$(w_1 + w_2) + (x_1 + x_2) = (y_1 + y_2) + (z_1 + z_2) = -(x_1 + x_2) + 2(y_1 + y_2)$$

Therefore, by definition, $(x_1 + x_2, y_1 + y_2, z_1 + z_2, w_1 + w_2) \in V$.

Thus, $V$ is closed under vector addition.

Furthermore, let $c \in \mathbb{R}$.

Multiplying the above equations by $c$, we obtain the following set of equations:

$$(cw_1) + (cx_1) = (cy_1) + (cz_1) = -(cx_1) + 2(cy_1)$$

Thus, by definition, $c(x_1, y_1, z_1, w_1) = (cx_1, cy_1, cz_1, cw_1) \in V$. Therefore, $V$ is closed under scalar multiplication.

Finally, $V$ is nonempty since setting $(x, y, z, w) = (0, 0, 0, 0)$ satisfies the equations and thus $(0, 0, 0, 0) \in V$.

Thus, $V$ is a subspace of $\mathbb{R}^4$.

**10.1.2. (2)**
Rewriting the equations:

$$w + x - y - z = 0$$

$$-x + y - z = 0$$

The equations correspond to the following augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & -1 & 0 \\ 0 & -1 & 1 & -1 & 0 \end{array}\right)$$

Reducing the matrix into RREF:

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & -1 & 0 \\ 0 & -1 & 1 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -2 & 0 \\ 0 & -1 & 1 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -2 & 0 \\ 0 & 1 & -1 & 1 & 0 \end{array}\right)$$

Thus, a parameterization for this system is:

$$w = 2s$$

$$x = r - s$$

$$y = r$$

$$z = s$$

Therefore:

$$(w, x, y, z) = r(0, 1, 1, 0) + s(2, -1, 0, 1)$$

Thus,

$$V = \text{span}\{(0, 1, 1, 0), (2, -1, 0, 1)\}$$

## 10.2. Problem 2

We can express this system as the following augmented matrix:

$$\left(\begin{array}{cccc|c} 0 & 10 & -4 & 1 & 0 \\ 1 & 4 & -1 & 1 & 0 \\ 3 & 2 & 1 & 2 & 0 \\ -2 & -8 & 2 & -2 & 0 \\ 1 & -6 & 3 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 0 & 10 & -4 & 1 & 0 \\ 1 & 4 & -1 & 1 & 0 \\ 3 & 2 & 1 & 2 & 0 \\ 1 & 4 & -1 & 1 & 0 \\ 1 & -6 & 3 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 0 & 10 & -4 & 1 & 0 \\ 1 & 4 & -1 & 1 & 0 \\ 3 & 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & -6 & 3 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 0 \\ 0 & 10 & -4 & 1 & 0 \\ 1 & 4 & -1 & 1 & 0 \\ 3 & 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 0 \\ 0 & 10 & -4 & 1 & 0 \\ 0 & 10 & -4 & 1 & 0 \\ 0 & 20 & -8 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 0 \\ 0 & 10 & -4 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 0 & 0 \\ 0 & 1 & -\frac{2}{5} & \frac{1}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & \frac{3}{5} & \frac{3}{5} & 0 \\ 0 & 1 & -\frac{2}{5} & \frac{1}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Thus, a parameterization for this system is:

$$\left(-\frac{3}{5}r - \frac{3}{5}s, \frac{2}{5}r - \frac{1}{10}s, r, s\right) = r\left(-\frac{3}{5}, \frac{2}{5}, 1, 0\right) + s\left(-\frac{3}{5}, -\frac{1}{10}, 0, 1\right)$$

Thus, the solution set is $\text{span}\left\{\left(-\frac{3}{5}, \frac{2}{5}, 1, 0\right), \left(-\frac{3}{5}, -\frac{1}{10}, 0, 1\right)\right\}$

## 10.3. Problem 3

### 10.3.1. (i)

For the four vectors to span all of $\mathbb{R}^4$, the RREF of the following matrix must be the identity matrix:

$$\begin{pmatrix} 0 & -a & 0 & 1 \\ 1 & 0 & a & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & a & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & -a & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & a & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & a & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The RREF of this matrix can never be the identity, as there is a row of zeros. Therefore, $\text{span}(S) \neq \mathbb{R}^4$

**10.3.2. (ii)**

Row-reducing again:

$$\left(\begin{array}{cccc|c} 0 & -a & 0 & 1 & x_1 \\ 1 & 0 & a & 0 & x_2 \\ 0 & 1 & 0 & 0 & x_3 \\ 0 & -1 & 0 & 1 & x_4 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & a & 0 & x_2 \\ 0 & 1 & 0 & 0 & x_3 \\ 0 & -a & 0 & 1 & x_1 \\ 0 & -1 & 0 & 1 & x_4 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & a & 0 & x_2 \\ 0 & 1 & 0 & 0 & x_3 \\ 0 & 0 & 0 & 1 & x_1 + ax_3 \\ 0 & 0 & 0 & 1 & x_4 + x_3 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & a & 0 & x_2 \\ 0 & 1 & 0 & 0 & x_3 \\ 0 & 0 & 0 & 1 & x_1 + ax_3 \\ 0 & 0 & 0 & 0 & x_4 - x_1 + (1-a)x_3 \end{array}\right)$$

Thus, for the vector to not be in the span, $x_4 - x_1 + (1-a)x_3 \neq 0$

The vector $(1, 0, 0, 0)$ is always not in the span.

## 10.4. Problem 4

### 10.4.1. (i)

We want to see if the vectors on the left are in the span of the vectors on the right. We can determine this by row-reducing the following augmented matrix:

$$\left(\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 2 & 1 & 1 & 3 & 0 & 0 & 3 & 1 \\ 1 & 4 & 1 & -1 & 2 & 2 & 6 & 2 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 0 & 7 & 1 & -5 & 4 & 2 & 3 & 3 \\ 0 & 7 & 1 & -5 & 4 & 3 & 6 & 3 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 0 & 7 & 1 & -5 & 4 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 0 & 7 & 1 & -5 & 4 & 2 & 3 & 3 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & 0 \end{array}\right)$$

We note that some of the vectors are not in the span, and thus the statement is false.

### 10.4.2. (ii)

This time, we just interchange the positions of the vectors.

$$\left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 6 & 2 & 1 & 4 & 1 & -1 \\ -4 & 5 & 9 & -1 & 0 & 0 & 3 & 6 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 7 & 9 & 1 & -2 & 6 & 3 & -2 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 7 & 9 & 1 & -2 & 6 & 3 & -2 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & -33 & -6 & -16 & -1 & -4 & -23 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & -33 & -6 & -16 & -1 & -4 & -23 \end{array}\right)$$

$$\left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & 0 & 5 & 6 & 10 & 7 & 10 \end{array}\right)$$

This statement is thus true.

#### 10.4.3. (iii)

From the calculation in (ii), this statement is true.

#### 10.4.4. (iv)

From the calculation in (i), this statement is false.

## 10.5. Problem 5

#### 10.5.1. (i)

False. $(0,0,1),(0,1,0),(0,1,1)$ is linearly dependent and thus does not form a spanning set of $\mathbb{R}^3$

#### 10.5.2. (ii)

False. Notice that $(1,0),(0,1)$ and $(1,1),(1,-1)$ both span $\mathbb{R}^2$, so the intersection of their spans is $\mathbb{R}^2$. But the span of their intersections is the span of the empty set, which is just the trivial subspace. This is thus false since the two expressions are not equal.

#### 10.5.3. (iii)

This is false because the RHS is not always a subspace (consider $S_1 = \{(1,0)\}$ and $S_2 = \{(0,1)\}$).

## 10.6. Problem 6

#### 10.6.1. (i)

We must verify the three axioms. Suppose $(a_1,b_1,c_1,d_1) \in V$ and $(a_2,b_2,c_2,d_2) \in V$. By definition, we have:

$$a_1 + b_1 = c_1 + d_1$$

$$a_2 + b_2 = c_2 + d_2$$

To verify the first axiom, add the two equations together to obtain:

$$a_1 + b_1 + a_2 + b_2 = c_1 + d_1 + c_2 + d_2$$

$$(a_1 + a_2) + (b_1 + b_2) = (c_1 + c_2) + (d_1 + d_2)$$

We have thus concluded:

$$(a_1,b_1,c_1,d_1) + (a_2,b_2,c_2,d_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2, d_1 + d_2) \in V$$

To verify the second axiom, multiply the first equation by $k$.

For any constant $k \in \mathbb{R}$, $(ka_1) + (kb_1) = (kc_1) + (kd_1)$. Thus, if $(a_1, b_1, c_1, d_1) \in V$, $k(a_1, b_1, c_1, d_1) \in V$.

Finally, $V$ is nonempty because $0 + 0 = 0 + 0$, thus $(0, 0, 0, 0) \in V$.

This shows that $V$ is a subspace of $\mathbb{R}^4$.

**10.6.2. (ii)**

$\text{span}(S) \subseteq V$ because all vectors in $S$ are in $V$.

To determine whether $V \subseteq \text{span}(S)$, we check that the following is always satisfiable given the constraint $a + b = c + d$:

$$\begin{pmatrix} 1 & 0 & 3 & a \\ 0 & -3 & 6 & b \\ 2 & 1 & 4 & c \\ -1 & -4 & 5 & d \end{pmatrix}$$

Reducing to row-echelon form:

$$\begin{pmatrix} 1 & 0 & 3 & a \\ 0 & -3 & 6 & b \\ 2 & 1 & 4 & c \\ -1 & -4 & 5 & d \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 3 & a \\ 0 & -3 & 6 & b \\ 0 & 1 & -2 & c - 2a \\ 0 & -4 & 8 & d + a \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 3 & a \\ 0 & 1 & -2 & c - 2a \\ 0 & 0 & 0 & b + 3c - 6a \\ 0 & 0 & 0 & d + a + 4c - 8a \end{pmatrix}$$

Note, however, that $(0, 1, 0, 1)$ is not in the span because $b + 3c - 6a = 1 + 3 \cdot 0 - 6 \cdot 0 = 1 \neq 0$ in this case, and thus, the matrix above is not satisfiable. Furthermore, $(0, 1, 0, 1) \in V$ by definition. Therefore, $V \not\subseteq \text{span}(S)$.

We also conclude that $\text{span}(S) \neq V$.

## 10.7. Problem 7

For this statement to be true, the vectors $\overrightarrow{w_1}$ and $\overrightarrow{w_2}$ must be in the span of the $\vec{v_i}$ s.

We can directly test this using standard matrix algorithms. Below, we have encoded the vectors $\vec{v_i}$ as column vectors to the left of the bar, while the vectors $\overrightarrow{w_i}$ are encoded as column vectors to the right of the bar.

$$\left(\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ -1 & 1 & a & 0 & 0 \\ 1 & 1 & -2 & 5 & -1 \end{array}\right)$$

Reducing this matrix using Gaussian Elimination:

$$\left(\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ -1 & 1 & a & 0 & 0 \\ 1 & 1 & -2 & 5 & -1 \end{array}\right)$$

$$\left(\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 2 & a+3 & 0 & 4 \\ 0 & 0 & -5 & 5 & -5 \end{array}\right)$$

$$\left(\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & a-1 & -2 & -2 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & a-1 & -2 & -2 \end{array}\right)$$

$$\left(\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & a-3 & -1-a \end{array}\right)$$

We note that this system is not satisfiable for any choice of $a$. Thus, this statement is never true.

## 10.8. Problem 8

### 10.8.1. (i)

Notice that we can build $\vec{v_i}$ in a telescoping fashion as follows:

$$\vec{v_i} = (\vec{v_i} - \overrightarrow{v_{i+1}}) + (\overrightarrow{v_{i+1}} - \overrightarrow{v_{i+2}}) + \cdots + \left(\overrightarrow{v_{m-1} - \overrightarrow{v_m}}\right) + \overrightarrow{v_m}$$

Thus, $\text{span}(\{\vec{v_1}, \vec{v_2}, ..., \overrightarrow{v_m}\}) \subseteq \text{span}\Big(\Big\{\vec{v_1} - \vec{v_2}, \vec{v_2} - \vec{v_3}, ..., \overrightarrow{v_{m-1} - \overrightarrow{v_m}}, \overrightarrow{v_m}\Big\}\Big)$

### 10.8.2. (ii)

We can build $\vec{v_i}$ by subtracting two of these prefix vector sums, as follows:

$$\vec{v_i} = \sum_{j=1}^{j=i} \vec{v_j} - \sum_{j=1}^{j=i-1} \vec{v_j}$$

Thus, $\text{span}(\{\vec{v_1}, \vec{v_2}, ..., \overrightarrow{v_m}\}) \subseteq \text{span}\left(\left\{\sum_{j=1}^{j=1} \vec{v_j}, \sum_{j=1}^{j=2} \vec{v_j}, ..., \sum_{j=m}^{j=1} \vec{v_j}\right\}\right)$

## 10.9. Problem 9

### 10.9.1. (a)

We want to determine when the following equations:

$$c_1(1, 2, 2, 3) + c_2(0, 3, 1, -1) = (1, -1, 1, 0)$$

$$c_1(1,2,2,3) + c_2(0,3,1,-1) = (1,-1,1,a)$$

Are satisfiable.

This can be captured through the system described by the following augmented matrix:

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 2 & 3 & -1 & 1 \\ 2 & 1 & 1 & -1 \\ 3 & -1 & 0 & a \end{array}\right)$$

Applying Gaussian Elimination to solve this system:

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 2 & 3 & -1 & 1 \\ 2 & 1 & 1 & -1 \\ 3 & -1 & 0 & a \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 3 & -3 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & -3 & a+3 \end{array}\right)$$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & a+4 \end{array}\right)$$

The only value for which one of the vectors is in the span is when $a = -4$.

**10.9.2. (b)**

To find $\text{span}(S_1)$ and $\text{span}(S_2)$, we can use matrix algorithms. For instance, the vector $(x_1, x_2, x_3, x_4)$ is in $\text{span}(S_1)$ when the following system is satisfiable:

$$\left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 2 & 3 & x_2 \\ 2 & 1 & x_3 \\ 3 & -1 & x_4 \end{array}\right)$$

We can also check when $(x_1, x_2, x_3, x_4) \in \text{span}(S_2)$ using the following augmented matrix: (in the actual exam, I would probably explicitly write out the equation corresponding to this matrix)

$$\left(\begin{array}{cc|c} 1 & -1 & x_1 \\ -1 & 1 & x_2 \\ 1 & -1 & x_3 \\ 0 & 2 & x_4 \end{array}\right)$$

Let us consider the first matrix and reduce it to row-echelon form:

$$\left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 0 & 3 & x_2 - 2x_1 \\ 0 & 1 & x_3 - 2x_1 \\ 0 & -1 & x_4 - 3x_1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 0 & 1 & x_3 - 2x_1 \\ 0 & 0 & x_2 - 2x_1 - 3x_3 + 6x_1 \\ 0 & 0 & x_4 - 3x_1 + x_3 - 2x_1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & x_1 \\ 0 & 1 & x_3 - 2x_1 \\ 0 & 0 & 4x_1 + x_2 - 3x_3 \\ 0 & 0 & -5x_1 + x_3 + x_4 \end{array}\right)$$

And let us do the same for the second matrix:

$$\left(\begin{array}{cc|c} 1 & -1 & x_1 \\ -1 & 1 & x_2 \\ 1 & -1 & x_3 \\ 0 & a & x_4 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & -1 & x_1 \\ 0 & 0 & x_2 + x_1 \\ 0 & 0 & x_3 - x_1 \\ 0 & a & x_4 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & -1 & x_1 \\ 0 & 2 & x_4 \\ 0 & 0 & x_2 + x_1 \\ 0 & 0 & x_3 - x_1 \end{array}\right)$$

For a vector to be in the intersection of the spans, both augmented matrices must be satisfiable. This leads us to write the following system of four linear equations:

$$4x_1 + x_2 - 3x_3 = 0$$

$$-5x_1 + x_3 + x_4 = 0$$

$$x_1 + x_2 = 0$$

$$-x_1 + x_3 = 0$$

Let us find a parameterization for this system of equations. We write the following augmented matrix corresponding to this system:

$$\left(\begin{array}{cccc|c} 4 & 1 & -3 & 0 & 0 \\ -5 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 4 & 1 & -3 & 0 & 0 \\ -5 & 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & -3 & -3 & 0 & 0 \\ 0 & 5 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & -3 & -3 & 0 & 0 \\ 0 & 5 & 1 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -4 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 1 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 1 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Thus, the following is a parameterization of the solution set of this system:

$$s(1, -1, 1, 4)$$

Therefore, $S_3 = \{(1, -1, 1, 4)\}$ works.

## 10.10. Problem 10

### 10.10.1. (i)

$V$ is not a subspace since it is not a homogenous system. Another easy way to see this is that the zero vector does not satisfy this system.

$W$ is a subspace because the system is homogenous.

$V \cap W$ is not a subspace because the zero vector is not in the set.

$V \cup W$ is not a subspace. First, notice that the vector $(1, 0, 0, 0, -1) \in V$ (it satisfies both equations). Thus, $(1, 0, 0, 0, -1) \in V \cup W$.

Furthermore, $(0, 1, 0, 1, 0) \in W$. Thus $(0, 1, 0, 1, 0) \in V \cup W$.

Now consider $(1, 0, 0, 0, -1) + (0, 1, 0, 1, 0) = (1, 1, 0, 1, -1)$. If $V \cup W$ is a subspace, it is necessary that this vector is in the set.

However:

$$2x_1 + x_2 - x_4 - 2x_5 = 2 \cdot 1 + 1 - 1 - 2 \cdot (-1) = 4 \neq 0$$

and

$$x_1 + x_2 - x_3 - x_4 - x_5 = 1 + 1 - 0 - 1 - (-1) = 2 \neq 0$$

.

Thus, $(1, 1, 0, 1, -1) \notin V \cup W$. Therefore, $V \cup W$ is not a subspace.

### 10.10.2. (ii)

A parameterization for $W$ is:

$$(r, -2r + s + 2t, s, t) = r(1, -2, 0, 0) + s(0, 1, 1, 0) + t(0, 2, 0, 1)$$

Therefore, $\{(1, -2, 0, 0), (0, 1, 1, 0), (0, 2, 0, 1)\}$ is a spanning set for $W$.

### 10.10.3. (iii)

We want to find a parameterization for the following system of linear equations:

$$x_1 - 2x_2 + 2x_3 - x_4 + x_5 = 0$$

$$x_1 + x_2 - x_3 - x_4 - x_5 = 2$$

$$2x_1 + x_2 - x_4 - 2x_5 = 0$$

This corresponds to the following augmented matrix:

$$\left(\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 1 & 1 & -1 & -1 & -1 & 2 \\ 2 & 1 & 0 & -1 & -2 & 0 \end{array}\right)$$

Applying Gaussian Elimination to this matrix:

$$\left(\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 1 & 1 & -1 & -1 & -1 & 2 \\ 2 & 1 & 0 & -1 & -2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 0 & 3 & -3 & 0 & -2 & 2 \\ 0 & 5 & -4 & 1 & -4 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 0 & 15 & -15 & 0 & -10 & 10 \\ 0 & 15 & -12 & 3 & -12 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 0 & 15 & -12 & 3 & -12 & 0 \\ 0 & 15 & -15 & 0 & -10 & 10 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 0 & 15 & -12 & 3 & -12 & 0 \\ 0 & 0 & -3 & -3 & 2 & 10 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 15 & -30 & 30 & -15 & 15 & 0 \\ 0 & 15 & -12 & 3 & -12 & 0 \\ 0 & 0 & -3 & -3 & 2 & 10 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 15 & -30 & 0 & -45 & 35 & 100 \\ 0 & 15 & 0 & 15 & -20 & -40 \\ 0 & 0 & -3 & -3 & 2 & 10 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 3 & -6 & 0 & -9 & 7 & 20 \\ 0 & 3 & 0 & 3 & -4 & -8 \\ 0 & 0 & -3 & -3 & 2 & 10 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 3 & 0 & 0 & -3 & -1 & 4 \\ 0 & 3 & 0 & 3 & -4 & -8 \\ 0 & 0 & -3 & -3 & 2 & 10 \end{array}\right)$$

The following is a parameterization for the solution set of this system:

$$\left(\frac{1}{3}(4+3r+s), \frac{1}{3}(-8-3r+4s), -\frac{1}{3}(10+3r-2s), r, s\right)$$

$$= r(1,-1,-1,1,0) + s\left(\frac{1}{3}, \frac{4}{3}, \frac{2}{3}, 0, 1\right) + \left(\frac{4}{3}, -\frac{8}{3}, -\frac{10}{3}, 0, 0\right)$$

Thus, $S = \left\{(1,-1,-1,1,0), \left(\frac{1}{3}, \frac{4}{3}, \frac{2}{3}, 0, 1\right), \left(\frac{4}{3}, -\frac{8}{3}, -\frac{10}{3}, 0, 0\right)\right\}$. Also, clearly $\text{span}(S) \neq \mathbb{R}^5$ since there are only three vectors in $S$.

# 11. Problem Set 10

## 11.1. Problem 1:

The vectors are linearly independent iff the only way to build the zero vector from these vectors is the trivial solution.

Setting up the augmented matrix, we get:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 3 & a & 1 & 0 \\ 0 & 0 & -a & 0 \\ -2 & -2 & 2a & 0 \end{array}\right)$$

Performing Gaussian Elimination:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & a-3 & 1 & 0 \\ 0 & 0 & -a & 0 \\ 0 & 0 & 2a & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & a-3 & 1 & 0 \\ 0 & 0 & -a & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

For there to be one unique solution, there must be no free parameters. Thus, the collection of vectors is linearly independent iff $a \neq 0, 3$.

## 11.2. Problem 2

### 11.2.1. (i)

Again, we wish to determine the values for which the augmented matrix with zeros to the right of the bar only has the trivial solution.

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ -2 & -3 & 1 & 0 \\ 0 & a & a^2 & 0 \end{array}\right)$$

Performing Gaussian Elimination:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & a & a^2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & a^2+a & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & a^2+a & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

For the vectors to be linearly independent, there must be no free parameters. Therefore, $a^2 + a \neq 0$, implying that the vectors are linearly independent if and only if $a \neq 0, -1$.

**11.2.2. (ii)**

For the collection of vectors to span the entire space, there must be no rows of zero in the row-echelon form of the following matrix:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -2 \\ a^2 & a & 0 \end{pmatrix}$$

The number of zero rows is $4 - \text{num_leading_entries} \geq 4 - \text{num columns} = 1$. Thus, there is at least one zero row, implying that the list of vectors does not span $\mathbb{R}^4$.

## 11.3. Problem 3

It suffices to express the zero vector as a nontrivial linear combination of the four vectors, where each vector has a nonzero coefficient. This will allow us to use algebra to express each vector in terms of the other three.

Specifically, we wish to solve for the following system:

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 & 0 \\ -1 & 3 & 0 & 1 & 0 \end{array}\right)$$

Applying Gauss-Jordan Elimination:

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 & 0 \\ 0 & 2 & 0 & 2 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 & 0 \\ 0 & 0 & -3 & 3 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 & 0 \\ 0 & 0 & -3 & 3 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right)$$

Supposing that the coefficients for the four vectors are $c_1, c_2, c_3, c_4$, respectively, we have found that:

$$c_1 = -2c_4$$

$$c_2 = -c_4$$

$$c_3 = c_4$$

These four relations give us the following equation:

$$-2c_4\begin{pmatrix}1\\0\\-1\end{pmatrix} - c_4\begin{pmatrix}-1\\2\\3\end{pmatrix} + c_4\begin{pmatrix}0\\3\\0\end{pmatrix} + c_4\begin{pmatrix}1\\-1\\1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

$$-2\begin{pmatrix}1\\0\\-1\end{pmatrix} - \begin{pmatrix}-1\\2\\3\end{pmatrix} + \begin{pmatrix}0\\3\\0\end{pmatrix} + \begin{pmatrix}1\\-1\\1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

Thus, we have the following four equations:

$$\begin{pmatrix}1\\0\\-1\end{pmatrix} = -\frac{1}{2}\begin{pmatrix}-1\\2\\3\end{pmatrix} + \frac{1}{2}\begin{pmatrix}0\\3\\0\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1\\-1\\1\end{pmatrix}$$

$$\begin{pmatrix}-1\\2\\3\end{pmatrix} = -2\begin{pmatrix}1\\0\\-1\end{pmatrix} + \begin{pmatrix}0\\3\\0\end{pmatrix} + \begin{pmatrix}1\\-1\\1\end{pmatrix}$$

$$\begin{pmatrix}0\\3\\0\end{pmatrix} = 2\begin{pmatrix}1\\0\\-1\end{pmatrix} + \begin{pmatrix}-1\\2\\3\end{pmatrix} - \begin{pmatrix}1\\-1\\1\end{pmatrix}$$

$$\begin{pmatrix}1\\-1\\1\end{pmatrix} = 2\begin{pmatrix}1\\0\\-1\end{pmatrix} + \begin{pmatrix}-1\\2\\3\end{pmatrix} - \begin{pmatrix}0\\3\\0\end{pmatrix}$$

## 11.4. Problem 4

### 11.4.1. (i)

$T'$ is linearly dependent since adding all three vectors (which is a nontrivial linear combination) results in the zero vector.

### 11.4.2. (ii)

This time, I claim that the list of vectors is linearly independent.

To prove this, we want to show that the only way to express the zero vector as a linear combination of the vectors in $T''$ is the trivial combination.

Suppose $c_1, c_2, c_3$ are real numbers satisfying:

$$c_1(\vec{v_1} - \vec{v_2}) + c_2(\vec{v_2} - \vec{v_3}) + c_3(\vec{v_1} + \vec{v_3}) = \vec{0}$$

$$(c_1 + c_3)\vec{v_1} + (c_2 - c_1)\vec{v_2} + (c_3 - c_2)\vec{v_3} = \vec{0}$$

Since $\vec{v_1}, \vec{v_2}, \vec{v_3}$ are linearly independent, we know that the only solution to the above equation is the trivial solution. Thus, we have:

$$c_1 + c_3 = c_2 - c_1 = c_3 - c_2 = 0$$

Which corresponds to the following augmented matrix:

$$\left(\begin{array}{ccc|c}1 & 0 & 1 & 0\\-1 & 1 & 0 & 0\\0 & -1 & 1 & 0\end{array}\right)$$

Notice that the system is homogenous.

Applying Gauss-Jordan Elimination, we get:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

Thus, the reduced-row echelon form of the matrix is the identity matrix. By the characterizations of invertibility, we thus know that the only solution is the trivial solution. Thus, $c_1 = c_2 = c_3$ is the only solution, implying that the original collection of vectors is linearly independent.

## 11.5. Problem 5

Since the list is linearly dependent, there is a set of real coefficients $c_1, c_2, c_3, ..., c_m, c_{m+1}$ satisfying:

1. $c_{m+1}\vec{w} + \sum_{i=1}^{m} c_i\vec{v_i} = \vec{0}$
2. $\exists i \in \mathbb{Z}, 1 \leq i \leq m+1 \wedge c_i \neq 0$

First, suppose, for contradiction, that $c_{m+1} = 0$. Since $\exists i \in \mathbb{Z}, 1 \leq i \leq m+1 \wedge c_i \neq 0$, we can conclude that $\exists i \in \mathbb{Z}, 1 \leq i \leq m \wedge c_i \neq 0$ satisfying:

$$c_{m+1}\vec{w} + \sum_{i=1}^{m} c_i\vec{v_i} = \vec{0}$$

Since $c_{m+1} = 0$, we have:

$$\sum_{i=1}^{m} c_i\vec{v_i} = \vec{0}$$

We have thus expressed the zero vector as a linear combination of the $\vec{v_i}$ s in a nontrivial way, implying that the $\vec{v_i}$ s are linearly dependent, contradiction.

Therefore, $c_{m+1} \neq 0$.

Rearranging the following equation:

$$c_{m+1}\vec{w} + \sum_{i=1}^{m} c_i\vec{v_i} = \vec{0}$$

$$\vec{w} = \sum_{i=1}^{m} -\left(\frac{c_i}{c_{m+1}}\right)\vec{v_i}$$

Since $c_{m+1} \neq 0$, $-\dfrac{c_i}{c_{m+1}} \in \mathbb{R}$. Thus, $\vec{w}$ is a linear combination of the $\vec{v_i}$ s, implying that:

$$\vec{w} \in \text{span}(\vec{v_1}, ..., \overrightarrow{v_m})$$

As needed. ■

## 11.6. (a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

## 11.7. (b)

Let us first show the forward direction.

Suppose that the matrix $A$ has determinant zero. Then, the matrix $A^T$ also has determinant zero and is thus not invertible. Then, by the characterizations of invertibility, a nonzero vector satisfies the following matrix equation:

$$A^T\vec{x} = \vec{0}$$

However, this means that the column vectors of $A^T$ are linearly dependent. We note that the column vectors of $A^T$ are just the row vectors of $A$, and thus the row vectors of $A$ are linearly dependent, as needed.

Next, let us show the backward direction.

Suppose that the row vectors of $A$ are linearly dependent. Therefore, the column vectors of $A^T$ are linearly dependent (since they are the same as the row vectors of $A$). This implies that the following matrix equation:

$$A^T\vec{x} = \vec{0}$$

Has a nontrivial solution $\vec{x} \neq \vec{0}$ (since we can express the zero vector as a nontrivial linear combination of the columns). By the characterizations of invertibility, this means that $A^T$ is not invertible and thus $\det(A) = \det(A^T) = 0$, as needed. ■

## 11.8. Problem 7

### 11.8.1. (a)

Again we want to check if we can express the zero vector as a nontrivial linear combnination of this list of vectors.

Let us construct the augmented matrix with the vectors as the columns of this matrix.

$$\left(\begin{array}{ccc|c} 1 & 7 & -2 & 0 \\ 4 & 10 & 1 & 0 \\ 2 & -4 & 5 & 0 \\ -3 & -1 & -4 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 7 & -2 & 0 \\ 0 & -18 & 9 & 0 \\ 0 & -18 & 9 & 0 \\ 0 & 20 & -10 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 7 & -2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Since the row-echelon form contains a free parameter, we can express the zero vector as a nontrivial linear combination of the three vectors on the list. Thus, the three vectors are linearly dependent.

**11.8.2. (b)**

To show that the vectors are linearly independent, we want to show that the following vector equation:

$$p\vec{x} + q\vec{y} = \vec{0}$$

Only has the trivial solution.

Expanding it out:

$$p(a\vec{u} + b\vec{v}) + q(c\vec{u} + d\vec{v}) = \vec{0}$$

$$(pa + qc)\vec{u} + (pb + qd)\vec{v} = \vec{0}$$

Since $\vec{u}$ and $\vec{v}$ are linearly independent, we have:

$$pa + qc = 0$$

$$pb + qd = 0$$

Which corresponds to the following matrix equation:

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} = \vec{0}$$

Since

$$\det\left(\begin{pmatrix} a & c \\ b & d \end{pmatrix}\right) = ad - bc \neq 0$$

$\begin{pmatrix} a & c \\ b & d \end{pmatrix}$ is invertible. Thus, by the characterizations of invertibility, the only solution to $\begin{pmatrix} a & c \\ b & d \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} = \vec{0}$ is $p = q = 0$.

Thus, the only solution to

$$p\vec{x} + q\vec{y} = \vec{0}$$

is the trivial solution, thereby proving that $\vec{x}$ and $\vec{y}$ are linearly independent, as needed. ■

## 11.9. Problem 8

### 11.9.1. (a)

$$\begin{aligned}\text{adj}(A) &= \begin{pmatrix} \begin{vmatrix} 0 & -2 \\ -4 & 1 \end{vmatrix} & -\begin{vmatrix} -1 & -2 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} -1 & 0 \\ 3 & -4 \end{vmatrix} \\ -\begin{vmatrix} 2 & -5 \\ -4 & 1 \end{vmatrix} & \begin{vmatrix} -3 & -5 \\ 3 & 1 \end{vmatrix} & -\begin{vmatrix} -3 & 2 \\ 3 & -4 \end{vmatrix} \\ \begin{vmatrix} 2 & -5 \\ 0 & -2 \end{vmatrix} & -\begin{vmatrix} -3 & -5 \\ -1 & -2 \end{vmatrix} & \begin{vmatrix} -3 & 2 \\ -1 & 0 \end{vmatrix} \end{pmatrix}^T \\ &= \begin{pmatrix} -8 & -5 & 4 \\ 18 & 12 & -6 \\ -4 & -1 & 2 \end{pmatrix}^T \\ &= \begin{pmatrix} -8 & 18 & -4 \\ -5 & 12 & -1 \\ 4 & -6 & 2 \end{pmatrix}\end{aligned}$$

### 11.9.2. (b): Mild Challenge

Recall that $\text{adj}(M)M = \det(M)I$, where $I$ is the identity matrix.

I claim that $\det(\text{adj}(M)) = 0$. Suppose not. Then, $\text{adj}(M)$ must be invertible since its determinant is nonzero.

Multiplying both sides of the matrix equation above by the inverse of the adjoint $\text{adj}(M)^{-1}$:

$$\text{adj}(M)^{-1}\ \text{adj}(M)M = \text{adj}(M)^{-1}\det(M)I$$

$$M = \text{adj}(M)^{-1} \cdot 0 \cdot I$$

$$M = \mathbf{0}$$

However, $M$ is nonzero, contradiction. Therefore, $\det(\text{adj}(M)) = 0$.

Further, $\det(\text{adj}(M)^T) = \det(\text{adj}(M)) = 0$. Thus, $\text{adj}(M)^T$ is not invertible. By the characterizations of invertibility, this implies that the following matrix-vector equation:

$$\text{adj}(M)^T\vec{x} = \vec{0}$$

Has a nontrivial solution. Equivalently, this means that there is a nontrivial way to build the zero vector as a linear combination of the column vectors of $\text{adj}(M)^T$. Thus, the column vectors of $\text{adj}(M)^T$ are linearly dependent. Therefore, the row vectors of $\text{adj}(M)$ are linearly dependent, as needed. ■

## 11.10. Problem 9

An equivalent way of saying that the associated vector $\vec{v}$ makes the resulting list linearly dependent upon insertion into the list is to say that $\vec{v}$ is in the span of the three vectors.

The vector is in the span of the three other vectors if the following augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 1 & 0 & 1 & b \\ 1 & 1 & 0 & c \\ 1 & 0 & -1 & d \\ 1 & 1 & 1 & e \end{array}\right)$$

Performing Gaussian Elimination:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & 0 & -2 & c-a \\ 0 & -1 & -3 & d-a \\ 0 & 0 & -1 & e-a \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & 0 & -2 & c-a \\ 0 & 0 & -2 & d-b \\ 0 & 0 & -1 & e-a \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & 0 & -1 & e-a \\ 0 & 0 & 0 & c+a-2e \\ 0 & 0 & 0 & d-b+2a-2e \end{array}\right)$$

Thus, $\vec{v}$ is on the span if and only if:

$$c+a-2e=0$$

$$d-b+2a-2e=0$$

both hold.

Thus, if at least one of these equations did not hold, then $\vec{v}$ would not be on the span and thus including it in the list would result in a linearly independent basis.

## 11.11. Problem 10

We want to show that the following matrix equation (with the columns organized into blocks):

$$(\vec{v_1}-\vec{v_2}\ \ \vec{v_2}-\vec{v_3}\ \ ...\ \ \overrightarrow{v_{n-1}}-\vec{v_n}\ \ \vec{v_n})\vec{x}=\vec{0}$$

Only has the trivial solution, as this implies that the only way to build the zero vector as a linear combination of these vectors is the trivial solution.

Notice that:

$$(\vec{v_1}-\vec{v_2}\ \ \vec{v_2}-\vec{v_3}\ \ ...\ \ \overrightarrow{v_{n-1}}-\vec{v_n}\ \ \vec{v_n}) = (\vec{v_1}\ \ \vec{v_2}\ \ \vec{v_3}\ \ ...\ \ \vec{v_n})\begin{pmatrix} 1 & 0 & 0 & 0 & ... & 0 \\ -1 & 1 & 0 & 0 & ... & 0 \\ 0 & -1 & 1 & 0 & ... & 0 \\ 0 & 0 & -1 & 1 & ... & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

Since $\begin{pmatrix} 1 & 0 & 0 & 0 & ... & 0 \\ -1 & 1 & 0 & 0 & ... & 0 \\ 0 & -1 & 1 & 0 & ... & 0 \\ 0 & 0 & -1 & 1 & ... & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$ is lower triangular, $\begin{vmatrix} 1 & 0 & 0 & 0 & ... & 0 \\ -1 & 1 & 0 & 0 & ... & 0 \\ 0 & -1 & 1 & 0 & ... & 0 \\ 0 & 0 & -1 & 1 & ... & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix} = 1$ (the product of the diagonal entries).

Furthermore, since $\vec{v_1}, \vec{v_2}, \vec{v_3}, ..., \vec{v_n}$ are linearly independent, the only way to build the zero vector as a linear combination of these vectors is the trivial solution. Equivalently, this means that the following matrix equation:

$$(\vec{v_1} \ \ \vec{v_2} \ \ \vec{v_3} \ \ ... \ \ \vec{v_n})\vec{x} = \vec{0}$$

Only has the zero vector as the solution. By the characterizations of invertibility, this means that $(\vec{v_1} \ \ \vec{v_2} \ \ \vec{v_3} \ \ ... \ \ \vec{v_n})$ is invertible. Thus, $\left|\vec{v_1} \ \ \vec{v_2} \ \ \vec{v_3} \ \ ... \ \ \vec{v_n}\right| \neq 0$.

By the zero-product property,

$$\left|\vec{v_1} - \vec{v_2} \ \ \vec{v_2} - \vec{v_3} \ \ ... \ \ \overrightarrow{v_{n-1}} - \vec{v_n} \ \ \vec{v_n}\right| = \left|\vec{v_1} \ \ \vec{v_2} \ \ \vec{v_3} \ \ ... \ \ \vec{v_n}\right| \begin{vmatrix} 1 & 0 & 0 & 0 & ... & 0 \\ -1 & 1 & 0 & 0 & ... & 0 \\ 0 & -1 & 1 & 0 & ... & 0 \\ 0 & 0 & -1 & 1 & ... & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix} \neq 0$$

Therefore, $\left|\vec{v_1} - \vec{v_2} \ \ \vec{v_2} - \vec{v_3} \ \ ... \ \ \overrightarrow{v_{n-1}} - \vec{v_n} \ \ \vec{v_n}\right|$ is invertible. By the characterizations of invertibility, the only solution to the following matrix equation:

$$(\vec{v_1} - \vec{v_2} \ \ \vec{v_2} - \vec{v_3} \ \ ... \ \ \overrightarrow{v_{n-1}} - \vec{v_n} \ \ \vec{v_n})\vec{x} = \vec{0}$$

Is the zero vector. This implies that the only way to build the zero vector as a linear combination of the column vectors is the trivial solution, implying that the column vectors — and thus the original list of vectors — is linearly independent, as needed. ■

## 11.12. Problem 11

### 11.12.1. (1)

Let us first show the forward direction.

Suppose that the vectors $\vec{v_1}, ..., \vec{v_p}$ are linearly independent. Thus, the only way to build the zero vector as a linear combination of these vectors is the trivial solution. This implies that the following matrix equation where the vectors are the columns vectors of the matrix:

$$\left(\vec{v_1} \ \ ... \ \ \vec{v_p}\right)\vec{x} = \vec{0}$$

only has the zero vector as its solution. By the characterizations of invertibility, this implies that $\left(\vec{v_1} \ \ ... \ \ \vec{v_p}\right)$ is invertible.

Therefore, $\left(P\vec{v_1} \ \ ... \ \ P\vec{v_p}\right) = P\left(\vec{v_1} \ \ ... \ \ \vec{v_p}\right)$ is also invertible.

By the characterizations of invertibility, the only solution to the following equation is the zero vector:

$$\left(P\vec{v_1} \ \ ... \ \ P\vec{v_p}\right)\vec{x} = \vec{0}$$

Therefore, the only way to build the zero vector as a linear combination of $P\vec{v_1}, ..., P\vec{v_p}$ is the trivial solution, implying that $P\vec{v_1}, ..., P\vec{v_p}$ is linearly independent.

To show the backward direction, assume $P\vec{v_1}, ..., P\vec{v_p}$ is linearly independent. Notice that $P^{-1}$ is also invertible.

By our work already done, $P^{-1}P\vec{v_1}, ..., P^{-1}P\vec{v_p}$ is also linearly independent. Simplifying, this means that $\vec{v_1}, ..., \vec{v_p}$ is linearly independent, as needed. ■

**11.12.2. (2)**

Let us first show the forward direction.

Suppose that $\vec{v_1}, ..., \vec{v_p}$ span $V$. Let $\vec{w} \in W$ be an arbitrary vector. By definition, there exists $\vec{v} \in V$ such that $\vec{w} = P\vec{v}$.

Since $\vec{v_1}, ..., \vec{v_p}$ span $V$ and $\vec{v} \in V$, there exist real $x_1, ..., x_p$ such that:

$$\vec{v} = \sum_{i=1}^{p} x_i \vec{v_i}$$

Thus:

$$\vec{w} = P\vec{v} = P\sum_{i=1}^{p} x_i \vec{v_i} = \sum_{i=1}^{p} x_i (P\vec{v_i})$$

Thus, $\vec{w}$ is a linear combination of $P\vec{v_1}, ..., P\vec{v_p}$.

Therefore, $P\vec{v_1}, ..., P\vec{v_p}$ span $W$.

Now let us show the backward direction.

Suppose $P\vec{v_1}, ..., P\vec{v_p}$ span $W$.

Let $\vec{v} \in V$ be some particular but arbitrarily-chosen element of $V$. By definition, $P\vec{v} \in W$. Thus, there exist real $x_1, ..., x_p$ such that:

$$P\vec{v} = \sum_{i=1}^{p} x_i (P\vec{v_i})$$

Therefore:

$$P\vec{v} = \sum_{i=1}^{p} x_i (P\vec{v_i}) = P\sum_{i=1}^{p} x_i \vec{v_i}$$

Since $P$ is invertible:

$$P^{-1}P\vec{v} = P^{-1}P\sum_{i=1}^{p} x_i \vec{v_i}$$

$$\vec{v} = \sum_{i=1}^{p} x_i \vec{v_i}$$

Thus, $\vec{v}$ is a linear combination of $\vec{v_1}, \vec{v_2}, \vec{v_3}, ..., \vec{v_p}$.

Therefore, $\vec{v_1}, \vec{v_2}, \vec{v_3}, ..., \vec{v_p}$ spans $V$, as needed. ■

# 12. Problem Set 11

## 12.1. Problem 1

For these vectors to form a basis, all vectors in $\mathbb{R}^4$ must be in the span of the vectors and the vectors must be linearly independent.

Therefore, the matrix consisting of these vectors as its columns must have the identity matrix as its RREF.

Applying Gauss-Jordan Elimination:

$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 5 & 0 \\ -1 & 3 & 2 & -4 \\ 2 & 1 & 0 & 8 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 5 & 0 \\ 0 & 5 & 2 & 0 \\ 0 & -3 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 5 & 0 \\ 0 & 5 & 2 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 5 & 2 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Clearly, the reduced row echelon form is not the identity matrix, therefore the vectors do not form a basis for $\mathbb{R}^4$.

## 12.2. Problem 2

We have:

$$\sum_{i=1}^{n} c_i \vec{v_i} = \sum_{i=1}^{n} d_i \vec{v_i}$$

$$\sum_{i=1}^{n} (c_i - d_i) \vec{v_i} = \vec{0}$$

Since the $\vec{v_i}$ s form a basis, the only solution to the equation above is if all of the coefficients were zero. Tehrefore, $c_i = d_i$, as needed ■.

## 12.3. Problem 3

(1) Is not a basis since its dimension is less than 3.

(2) Is a basis. We can check this by performing Gauss-Jordsan Elimination on the matrix containing these three vectors as its columns and checking that the result is the identity matrix.

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ -1 & 3 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ 0 & 2 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Thus, the three vectors are linearly independent.

To obtain the coordinate vector of $(1, 2, 3)$ with respect to the basis, we can do:

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ -1 & 3 & 0 & 3 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 2 & 0 & 4 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 2 & 0 & 4 \\ 0 & 2 & 3 & 2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & 3 & -2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -\frac{2}{3} \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -\frac{2}{3} \end{array}\right)$$

Thus, $\left(3, 2, -\frac{2}{3}\right)$ is the coordinate vector for $(1, 2, 3)$ with respect to the basis.

(3) Is not a basis because the entries are not linearly dependent. Specifically, adding the first and second vectors, then multiplying the result by $\frac{2}{3}$ results in the third vector.

(4) Is not a basis because there are four vectors, which is one more than the dimension of $\mathbb{R}^3$. Thus, at least one of the vectors can be expressed as a linear combination of the other ones, implying that this list of vectors is not linearly dependent and is thus not a basis.

## 12.4. Problem 4

Writing all these vectors as the columns of a matrix:

$$\begin{pmatrix} 1 & 4 & 0 & 5 & 2 \\ 2 & 1 & 0 & 3 & 0 \\ -1 & 3 & 0 & 2 & 2 \\ 1 & -1 & 1 & -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 & 5 & 2 \\ 0 & -7 & 0 & -7 & -4 \\ 0 & 7 & 0 & 7 & 4 \\ 0 & -5 & 1 & -6 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 & 5 & 2 \\ 0 & -7 & 0 & -7 & -4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -5 & 1 & -6 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 & 5 & 2 \\ 0 & 35 & 0 & 35 & 20 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -35 & 7 & -42 & -7 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 & 5 & 2 \\ 0 & 35 & 0 & 35 & 20 \\ 0 & -35 & 7 & -42 & -7 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 & 5 & 2 \\ 0 & 35 & 0 & 35 & 20 \\ 0 & 0 & 7 & -7 & 13 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Since there are three leading entries, the dimension of $W$ is 3, formed by the vectors on the columns with leading entries, namely:

$$(1, 2, -1, 1), (4, 1, 3, -1), (0, 0, 0, 1)$$

## 12.5. Problem 5

The rank of a matrix is equal to the number of leading entries in a row-echelon form. (Note: also the number of nonzero rows)

Reducing $A$, we get:

$$\begin{pmatrix} 1 & 43 & -37 \\ 0 & -12 & 12 \\ 0 & -12 & 12 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 43 & -37 \\ 0 & -12 & 12 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus, $\text{rank}(A) = 2$.

Further, reducing $B$, we get:

$$\begin{pmatrix} 1 & 7 & -1 \\ 2 & 2 & 10 \\ 3 & 9 & 9 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 7 & -1 \\ 1 & 1 & 5 \\ 1 & 3 & 3 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 7 & -1 \\ 1 & 1 & 5 \\ 1 & 3 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 6 & -2 \\ 0 & 0 & 4 \\ 0 & 2 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 6 & -2 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 6 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 6 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 6 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 6 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus, $\text{rank}(B) = 3$.

## 12.6. Problem 6

### 12.6.1. (i)

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$$

### 12.6.2. (ii)

No such matrix exists because the dimensions of the column and row spaces must match.

## 12.7. Problem 7

The row space can be obtained by picking the nonzero rows of the row echelon form.

The column space can be obtained by picking the column vectors corresponding to the leading entries of the row echelon form.

Finally, the null space can be obtained by solving for when the weighted sum of the column vectors equals zero. Specifically, it can be obtained by considering the augmented matrix $\left(\boldsymbol{A} \middle| \vec{0}\right)$ and then finding a parameterization for the set of solutions.

In all three cases, we want to compute the row-echelon form of the matrix.

### 12.7.1. (i)

Computing the reduced row-echelon forms of the matrices and finding a parameterization for the solution set of $M\vec{x} = \vec{0}$.

Matrix $A$:

$$\begin{pmatrix} 7 & -2 & -1 \\ 4 & 1 & -3 \\ 1 & -1 & 0 \\ 8 & -3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 7 & -2 & -1 \\ 4 & 1 & -3 \\ 8 & -3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 5 & -1 \\ 0 & 5 & -3 \\ 0 & 5 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 5 & -1 \\ 0 & 0 & -2 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 5 & -1 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The parameterization is (we omit the set-builder notation for brevity, note that it is implicitly there):

$$(0, 0, 0)$$

Matrix $B$:

$$\begin{pmatrix} 8 & 8 & 8 & 38 \\ -8 & 8 & 8 & 38 \\ 8 & -8 & 8 & 38 \\ 8 & 8 & -8 & -38 \end{pmatrix}$$

$$\begin{pmatrix} 8 & 8 & 8 & 38 \\ 0 & 16 & 16 & 76 \\ 0 & -16 & 0 & 0 \\ 0 & 0 & -16 & -76 \end{pmatrix}$$

$$\begin{pmatrix} 8 & 8 & 8 & 38 \\ 0 & 0 & 0 & 0 \\ 0 & -16 & 0 & 0 \\ 0 & 0 & -16 & -76 \end{pmatrix}$$

$$\begin{pmatrix} 8 & 8 & 8 & 38 \\ 0 & -16 & 0 & 0 \\ 0 & 0 & -16 & -76 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 8 & 0 & 0 & 0 \\ 0 & -16 & 0 & 0 \\ 0 & 0 & -16 & -76 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{19}{4} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The parameterization is:

$$\left(0, 0, -\frac{19}{4}r, r\right) = r\left(0, 0, -\frac{19}{4}, 1\right)$$

Matrix $C$:

$$\begin{pmatrix} 1 & 2 & 0 & -3 & -1 \\ 2 & 4 & 1 & -8 & 3 \\ -1 & -2 & -1 & 5 & -4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & -3 & -1 \\ 0 & 0 & 1 & -2 & 5 \\ 0 & 0 & -1 & 2 & -5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & -3 & -1 \\ 0 & 0 & 1 & -2 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The parameterization is:

$$(-2r + 3s + t, r, 2s - 5t, s, t) = r(-2, 1, 0, 0, 0) + s(3, 0, 2, 1, 0) + t(1, 0, -5, 0, 1)$$

We thus find the following bases for the row, column, and null spaces of these matrices:

1. $A$
   1. Row Space: $(1, 0, 0), (0, 1, 0), (0, 0, 1)$
   2. Column Space: $(7, 4, 1, 8), (-2, 1, -1, -3), (-1, -3, 0, 2)$
   3. Null Space: $\varnothing$
2. $B$
   1. Row Space: $(1, 0, 0, 0), (0, 1, 0, 0), \left(0, 0, 1, \frac{19}{4}\right)$
   2. Column Space: $(8, -8, 8, 8), (8, 8, -8, 8), (8, 8, 8, -8)$
   3. Null Space: $\left(0, 0, -\frac{19}{4}, 1\right)$
3. $C$
   1. Row Space: $(1, 2, 0, -3, -1), (0, 0, 1, -2, 5)$
   2. Column Space: $(1, 2, -1), (0, 1, -1)$
   3. Null Space: $(-2, 1, 0, 0, 0), (3, 0, 2, 1, 0), (1, 0, -5, 0, 1)$

#### 12.7.2. (ii)

| $M$ | Rank | Nullity | Rank + Nullity | Num Columns |
|---|---|---|---|---|
| $A$ | 3 | 0 | 3 | 3 |
| $B$ | 3 | 1 | 4 | 4 |
| $C$ | 2 | 3 | 5 | 5 |

Notice that rank + nullity always equals the number of columns. This is an example of the Rank-Nullity theorem in action.

## 12.8. Problem 8

#### 12.8.1. 1

Basis: $(1,0,0,0),(0,0,1,0)$

Dimension: 2

Note: In the actual exam do check that the vectors are linearly independent. If not, discard some of them.

#### 12.8.2. 2

Basis: $(1,0,0,1),(0,1,1,0)$

Dimension: 2

#### 12.8.3. 3

Basis: $(12,6,4,3)$

Dimension: 1

#### 12.8.4. 4

This can be expressed as the following augmented matrix:

$$\left(\begin{array}{cccc|c} 2 & 3 & 1 & 1 & 0 \\ -3 & 1 & 4 & -7 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 1 & 0 \\ -3 & 1 & 4 & -7 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 7 & 7 & -7 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

A parameterization for the solution set is:

$$(r-2s, -r+s, r, s) = r(1,-1,1,0) + s(-2,1,0,1)$$

Basis: $(1,-1,1,0), (-2,1,0,1)$

Dimension: 2

**12.8.5. 5**

This can be expressed as the following augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 0 & 0 \\ 1 & -1 & 0 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 0 & 0 \\ 0 & -2 & 1 & -1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} & 0 \end{array}\right)$$

A parameterization for the solution set is:

$$\left(\frac{1}{2}r + \frac{1}{2}s, \frac{1}{2}r - \frac{1}{2}s, r, s\right) = r\left(\frac{1}{2}, \frac{1}{2}, 1, 0\right) + s\left(\frac{1}{2}, -\frac{1}{2}, 0, 1\right)$$

Basis: $\left(\frac{1}{2}, \frac{1}{2}, 1, 0\right), \left(\frac{1}{2}, -\frac{1}{2}, 0, 1\right)$

Dimension: 2

## 12.9. Problem 9

Notice that $X$ must be a $101 \times 1$ matrix. In other words, $X$ is isomorphic to a column vector with 101 entries. Similarly, $C$ is also isomorphic to a column vector with 100 entries.

This is, therefore, equivalent to the following augmented matrix:

$$(A \,|\, \vec{c})$$

Since the rank of $A$ is 100, its nullity must be $101 - 100 = 1$. Therefore, the row-echelon form of the matrix $A$ always has one free parameter. Therefore, there is no $C$ for which a unique $X$ satisfies the equation.

Note: You can also argue by noting that, since the nullity is greater than 0, there is a nonzero matrix $B$ so that $AB = \mathbf{0}$. From there it is easy to see that if $AX = C$, then $A(X + B) = C$

too, implying nonuniqueness of solutions. This argument is generalizable, which is its main appeal.

## 12.10. Problem 10

### 12.10.1. (a)

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$$

### 12.10.2. (b)

Since $A$ has rank 1, we can form a basis for the row space that contains exactly one vector.

Suppose the basis vector for the row space of $A$ is row vector $\vec{y}$.

Therefore, the rows of matrix $A$ are scalar multiples of $\vec{y}$. That is, it is of the following form:

$$A = \begin{pmatrix} x_1\vec{y} \\ x_2\vec{y} \\ \vdots \\ x_m\vec{y} \end{pmatrix}$$

Thus:

$$A = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} (\vec{y})$$

Where $(\vec{y})$ is a row vector (isomorphic to a matrix with one row).

We can then set $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$ and $Y = (\vec{y})$ to achieve the desired result.

## 12.11. Problem 11

First, it is clear that the two vectors are linearly independent since neither of them are constant multiples of the other.

Applying Gauss-Jordan Elimination on the matrix:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array}\right)$$

A parameterization for the solution set of this system is thus:

$$(s, -r - s, r, s) = r(0, -1, 1, 0) + s(1, -1, 0, 1)$$

Therefore, the vectors $(0, -1, 1, 0), (1, -1, 0, 1)$ form a spanning set for the solution set.

Notice that $(0,-1,1,0) = -\frac{1}{2}(1,0,-1,1) + \frac{1}{2}(1,-2,1,1)$ and $(1,-1,0,1) = \frac{1}{2}(1,0,-1,1) + \frac{1}{2}(1,-2,1,1)$.

Further, $(1,-2,1,1) = (0,-1,1,0) + (1,-1,0,1)$ and $(1,0,-1,1) = (1,-1,0,1) - (0,-1,1,0)$.

We have thus shown that $\text{span}(\{(1,-2,1,1),(1,0,-1,1)\}) = \text{span}(\{(0,-1,1,0),(1,-1,0,1)\})$.

This implies that $(1,0,-1,1),(1,-2,1,1)$ form a spanning set for the solution set of the system. ■

## 12.12. Problem 12

### 12.12.1. (a)

Substituting, we get the equations:

$$c_0 + c_1 + c_2 + c_3 = a$$

$$c_0 - c_1 + c_2 - c_3 = b$$

This corresponds to the following augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & a \\ 1 & -1 & 1 & -1 & b \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & a \\ 0 & -2 & 0 & -2 & b-a \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & a \\ 0 & 1 & 0 & 1 & \frac{1}{2}a - \frac{1}{2}b \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & \frac{1}{2}a + \frac{1}{2}b \\ 0 & 1 & 0 & 1 & \frac{1}{2}a - \frac{1}{2}b \end{array}\right)$$

Therefore, the following set of polynomials satisfy both equations:

$$\left\{\left(\frac{1}{2}a + \frac{1}{2}b - r\right) + \left(\frac{1}{2}a - \frac{1}{2}b - s\right)x + rx^2 + sx^3 \mid r,s \in \mathbb{R}\right\}$$

### 12.12.2. (b)

Substituting, we get:

$$c_0 - c_1x + c_2x^2 - c_3x^3 = -c_0 - c_1x - c_2x^2 - c_3x^3$$

$$c_0 + c_2x^2 = -c_0 - c_2x^2$$

$$2c_0 + 2c_2x^2 = 0$$

$$c_0 + c_2x^2 = 0$$

Thus, $c_0 = c_2 = 0$.

We therefore have $f(x) = c_1x + c_3x^3$.

Notice that, for any $c_1, c_3 \in \mathbb{R}$, $f(-x) = -c_1x - c_3x^3 = -f(x)$.

Therefore, $r(0,1,0,0) + s(0,0,0,1)$ is a parameterization of the set of 4-tuples. Notice that:

$$(r_1(0,1,0,0)+s_1(0,0,0,1))+(r_2(0,1,0,0)+s_2(0,0,0,1))=(r_1+r_2)(0,1,0,0)+(s_1+s_2)(0,0,0,1)$$

So this set is closed under addition.

Further notice:

$$c(r(0,1,0,0)+s(0,0,0,1))=(cr)(0,1,0,0)+(cs)(0,0,0,1)$$

Thus, this set is also closed under scalar multiplication.

Finally, it is clearly nonempty since $(0,0,0,0)$ is in the set.

Therefore, the set forms a subspace of $\mathbb{R}^4$ of dimension 2.

Note: Alternatively, you can **appeal to the fact that the system $c_0 = c_2 = 0$ is homogenous**.

## 12.13. Problem 13

### 12.13.1. (a)

An equivalent way of saying that the rank of the matrix is less than 3 is to say that at most two of the rows in its reduced row-echelon form are nonzero. Therefore, its RREF is not the identity matrix, implying that any row-echelon form has at least one row of zeroes.

Let us apply gaussian elimination.

$$\begin{pmatrix} 2 & 3 & a \\ 1 & 3 & 1 \\ 1 & 2 & a \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 1 \\ 2 & 3 & a \\ 1 & 2 & a \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 1 \\ 0 & -3 & a-2 \\ 0 & -1 & a-1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 1 \\ 0 & -1 & a-1 \\ 0 & -3 & a-2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 1-a \\ 0 & -3 & a-2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 1-a \\ 0 & 0 & a-2+3-3a \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 1-a \\ 0 & 0 & -2a+1 \end{pmatrix}$$

The only way for one of the rows to be zero is when $a = \frac{1}{2}$. Thus, the matrix has a rank less than 3 if and only if $a = \frac{1}{2}$.

**12.13.2. (b)**

This is true. We can show this by proving the contrapositive. Suppose that the list of rows of the matrix are linearly independent. Then, the rows form a basis for the row space of the matrix. This implies that the rank of the matrix is 3.

Taking the contrapositive, if the rank were less than 3, then the rows of the matrix would be linearly dependent.

**12.13.3. (c)**

When $a = \frac{1}{2}$, the row vectors are: $\left(2, 3, \frac{1}{2}\right), (1, 3, 1), \left(1, 2, \frac{1}{2}\right)$.

We can express the last vector as a linear combination of the first two as follows:

$$\left(1, 2, \frac{1}{2}\right) = \frac{1}{3}\left(2, 3, \frac{1}{2}\right) + \frac{1}{3}(1, 3, 1)$$

## 12.14. Problem 14

**12.14.1. (a)**

Notice that $\vec{x} = (x_1, x_2, x_3, x_4) = r(1, -1, 0, 2) + s(0, 1, 1, -3) + t(1, 0, 1, -1)$.

The sum of two such elements is:

$$0, 1, 1, -3) + t_1(1, 0, 1, -1)) + (r_2(1, -1, 0, 2) + s_2(0, 1, 1, -3) + t_2(1, 0, 1, -1)) = (r_1 + r_2)(1, -1, 0, 2) + (s_1 + s_2)(0, 1, 1, -3) +$$

Which is yet another element of the solution set of $X$. Thus, the solutions of $X$ are closed under vector addition.

It is also closed under scalar multiplication, as the following computation shows:

$$c(r(1, -1, 0, 2) + s(0, 1, 1, -3) + t(1, 0, 1, -1)) = (cr)(1, -1, 0, 2) + (cs)(0, 1, 1, -3) + (ct)(1, 0, 1, -1)$$

And finally, it is nonempty, as $(0, 0, 0, 0)$ is a solution (set $r = s = t = 0$).

**12.14.2. (b)**

The system is a homogenous system because setting $r = s = t = 0$ results in the zero vector. Thus, the zero vector is part of the solution set of $X$, implying that it is homogenous.

**12.14.3. (c)**

We know that $(1, -1, 0, 2), (0, 1, 1, -3), (1, 0, 1, -1)$ spans the solution set of $X$. It remains to be seen that this spanning set is actually linearly independent (if it is, then it forms a basis of the solution set of $X$).

We can directly verify this by row-reducing the following matrix consisting of the aforementioned vectors as columns:

$$\begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \\ 2 & -3 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & -3 & -3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Since there are free parameters, the three vectors are not linearly independent. Picking only the column vectors corresponding to the columns with leading entries, we find a basis for the solution set:

$$(1, -1, 0, 2), (0, 1, 1, -3)$$

Thus, the dimension of the subspace $S$ is 2.

## 12.15. Problem 15

### 12.15.1. (a)

Reducing the matrix into row-echelon form:

$$\begin{pmatrix} 1 & 1 & 2 & 0 \\ -1 & 0 & 3 & 1 \\ 2 & 3 & 9 & 1 \\ 0 & 2 & 10 & 2 \\ 5 & 1 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 5 & 1 \\ 0 & 1 & 5 & 1 \\ 0 & 2 & 10 & 2 \\ 0 & -4 & -9 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 5 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 11 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 5 & 1 \\ 0 & 0 & 11 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since there are three nonzero rows, the rank of $A$ is 3.

### 12.15.2. (b)

This is equivalent to finding a basis for the column space of the transpose of the matrix.

Taking the transpose of the matrix:

$$A^T = \begin{pmatrix} 1 & -1 & 2 & 0 & 5 \\ 1 & 0 & 3 & 2 & 1 \\ 2 & 3 & 9 & 10 & 1 \\ 0 & 1 & 1 & 2 & 1 \end{pmatrix}$$

And row-reducing:

$$\begin{pmatrix} 1 & -1 & 2 & 0 & 5 \\ 1 & 0 & 3 & 2 & 1 \\ 2 & 3 & 9 & 10 & 1 \\ 0 & 1 & 1 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 2 & -4 \\ 0 & 5 & 5 & 10 & -9 \\ 0 & 1 & 1 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 2 & -4 \\ 0 & 0 & 0 & 0 & 11 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 2 & 0 & 5 \\ 0 & 1 & 1 & 2 & -4 \\ 0 & 0 & 0 & 0 & 11 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Therefore, the vectors in the first, second, and fifth columns of $A^T$ form a basis for the column space of $A^T$.

Equivalently, the vectors in the first, second, and fifth rows of $A$ form a basis for the rows space of $A$.

In other words, the vectors $(1, 1, 2, 0), (-1, 0, 3, 1), (5, 1, 1, 1)$ form a basis for the row space of $A$.

**(Challenging Problems Begin Here)**

## 12.16. Problem 16

### 12.16.1. (a)

This corresponds to the following augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right)$$

Row-reducing:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 3 & 2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & \frac{2}{3} & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & 0 \\ 0 & 1 & \frac{2}{3} & 0 \end{array}\right)$$

Thus, a parameterization for the set of solutions is:

$$\left(-\frac{5}{3}s, -\frac{2}{3}s, s\right)$$

**12.16.2. (b)**
Order the vectors of $Z$ in some way. Let the $j$th entry of the $i$th vector in $Z$ be $z_{ij}$.

Consider the $m \times n$ matrix $Z' = (z_{ij})$.

Notice that the set of vectors that have a dot product of zero with all of the vectors of $Z$ are elements of the following augmented matrix:

$$(Z' \,|\, \mathbf{0})$$

This is because the equation $Z'\vec{v} = \vec{0}$ exactly characterizes the orthogonality condition imposed within the set $Z^\perp$: Matrix multiplication represents taking dot products of columns of the right matrix and rows of the left matrix.

Crucially, the augmented matrix formed is always **homogenous**. Thus, using what we have proved in class, the set of solutions forms a subspace of $\mathbb{R}^n$, as needed.

**12.16.3. (c)**
First, note that $Z$ froms the rows of $Z'$. Thus, $\dim(\text{span}(Z)) = \text{rank}(Z')$ by definition (since the rank of a matrix is the dimension of the space spanned by its rows).

Furthermore, $Z^\perp$ is composed of the vectors corresponding to the solution set of:

$$(Z' \,|\, \mathbf{0})$$

But this is just the null space of $Z'$. Thus, $\dim(Z^\perp) = \text{nullity}(Z')$.

Finally, by the rank-nullity theorem, we conclude:

$$\dim(\text{span}(Z)) + \dim(Z^\perp) = \text{rank}(Z') + \text{nullity}(Z') = m$$

As needed. ■

## 12.17. Problem 17

**12.17.1. (i)**
Let us row-reduce the matrix with these vectors as columns:

$$\begin{pmatrix} 1 & 2 & 0 & 2 \\ -2 & -5 & 5 & 1 \\ 0 & -3 & 15 & 15 \\ 0 & -2 & 10 & 8 \\ 3 & 6 & 0 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & -1 & 5 & 5 \\ 0 & -3 & 15 & 15 \\ 0 & -2 & 10 & 8 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & -1 & 5 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & -1 & 5 & 5 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Thus, the vectors $(1, -2, 0, 0, 3), (2, -5, -3, -2, 6), (2, 1, 15, 8, 6)$ form a basis for $V$.

**12.17.2. (ii)**

I claim that $(1, -2, 0, 0, 3), (2, -5, -3, -2, 6), (2, 1, 15, 8, 6), (0, 0, 1, 0, 0), (0, 0, 0, 0, 1)$ is an extension of the basis for $V$ that spans the entirety of $\mathbb{R}^5$.

To check that these vectors span the entirety of $\mathbb{R}^5$, it suffices to check that the following matrix:

$$\begin{pmatrix} 1 & 2 & 2 & 0 & 0 \\ -2 & -5 & 1 & 0 & 0 \\ 0 & -3 & 15 & 1 & 0 \\ 0 & -2 & 8 & 0 & 0 \\ 3 & 6 & 6 & 0 & 1 \end{pmatrix}$$

has the identity matrix as its RREF.

Performing Gauss-Jordan Elimination:

$$\begin{pmatrix} 1 & 2 & 2 & 0 & 0 \\ 0 & -1 & 5 & 0 & 0 \\ 0 & -3 & 15 & 0 & 0 \\ 0 & -2 & 8 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 2 & 0 & 0 \\ 0 & -1 & 5 & 0 & 0 \\ 0 & -3 & 15 & 1 & 0 \\ 0 & -2 & 8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 2 & 0 & 0 \\ 0 & -1 & 5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 2 & 0 & 0 \\ 0 & -1 & 5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 2 & 0 & 0 \\ 0 & -1 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

## 12.18. Problem 18

### 12.18.1. (a)

We can apply Gaussian Elimination on the matrix consisting of these vectors as columns:

$$\begin{pmatrix} 1 & 3 & -1 \\ 3 & -1 & 2 \\ 1 & 1 & 0 \\ 3 & -1 & 2 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & -1 \\ 3 & -1 & 2 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & -1 \\ 0 & -10 & 5 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & -1 \\ 0 & -10 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore, $(1, 3, 1, 3, 1), (3, -1, 1, -1, 1)$ forms a basis for $\text{span}\{\vec{u}, \vec{v}, \vec{w}\}$.

**12.18.2. (b)**

We know that a parameterization for this set of solutions is:

$$r(1, 3, 1, 3, 1) + s(3, -1, 1, -1, 1)$$

We want to know how the variables relate to one another. Notice that each variable is associated with a 2d vector — for instance, $x_1 \sim (1, 3)$ because $x_1 = r + 3s$.

To see how the variables relate to one another, we set-up the following augmented matrix and reduce to determine, explicitly, the linear dependence between the variables (we know that any solution must correspond to a satisfiable system, so for any rows with all zeros, the numbers to the right of the bar must equal zero. This will give us the equations needed.):

$$\left(\begin{array}{cc|c} 1 & 3 & a_1 \\ 3 & -1 & a_2 \\ 1 & 1 & a_3 \\ 3 & -1 & a_4 \\ 1 & 1 & a_5 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 3 & a_1 \\ 3 & -1 & a_2 \\ 1 & 1 & a_3 \\ 0 & 0 & a_4 - a_2 \\ 0 & 0 & a_5 - a_3 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 3 & a_1 \\ 0 & -10 & a_2 - 3a_1 \\ 0 & -2 & a_3 - a_1 \\ 0 & 0 & a_4 - a_2 \\ 0 & 0 & a_5 - a_3 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 3 & a_1 \\ 0 & -10 & a_2 - 3a_1 \\ 0 & 10 & 5a_1 - 5a_3 \\ 0 & 0 & a_4 - a_2 \\ 0 & 0 & a_5 - a_3 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 3 & a_1 \\ 0 & -10 & a_2 - 3a_1 \\ 0 & 0 & 2a_1 - 5a_3 + a_2 \\ 0 & 0 & a_4 - a_2 \\ 0 & 0 & a_5 - a_3 \end{array}\right)$$

Therefore, the solution set of the following system of linear equations is equal to the span of the three vectors:

$$2x_1 + x_2 - 5x_3 = 0$$

$$x_4 - x_2 = 0$$

$$x_5 - x_3 = 0$$

## 12.19. Problem 19

We proceed by induction on the length of the list. Let $P(k)$ be the statement:

$P(k)$ : Any list of vectors with length at most $k$ satisfying the property is linearly independent

$P(1)$ is clear as any list containing a single nonzero vector is linearly independent.

Suppose $P(k-1)$. We want to show $P(k)$ is true.

Consider a particular but arbitrarily chosen list of $k$ nonzero vectors $\vec{v_1}, \vec{v_2}, \vec{v_3}, ..., \vec{v_k}$.

By the induction hypothesis, we know that $\vec{v_2}, \vec{v_2}, \vec{v_3}, ..., \vec{v_k}$ is linearly independent. Equivalently, the only solution to the following equation:

$$\sum_{i=2}^{k} c_i \vec{v_i} = \vec{0}$$

Is the trivial one.

We want to show that the only solution to the following equation is also the trivial one:

$$\sum_{i=1}^{k} c_i \vec{v_i} = \vec{0}$$

Let the first nonzero index of the $i$th vector be $f(i)$. Therefore $\forall j \in \mathbb{Z}_{\geq 0}, j < f(i) \rightarrow (\vec{v_i})_j = 0$, and $(\vec{v_i})_{f(i)} \neq 0$. Also, $i < j \rightarrow f(i) < f(j)$.

Notice that for all integers $i$, when $1 < i$, $f(1) < f(i)$. Further, $\vec{v}_{i_{f(1)}} = 0$.

By the definition of vector equality, for the following to be satisfied:

$$\sum_{i=1}^{k} c_i \vec{v_i} = \vec{0}$$

It is necessary that:

$$\left(\sum_{i=1}^{k} c_i \vec{v_i}\right)_{f(1)} = 0$$

Performing some manipulations:

$$\left(\sum_{i=1}^{k} c_i \vec{v_i}\right)_{f(1)} = 0$$

$$\sum_{i=1}^{k} c_i (\vec{v_i})_{f(1)} = 0$$

$$c_1(\vec{v_1})_{f(1)} + \sum_{i=2}^{k} c_i (\vec{v_i})_{f(1)} = 0$$

$$c_1(\vec{v_1})_{f(1)} + \sum_{i=2}^{k} c_i \cdot 0 = 0$$

$$c_1(\vec{v_1})_{f(1)} = 0$$

Since $(\vec{v_1})_{f(1)} \neq 0$, by the zero-product property, $c_1 = 0$.

Therefore, $\sum_{i=1}^{k} c_i \vec{v_i} = \sum_{i=2}^{k} c_i \vec{v_i} = \vec{0}$. From our induction hypothesis, this implies that $c_2 = c_3 = \cdots = c_k = 0$. We have thus shown that the only solution is:

$$c_1 = c_2 = c_3 = \cdots = c_k = 0$$

Which is what we wanted to show.

This completes the induction proof. ■

## 12.20. Problem 20

Without loss of generality, let us assume that $m \leq n$ — if $m \geq n$, we can apply the argument below in reverse — instead of turning $B$ to $B'$, we instead turn $B'$ to $B$.

We proceed by induction on $0 \leq i \leq m$, where the statement $P(i)$ is:

$$P(i) : \exists S \subset B, |S| = m - i \wedge \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup S) = V$$

First, the base case states that:

$$P(0) : \exists S \subset B, |S| = m - 0 \wedge \text{span}(\varnothing \cup S) = V$$

This is trivially true because we can set $S = B$.

Next, the inductive step. Suppose that $P(i-1)$ were true for some particular but arbitrarily chosen $i$ such that $1 \leq i \leq m$. We are given:

$$P(i-1) : \exists S \subset B, |S| = m - i + 1 \wedge \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S) = V$$

We want to show:

$$P(i) : \exists S \subset B, |S| = m - i \wedge \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup S) = V$$

Let the set $S'$ be a set satisfying the property:

$$|S'| = m - i + 1 \wedge \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S') = V$$

Number the elements of the set $S'$ as: $\vec{v_1}, \vec{v_2}, ..., \overrightarrow{v_{m-i+1}}$.

Since $\text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S') = V$ and $\overrightarrow{w_i} \in V$, we can express $\overrightarrow{w_i}$ as a linear combination of the vectors in the spanning set. That is, there exist real coefficients satisfying the following equation:

$$\overrightarrow{w_i} = \sum_{j=1}^{j=i-1} a_j \overrightarrow{w_j} + \sum_{j=1}^{j=m-i+1} b_j \vec{v_j}$$

I claim that there exists an integer $j$ satisfying both $1 \leq j \leq m - i + 1$ and $b_j \neq 0$. Suppose not. Then, the equation above simplifies to:

$$\overrightarrow{w_i} = \sum_{j=1}^{j=i-1} a_j \overrightarrow{w_j}$$

Implying that $\{\overrightarrow{w_1}, \overrightarrow{w_2}, ..., \overrightarrow{w_i}\}$ is linearly dependent.

This is a contradiction, since we have assumed that $B'$ is a basis and is thus linearly independent. Therefore, there exists a $j$ such that $b_j \neq 0$.

Suppose $j^*$ were this integer. That is, suppose $b_{j^*} \neq 0$.

Further let $N$ be the set such that $N = S' - \{\overrightarrow{v_{j^*}}\}$.

I claim that

$$\text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup N) = \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S') = V$$

To show this, we show that every vector in $\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup N$ can be expressed as a linear combination of the vectors in $\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S'$ and vice versa.

For the vectors $\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\}$ and the vectors $\vec{v_j}$ for $j \neq j^*$, this is trivial as they are in both spanning sets. It thus suffices to show that $\overrightarrow{v_{j^*}} \in \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup N)$ and $\overrightarrow{w_i} \in \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S')$.

Recall the following equation:

$$\overrightarrow{w_i} = \sum_{j=1}^{j=i-1} a_j \overrightarrow{w_j} + \sum_{j=1}^{j=m-i+1} b_j \vec{v_j}$$

We can rewrite it as follows:

$$\overrightarrow{w_i} = b_{j^*} \overrightarrow{v_{j^*}} + \sum_{j=1}^{j=i-1} a_j \overrightarrow{w_j} + \sum_{\substack{1 \leq j \leq m-i+1 \\ j \neq j^*}} b_j \vec{v_j}$$

Thus, $\overrightarrow{w_i} \in \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S')$.

Since $b_{j^*} \neq 0$, we can rearrange the above equation as:

$$b_{j^*} \overrightarrow{v_{j^*}} = \overrightarrow{w_i} - \sum_{j=1}^{j=i-1} a_j \overrightarrow{w_j} - \sum_{\substack{1 \leq j \leq m-i+1 \\ j \neq j^*}} b_j \vec{v_j}$$

$$\overrightarrow{v_{j^*}} = \left(\frac{1}{b_{j^*}}\right)\overrightarrow{w_i} - \sum_{j=1}^{j=i-1}\left(\frac{a_j}{b_{j^*}}\right)\overrightarrow{w_j} - \sum_{\substack{1\le j\le m-i+1 \\ j\neq j^*}}\left(\frac{b_j}{b_{j^*}}\right)\vec{v_j}$$

$$\overrightarrow{v_{j^*}} = \left(\frac{1}{b_{j^*}}\right)\overrightarrow{w_i} + \sum_{j=1}^{j=i-1}\left(-\frac{a_j}{b_{j^*}}\right)\overrightarrow{w_j} + \sum_{\substack{1\le j\le m-i+1 \\ j\neq j^*}}\left(-\frac{b_j}{b_{j^*}}\right)\vec{v_j}$$

Thus, $\overrightarrow{v_{j^*}} \in \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup N)$.

We therefore conclude that

$$\text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup N) = \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_{i-1}}\} \cup S') = V$$

Further, since $N$ is obtained by removing $\overrightarrow{v_{j^*}}$ from $S'$, $|N| = |S'| - 1 = (m - i + 1) - 1 = m - i$. And finally, $N \subset S \subset B$

We have thus shown, by construction, that:

$$\exists S \subset B, |S| = m - i \wedge \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup S) = V$$

Explicitly, $S = N$ satisfies this condition.

Therefore, $P(i)$ is true for integers $1 \le i \le m$.

Consider $P(m)$. It states that:

$$P(m) : \exists S \subset B, |S| = m - m = 0 \wedge \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_m}\} \cup S) = V$$

Since $|S| = 0$, $S = \varnothing$. Therefore:

$$\text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_m}\}) = \text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_i}\} \cup S) = V$$

Now, I claim that $n = m$. Suppose not. Then, $m < n$.

Since $\text{span}(\{\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_m}\}) \in V$ and $\overrightarrow{w_n} \in V$, we can express $\overrightarrow{w_n}$ as a linear combination of the vectors $\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_m}$ (and note that since $m < n$, $\overrightarrow{w_n}$ is not in this list). But this implies that $\overrightarrow{w_1}, \overrightarrow{w_2}, \overrightarrow{w_3}, ..., \overrightarrow{w_n}$ is linearly dependent and thus is not a basis, contradiction.

Therefore, $n = m$, as needed. ■

## 12.21. Problem 21

### 12.21.1. (a)

This is true. Suppose we express $\vec{x}$ using two sets of coefficients:

$$\vec{x} = \sum_{i=1}^{n} c_i\vec{v_i} = \sum_{i=1}^{n} d_i\vec{v_i}$$

Thus,

$$\sum_{i=1}^{n}(c_i - d_i)\vec{v_i} = \vec{0}$$

Since the $\vec{v_i}$ s form a basis, they must be linearly independent. Equivalently, this means that the only way to express $\vec{0}$ as a linear combination of these vectors is the trivial solution.

This implies that $c_i - d_i = 0$ and thus $c_i = d_i$.

Therefore, there is a unique set of coefficients so that:

$$\vec{x} = \sum_{i=1}^{n} c_i \vec{v_i}$$

**12.21.2. (b)**
Let $A = \left(a_{i,j}\right)$ be an $n \times n$ matrix.

We can thus express the equation:

$$\vec{x} = \sum_{i=1}^{n} c_i \vec{v_i}$$

As follows:

$$A^T \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_n \end{pmatrix} = \vec{x}$$

Since the vectors form a basis of $\mathbb{R}^n$ and also comprise the columns of $A^T$, $A^T$ has rank $n$. Therefore, its RREF has $n$ leading entries, implying that its RREF is the identity matrix. This implies, by the characterizations of invertibility, that $A^T$ is invertible. Thus, $\det(A^T) \neq 0$.

By Cramer's Rule, we can thus solve for the $x_i$ s:

$$x_i = \frac{\begin{vmatrix} (\text{col } 1) & (\text{col } 2) & (\text{col } 3) & \cdots & (\text{col } i) & \cdots & (\text{col } n) \\ a_{1,1} & a_{2,1} & a_{3,1} & \cdots & x_1 & \cdots & a_{n,1} \\ a_{1,2} & a_{2,2} & a_{3,2} & \cdots & x_2 & \cdots & a_{n,2} \\ a_{1,3} & a_{2,3} & a_{3,3} & \cdots & x_3 & \cdots & a_{n,3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & a_{3,n} & \cdots & x_n & \cdots & a_{n,n} \end{vmatrix}}{\begin{vmatrix} a_{1,1} & a_{2,1} & a_{3,1} & \cdots & a_{n,1} \\ a_{1,2} & a_{2,2} & a_{3,2} & \cdots & a_{n,2} \\ a_{1,3} & a_{2,3} & a_{3,3} & \cdots & a_{n,3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & a_{3,n} & \cdots & a_{n,n} \end{vmatrix}}$$

## 12.22. Problem 22

**12.22.1. (a)**

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

This matrix has rank one, so we expect all of the $2-$ and $3-$ minors are zero. Notice that any such minor matrix contains at least one row of zeros, so indeed the determinant of any such minor matrix is zero.

**12.22.2. (b)**

**12.22.2.1. Forward Direction**
Let us first prove the forward direction.

Suppose that the rank of $X$ is less than $k$. Thus, picking any $k$ vectors as rows results in a list of row vectors that are linearly dependent. Further, these $k$ vectors form a $k \times n$ submatrix of $X$. Call this submatrix $X'$. Thus, the row vectors of $X'$ are linearly dependent.

These $k$ vectors all live in $\mathbb{R}^n$. Now, further suppose we pick $k$ entries from each vector, such that the entries picked are along the same dimension. This also constructs a particular but arbitrarily chosen $k$-minor of the matrix, as well as a list of $k$ row vectors that live in $\mathbb{R}^k$. Call the resulting $k$-minor matrix $X''$.

We note that the act of picking $k$ entries of the vector could be represented as a projection matrix $P$. Formally, suppose that for the $i$th entry of the smaller vector, we pick entry $f(i)$ of the original vector.

We construct the projection matrix $P$ as follows. Let $P$ be a $k \times n$ matrix. Then, an entry $(P)_{i,j}$ of $P$ is one iff $j = f(i)$ and zero otherwise.

Let the row vectors of $X'$ be numbered as $\vec{v_1}, \vec{v_2}, \vec{v_3}, ..., \vec{v_k}$. The corresponding row vectors in $X''$ are $P\vec{v_1}, P\vec{v_2}, P\vec{v_3}, ..., P\vec{v_k}$.

Since the row vectors of $X'$ are linearly dependent, there is a nontrivial linear combination that results in the zero vector:

$$\sum_{i=1}^{k} c_i \vec{v_i} = \vec{0}_{\mathbb{R}^n}$$

Left-multiplying both sides by the projection matrix $P$, and we find the zero vector of $\mathbb{R}^k$ expressed as a nontrivial linear combination of the row vectors of $X''$:

$$P \sum_{i=1}^{k} c_i \vec{v_i} = P\vec{0}_{\mathbb{R}^n}$$

$$\sum_{i=1}^{k} c_i (P\vec{v_i}) = \vec{0}_{\mathbb{R}^k}$$

Thus, the rank of $X''$ is less than $k$, implying that its RREF contains at least one zero. Therefore, $X''$ is singular and thus $\det(X'') = 0$.

However, since $X''$ were arbitrarily-chosen, all $k$-minor matrices must have determinant zero, as needed.

#### 12.22.2.2. Backward Direction

Now, let us show the backward direction.

The proof is by contradiction.

We want to show that if all the $k$-minors of $X$ are zero, then its rank must be less than $k$.

Suppose not. Then, all the $k$-minors of $X$ are zero but its rank is at least $k$.

Consider the RREF of $X$. It must have at least $k$ leading entries because the rank of $X$ is at least $k$.

Suppose the $i$th column of $X$ with a leading entry were column $c(i)$. Now, consider the $m \times k$ submatrix formed by only picking columns $c(1), c(2), c(3), ..., c(k)$ of $X$. Let this submatrix be $X'$.

Clearly, the rank of $X'$ is $k$ because we have picked the columns with leading entries.

Now consider the RREF of $X'^T$. It must have exactly $k$ leading entries because the rank of $X'^T$ is $k$. Suppose the $i$th column of $X'^T$ with a leading entry were column $r(i)$ (note that this is row $r(i)$ of $X'$). Consider the $k \times k$ submatrix of $X'^T$ formed by only picking columns $r(1), r(2), r(3), ..., r(k)$ of $X'^T$. Let this submatrix be $X''$.

Clearly, the rank of $X''$ is $k$ because we have picked the columns with leading entries. Since its rank is $k$, the RREF of $X''$ must have exactly $k$ leading entries, but since $X''$ is a $k \times k$ matrix, this simply implies that the RREF of $X''$ is the identity matrix. Thus, $X''$ is invertible and $\det(X'') \neq 0$. Therefore, $\det(X''^T) \neq 0$.

Further, $X''^T$ is a $k$-minor matrix of $X$ whose determinant is nonzero, contradiction.

We thus conclude the result. ■